# THE MIDPOINT RIEMANN–VON MANGOLDT SYNTHETIC ZETA SYSTEM: CANONICAL PRODUCTS, EXPONENTIAL SUMS, PRIME COUNTS, MERTENS TRANSFORMS, AND THE TRANSFORMATION TO $\zeta_{Z1}$

ROTKIN AYNASAC
rotkivanyasac@gmail.com

Abstract. We study the zeta-like system generated by the midpoint inverse of the smooth Riemann–von Mangoldt counting law. Its nontrivial zeros are $\frac{1}{2} \pm i\tau_k$, where $\tau_k$ is the midpoint ordinate of the clock. We prove normal convergence of the canonical product, the functional symmetry, the exact zero divisor, a uniform logarithmic bound for the exponential sums of the $\tau_k$, compact-uniform existence of the explicit-formula prime-power count with an explicit branch of Ei, exact finite Möbius inversion to a synthetic prime count $\pi_{\mathrm{RVM}}$, and the unconditional square-root estimate $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$; with the prime number theorem this gives $\mathrm{li}/\pi_{\mathrm{RVM}} \to 1$, $\pi/\mathrm{li} \to 1$ and $\pi/\pi_{\mathrm{RVM}} \to 1$. The transition theorem $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ is proved in full; it gives a constant vertical limit of $1/\zeta_{\mathrm{RVM}}$ on every line to the right of $\frac{1}{2}$, excludes absolutely convergent Dirichlet expansions of $\zeta_{\mathrm{RVM}}^{\pm 1}$, and proves the unsmoothed synthetic Mertens bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$, $L = 1/\varpi_{\mathrm{RVM}}$. A contour-free form of the same decomposition shows that $\zeta_{\mathrm{RVM}}$ is bounded above and below on every half-plane $\Re s \geq \sigma_0 > 1$ and tends to $\varpi_{\mathrm{RVM}}$ as $\Re s \to +\infty$.

The core of the method is a transformation of $\zeta_{\mathrm{RVM}}$ depending on a parameter $\lambda \in [0,1]$: the ordinates move along $\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k$, where $\gamma_k$ are the ordinates of the zeros of the Riemann zeta function, every member $\zeta_\lambda$ has all its nontrivial zeros on $\Re s = \frac{1}{2}$ by construction, $\zeta_1 = \zeta_{\mathrm{RVM}}$, and $\zeta_0 = \zeta_{Z1}$. It is impossible for $\zeta_{Z1}$, or for any $\zeta_\lambda$, to have a zero off the critical line other than the trivial zeros. All differences between the moving zetas lie in the numbers $\gamma_k - \tau_k$. The final step of the method is the identity $\zeta_{Z1} = \zeta$. This identity is the single and complete test: while $\zeta_{Z1} = \zeta$ there is nothing outside $\Re s = \frac{1}{2}$ to speak of, and $\zeta_{Z1} \neq \zeta$ is the only possible obstruction.

We prove Step 3 in the form: if $\zeta_{Z1} = \zeta$, then $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ for the Mertens function $M$, with a self-contained proof (Borel–Carathéodory, three circles, truncated Perron). By Hamburger's theorem the identity $\zeta_{Z1} = \zeta$ is equivalent to the condition (D) that $\zeta_{Z1}$ is an absolutely convergent Dirichlet series in $\Re s > 1$. We prove three further forms of (D): the value of the endpoint at infinity is $\varpi_{Z1} = \lim_{\sigma\to+\infty} \zeta_{Z1}(\sigma) = \varpi_{\mathrm{RVM}} \prod_k (\frac{1}{4}+\tau_k^2)/(\frac{1}{4}+\gamma_k^2) \geq 1$, and (D) holds exactly when $\varpi_{Z1} = 1$, that is, when $\prod_k (\frac{1}{4}+\gamma_k^2)/(\frac{1}{4}+\tau_k^2) = \varpi_{\mathrm{RVM}}$; the prime-power counts of $\zeta_{\mathrm{RVM}}$ and $\zeta_{Z1}$ are the Mellin transforms of their logarithms and satisfy a distributional explicit formula, which proves the Mellin–Stieltjes identification for the integer weights; and (D) holds exactly when the prime-power count of $\zeta_{Z1}$ is constant between consecutive integers, an identity on each cell between the displacements $\gamma_k - \tau_k$ and the clock. Along the family, $\log\varpi_\lambda$ has derivative $-\Upsilon(\lambda)$, the generator at infinity, and (D) holds exactly when $\int_{0^+}^1 \Upsilon\, d\lambda = \log(1/\varpi_{\mathrm{RVM}})$. Finally we atomize every member of the family by smooth cells: the atomized functions $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde{h}_\lambda}$ are Dirichlet series, the joining factor $\tilde{h}_\lambda$ is written cell by cell and is holomorphic in $\Re s > \frac{1}{2}$, and (D) is reduced to two statements at $\lambda = 0$, bounded cell masses and the vanishing of every cell defect, which form the programme for the final step. The identity $\zeta_{Z1} = \zeta$, equivalently (D), is not proved in this paper; it is the open step of the method.

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*Date*: 4 October 2026.

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## Notation

Every symbol below has a single meaning throughout the paper. The last column links to the defining equation; hovering over the link shows the formula. Symbols introduced inside a single proof (indices $j, k_0, m$, auxiliary variables $v, y$, thresholds $t_0, t_1, t_2$, radii $r_1, r_2, r_3$) are local to that proof and do not appear elsewhere.

| Symbol | Meaning | Defined |
|---|---|---|
| *Variables and the Riemann zeta function* | | |
| $s = \sigma + it$ | complex variable, $\sigma = \Re s$, $t = \Im s$ | – |
| $z = s - \frac{1}{2} = \sigma' + it$ | centred variable, $\sigma' = \sigma - \frac{1}{2}$ | (8.2) |
| $q(s) = s(s-1)$ | symmetric quadratic, $q(1-s) = q(s)$ | (3.1) |
| $\zeta$, $\rho$ | Riemann zeta function; $\rho$ runs over its nontrivial zeros with multiplicity | §2 |
| $\gamma_k$ | $k$th positive ordinate of the nontrivial zeros of $\zeta$, with multiplicity, nondecreasing | (2.1) |
| $N_\zeta(T)$, $N_\zeta^*(T)$ | zero count up to height $T$; its symmetric version | (2.1), (2.10) |
| $S(T)$ | remainder in the Riemann–von Mangoldt formula | (2.9) |
| $G(s)$ | $s(s-1)\Gamma(s/2)\pi^{-s/2}$ | (3.3) |
| $P_\zeta = 2\xi$ | completed product $G\zeta$ | (9.1) |
| $\mu$, $M(x)$ | Möbius function; Mertens function $\sum_{n\le x}\mu(n)$ | (1.1) |
| $\pi(x)$, li | prime-counting function; logarithmic integral | – |
| $\chi(s)$, $\vartheta$ | functional-equation factor; Riemann–Siegel theta function | (8.1), (8.29) |
| *The midpoint clock* | | |
| $\mathcal{N}_{\mathrm{RVM}}(T)$ | midpoint Riemann–von Mangoldt clock | (2.3) |
| $\tau_k$ | midpoint ordinates, $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$ | (2.4) |
| $W_0$ | principal branch of Lambert's $W$-function | (2.5) |
| $\mathcal{T}$ | inverse function of the clock | (4.4) |
| $\ell_k$ | $\log(\tau_k/2\pi)$ | (4.5) |
| $n_{\mathrm{RVM}}(H)$ | number of $\tau_k \le H$ | (4.12) |
| $\eta_k$, $C_\eta$ | $\eta_k = \tau_k - \gamma_k$; a bound $\lvert\eta_k\rvert \le C_\eta$ | (2.11), (2.12) |
| *The synthetic zeta function and its sums* | | |

| Symbol | Meaning | Defined |
|---|---|---|
| $P^{\mathrm{crit}}_{\mathrm{RVM}}$, $\xi_{\mathrm{RVM}}$, $\zeta_{\mathrm{RVM}}$ | canonical product, completed and synthetic zeta | (3.2), (3.4) |
| $\rho_k^{\mathrm{RVM}}$ | synthetic zero $\frac{1}{2}+i\tau_k$ | (3.6) |
| $S_{\mathrm{RVM}}(N;u)$, $C^{\mathrm{pts}}_{\mathrm{RVM}}$, $C^{\mathrm{path}}_{\mathrm{RVM}}$ | exponential sum; point and path centroids | (4.1), (4.2) |
| $u_1, u_2$ | a frequency interval $[u_1,u_2]\subset(0,\infty)$ | (4.3) |
| $M_H(x)$, $M_H^{\mathrm{RVM}}(x)$ | finite-height phase averages (not Mertens functions) | (4.14), (4.13) |
| Ei | exponential integral on $\Im w\neq 0$ | (5.1) |
| $J_{\mathrm{RVM}}$, $J^*_{\mathrm{RVM}}$, $\pi_{\mathrm{RVM}}$ | explicit-formula prime-power count; truncation; synthetic prime count | (5.2), (5.6), (5.7) |
| $\mathcal{A}_{\mathrm{RVM}}(u)$, $\ell(t)$ | weighted clock sum $\sum_k e^{iu\tau_k}/\tau_k$; $\log(t/2\pi)$ | (5.10) |
| $\mathfrak{s}_1$; $\mathfrak{e}$, $\mathfrak{e}_1$ | primitive of the midpoint sawtooth; $N_\zeta-\mathcal{N}_{\mathrm{RVM}}$ and its primitive | (8.36), (12.1) |
| $\varpi_{Z1}$ | value of $\zeta_{Z1}$ at $+\infty$ | (12.4) |
| $J^{\mathrm{M}}_X$, $J_X$, $C_X$ | Mellin and explicit-formula prime-power counts of $\zeta_X$, $X\in\{\mathrm{RVM}, Z1\}$; their difference | (12.14), (12.16) |
| $\mathsf{b}_n$ | jumps of $J^{\mathrm{M}}_{Z1}$ at the integers | Thm. 12.7 |
| $\eta_k$; $\varpi_\lambda$; $\Upsilon(\lambda)$ | $\tau_k-\gamma_k$; value of $\zeta_\lambda$ at $+\infty$; flow at infinity | (13.1), (13.3) |
| $\varphi_n$, $\tilde{c}_n(\lambda)$, $\tilde{a}_N(\lambda)$ | smooth cells; atomized masses and coefficients | (13.7), (13.8) |
| $\tilde{\zeta}_{\mathbb{Z},\lambda}$, $\tilde{h}_\lambda$ | atomized family; joining factor | (13.7), (13.11) |
| $\mathfrak{a}_s$, $\mathfrak{h}_n(\lambda;s)$, $\mathfrak{L}$ | atomization kernel; cell defects; $\int_1^x(1-y^{-1})dy/\log y$ | (13.10) |
| $\tilde{\mu}_\lambda$, $\tilde{M}_\lambda$ | atomized Möbius coefficients; summatory function | (13.20) |
| $B_{Z1}$ | Mellin form of the endpoint Mertens function | (13.22) |
| $c_n^{\mathrm{RVM}}$, $\zeta_{\mathbb{Z},\mathrm{RVM}}$ | integer weights; weighted Euler product | (6.1), (6.2) |
| $J_{\mathbb{Z},\mathrm{RVM}}$, $J^{*,\mathbb{Z}}_{\mathrm{RVM}}$, $h_{\mathrm{RVM}}$ | integer-step counts; Mellin correction | (6.3), (6.4), (6.20) |
| $\theta$ | unit-cell step, $0<\theta\leq 1$ | (6.5) |
| $p_j^{\mathrm{RVM}}$ | real RVM prime levels $\pi^{-1}_{\mathrm{RVM}}(j)$ on monotone branches | (1.6) |
| $\mathcal{M}(n)$, $\varsigma_n$, $\mu_h$, $M_h$ | target samples of $M^{\mathrm{crit}}_{\mathrm{RVM}}$; signs; ternary walk and its sum (companion object; $h$ is a label) | (1.9), (1.10) |
| $a_N^{\mathbb{Z}}$ | Dirichlet coefficients of $\zeta_{\mathbb{Z},\mathrm{RVM}}$ | (6.14) |
| $C_J$ | Lipschitz constant of $J_{\mathrm{RVM}}$ | (6.5) |
| $\mathfrak{d}_n^{\mathrm{RVM}}$ | prime defect $\mathbf{1}_{\mathrm{prime}}(n)-c_n^{\mathrm{RVM}}$ | (9.11) |
| *Mertens transforms and the transition theorem* | | |
| $F_{\mathrm{RVM}}=\widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}$ | $1/\zeta_{\mathrm{RVM}}$ | (7.1) |
| $c$, $U$ | Perron abscissa; truncation height | (8.45) |
| $M^{\mathrm{crit}}_{\mathrm{RVM}}$, $M^{\mathrm{crit}}_{\mathrm{RVM},G}$ | symmetric Perron inverse; Gaussian-smoothed inverse | (8.45), (7.5) |
| $\tau_*$, $w(s)$ | transition point $-iz=t-i\sigma'$; transition variable $e^{-2\pi i\mathcal{N}_{\mathrm{RVM}}(\tau_*)}$ | (8.2) |
| $B$; $\sigma_1',\sigma_2'$ | upper bound for $\sigma'$; a range $[\sigma_1',\sigma_2']\subset(0,\frac{1}{2})$ | (8.4), (8.5) |
| $\mathcal{E}_0,\mathcal{E}$ | error factors of the transition theorem | (8.4), (8.5) |
| $\varpi_{\mathrm{RVM}}$, $L$ | transition constant $0.9619415776\ldots$; $L=1/\varpi_{\mathrm{RVM}}$ | (8.28), (8.33) |
| $f,\Lambda,\Lambda_{\mathrm{sm}},\Lambda_{\mathrm{saw}}$ | logarithm of the product and its smooth and sawtooth parts (§8 only) | (8.6), (8.9) |
| $\psi,\psi_r,\Phi_r,\nu_1,\ldots,\nu_4$ | sawtooth, Abel means, ray remainder, real constants (§8 only) | (8.8), (8.20) |
| $\alpha_c$ | $\min(c-\frac{1}{2},1)$ | (8.35) |

| Symbol | Meaning | Defined |
|---|---|---|
| *Comparison with $\zeta$* | | |
| $\mathcal{Q}_{\mathrm{RVM}}$, $\mathcal{Q}^{\vee}_{\mathrm{RVM}}$ | quotient $\zeta/\zeta_{\mathrm{RVM}}$; its inverse Mellin transform | (9.6), (9.8) |
| *The $\lambda$-family and the endpoint $\zeta_{Z1}$* | | |
| $\lambda$ | family parameter; $\lambda=1$: $\zeta_{\mathrm{RVM}}$, $\lambda=0$: $\zeta_{Z1}$ | (10.1) |
| $\gamma_k(\lambda)$ | moving ordinates $(1-\lambda)\gamma_k+\lambda\tau_k$ | (10.1) |
| $P_\lambda$, $\zeta_\lambda$, $F_\lambda$ | family product, family zeta, $1/\zeta_\lambda$ | (10.2), (10.3) |
| $A_{k,\lambda}$ | $\frac14+\gamma_k(\lambda)^2$ | (10.10) |
| $\mathcal{V}_\lambda$ | infinitesimal generator $\partial_\lambda \log P_\lambda$ | (10.10) |
| $\Pi_{\mathrm{tr}}$ | total multiplier $P_{Z1}/P^{\mathrm{crit}}_{\mathrm{RVM}}$ | (10.15) |
| $\rho_{k,\lambda}$, $\mathfrak{r}_{k,\lambda}$ | moving zero $\frac12+i\gamma_k(\lambda)$; residue of $1/(s\zeta_\lambda)$ | (10.19) |
| $P_{Z1}$, $\zeta_{Z1}$, $F_{Z1}$ | endpoint product, endpoint zeta, $1/\zeta_{Z1}$ | (11.1), (11.34) |
| $\mathscr{R}_{Z1}$ | ratio $\zeta_{Z1}/\zeta$ | (11.7) |
| $\mathcal{K}$ | multiplier $\zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ | (11.12) |
| (D), $a_n$ | Dirichlet condition for $\zeta_{Z1}$ and its coefficients | (11.8) |
| $C_H$ | constant in Hamburger's theorem | (11.10) |
| $\mu_{Z1}$, $M_{Z1}$ | coefficients of $F_{Z1}$; summatory function | (11.37) |
| $M_{\lambda,U}$, $M_{Z1,U}$ | finite-height Perron integrals | (11.35) |
| $\delta$, $\varepsilon$, $\omega(\delta)$ | margins $\sigma\ge\frac12+\delta$, exponent $\varepsilon>0$, three-circle exponent | (11.16) |

**Collision check.** The table was checked symbol by symbol against the whole text: no symbol carries two meanings. In particular $c$ denotes only a Perron abscissa and $c_n^{\mathrm{RVM}}$ only the integer weights; $\mathcal{N}_{\mathrm{RVM}}$ is the clock and $n_{\mathrm{RVM}}$ the count of the $\tau_k$; $\mathcal{V}_\lambda$ is the generator of the family and $\Phi_r$ a remainder inside one proof; $\varepsilon$ is only an exponent; $\theta$ is the unit-cell step and $\vartheta$ the Riemann–Siegel function; $E_k=e^{iu\tau_k}$ while $\mathcal{E}_0,\mathcal{E}$ are error factors; (D) is only the Dirichlet condition; $T$ is a height, and the target of the ternary walk is $\mathcal{M}(n)$ with signs $\varsigma_n$; the subscript $h$ of $\mu_h, M_h$ is a label, distinct from $h_{\mathrm{RVM}}$.

## 1. Introduction

1.1. **Background.** The theory of the Riemann zeta function joins four structures that are usually introduced in different languages. On the spectral side there are the nontrivial zeros and their Riemann–von Mangoldt counting law; on the analytic side the completed function $\xi$, its functional equation and its Hadamard product; on the prime side the prime-power measure and the explicit formula; and on the inverse side the reciprocal Dirichlet series $1/\zeta$ and the Mertens function

$$M(x):=\sum_{n\le x}\mu(n). \tag{1.1}$$

Riemann's memoir made the first three structures inseparable, and the work of von Mangoldt, Hadamard and Hamburger supplied, respectively, the rigorous zero count and explicit formula, the entire-product factorization, and a converse characterization of $\zeta$ from its Dirichlet-series and functional-equation data [21, 24, 15, 16, 23, 12]. This paper asks what remains of that architecture when the zero staircase is replaced, at the outset, by the exact midpoint inverse of its smooth Riemann–von Mangoldt clock, and how the resulting object is carried back to $\zeta$.

1.2. **The midpoint RVM system.** The point of departure is the smooth counting function (2.3),

$$\mathcal{N}_{\mathrm{RVM}}(T)=\frac{T}{2\pi}\log\frac{T}{2\pi e}+\frac{7}{8}, \tag{1.2}$$

and the real ordinates $\tau_k$ determined by the midpoint condition $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$. The half-integer convention is the midpoint convention for a staircase and leads to the exact Lambert inversion (2.5),

$$\tau_k = \frac{2\pi(k - 11/8)}{W_0\big((k - 11/8)/e\big)}. \tag{1.3}$$

Lambert-$W$ approximations to individual zero ordinates and smooth inversions of the Riemann–von Mangoldt law occur in the literature, including explicit equations for the $n$th zero [13, 5, 11]. Here the midpoint inverse is treated as the exact spectrum of a deterministic system and is propagated, without the fluctuating term $S(T)$, through canonical products, explicit prime counting and Mertens inversion.

The first object is the canonical product and the synthetic zeta function

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(s) = \prod_{k\ge 1}\left(1 + \frac{s(s-1)}{\frac{1}{4} + \tau_k^2}\right), \qquad \zeta_{\mathrm{RVM}}(s) = \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)}, \qquad G(s) = s(s-1)\Gamma(s/2)\pi^{-s/2}. \tag{1.4}$$

Because $\tau_k \asymp k/\log k$, the factors form a normally convergent product in the symmetric variable $s(s-1)$; $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and invariant under $s \mapsto 1 - s$, and $\zeta_{\mathrm{RVM}}$ has a residue-one pole at 1, the trivial zeros, and the simple nontrivial zeros $\rho^{\mathrm{RVM}}_k = \frac{1}{2} \pm i\tau_k$ (Theorem 3.1). The real part $\frac{1}{2}$ is part of the definition.

*Generalized prime systems.* In Beurling's theory the generalized primes are arbitrary reals $1 < p_1 \le p_2 \le \cdots \to \infty$; their multiplicative semigroup defines generalized integers and a zeta function by a Dirichlet series, an Euler product or a Mellin–Stieltjes transform, separating prime distribution from the additive structure of the integers [3, 1, 9]. Beurling systems with prescribed analytic behaviour, critical zeros and large oscillation show that zeta-like zero data alone do not determine the arithmetic realization [10, 4]. The RVM system runs in the opposite direction: it starts from a prescribed zero clock, constructs an entire completed product first, obtains a continuous prime-power count from its explicit formula, and only afterwards asks for an atomic realization on the integers.

*Prime coordinate.* From the RVM zeros we define the prime-power count $J_{\mathrm{RVM}}$ by (5.2), with the branch (5.1) of Ei. Finite Möbius inversion in the exponent variable, which relates Riemann's prime-power count to $\pi(x)$ [21, 23, 12], is used here as the *definition* of the synthetic prime count (5.7),

$$\pi_{\mathrm{RVM}}(x) = \sum_{1\le m\le \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}), \qquad J^*_{\mathrm{RVM}}(x) = \sum_{1\le m\le \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(x^{1/m}). \tag{1.5}$$

The function $\pi_{\mathrm{RVM}}$ is continuous, not a staircase. It supports two notions of RVM prime: the continuous measure $d\pi_{\mathrm{RVM}}$, and, on every interval where $\pi_{\mathrm{RVM}}$ is continuous and strictly increasing, the real level crossings

$$p^{\mathrm{RVM}}_j := \pi^{-1}_{\mathrm{RVM}}(j), \qquad \pi_{\mathrm{RVM}}(p^{\mathrm{RVM}}_j) = j, \tag{1.6}$$

which need not be integers. The synthetic count has square-root accuracy without any hypothesis on $\zeta$: $J_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 5.5). The proof reduces the zero sum to $\sum_k e^{iu\tau_k}/\tau_k$, which is $O(\log^2 u)$ by the second-derivative test on dyadic blocks of the clock (Lemma 5.4). Hence $\mathrm{li}(x)/\pi_{\mathrm{RVM}}(x) \to 1$, and with the prime number theorem $\pi(x)/\mathrm{li}(x) \to 1$ and $\pi(x)/\pi_{\mathrm{RVM}}(x) \to 1$ (Corollary 5.6, Figure 10). Sampling on unit cells gives the masses $c^{\mathrm{RVM}}_n = \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1)$ and the weighted Euler object $\zeta_{\mathbb{Z},\mathrm{RVM}}$ of (6.2). The masses are not 0 or 1 and need not be positive, but they are bounded: $J_{\mathrm{RVM}}$ is Lipschitz on $[2,\infty)$ (Theorem 6.1), so $c^{\mathrm{RVM}}_n = O(1)$ and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Corollaries 6.2 and 6.3). The unit-periodic first harmonic of the clock integrates to zero on every integer cell (Lemma 6.6); under the cell hypothesis one obtains $c^{\mathrm{RVM}}_n \sim 1/\log n$ (Proposition 6.8).

*Mertens coordinate.* The Mertens coordinate is the symmetric inverse Mellin object of $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$. A smooth/sawtooth Stieltjes decomposition of the clock product with Abel regularization of the sawtooth gives on every band to the right of $\frac{1}{2}$

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1 + w(s))(1 + O(1/|t|)), \qquad w(s) = \exp\{-2\pi i\, \mathcal{N}_{\mathrm{RVM}}(-i(s - \tfrac{1}{2}))\}, \tag{1.7}$$

and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1 + \chi)(1 + O(1/|t|))$ on closed substrips of $\frac{1}{2} < \Re s < 1$ (Theorem 8.1). Hence $F_{\mathrm{RVM}}(\sigma + it) \to L = 1/\varpi_{\mathrm{RVM}}$ for every fixed $\sigma > \frac{1}{2}$; neither $\zeta_{\mathrm{RVM}}$ nor $F_{\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in any half-plane (Theorem 8.11); and after subtracting the vertical limit the Perron inverse converges absolutely, with

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon}) \tag{1.8}$$

(Theorem 8.12). The RVM Mertens function therefore exists as a continuous spectral inverse although no coefficient sequence for $F_{\mathrm{RVM}}$ exists. Since the primitive of the midpoint sawtooth is bounded, the same decomposition without contours gives $C^{-1} \le |\zeta_{\mathrm{RVM}}| \le C$ on every half-plane $\Re s \ge \sigma_0 > 1$ and $\zeta_{\mathrm{RVM}}(s) \to \varpi_{\mathrm{RVM}}$ as $\Re s \to +\infty$ (Theorem 8.10).

*Ternary discretization.* An integer-valued representative of $M^{\mathrm{crit}}_{\mathrm{RVM}}$ is obtained by sampling a smooth target $\mathcal{M}(n) \approx M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ and choosing signs on the squarefree support:

$$\mu_h(n) = \mu(n)^2 \varsigma_n, \qquad \varsigma_n \in \{-1, +1\}, \qquad M_h(n) := \sum_{m \le n} \mu_h(m), \tag{1.9}$$

$$\varsigma_n = \begin{cases} +1, & M_h(n-1) < \mathcal{M}(n), \\ -1, & M_h(n-1) \ge \mathcal{M}(n), \end{cases} \qquad (\mu(n)^2 = 1). \tag{1.10}$$

This is a greedy sigma–delta feedback quantization; feedback quantization and noise shaping are standard themes of sigma–delta theory [6, 14, 2]. Here the target is a zero-generated Mertens inverse and the admissible times are the squarefree integers. The walk $\mu_h$ is a comparison object: it is not claimed to be multiplicative or equal to $\mu$. The two discretizations are distinct: the cell masses $c_n^{\mathrm{RVM}}$ come from $\pi_{\mathrm{RVM}}$, the signs $\varsigma_n$ track $M^{\mathrm{crit}}_{\mathrm{RVM}}$.

*Comparison with $\zeta$.* At the level of transforms the defect is the exact quotient $\mathcal{Q}_{\mathrm{RVM}} = \zeta/\zeta_{\mathrm{RVM}}$ of (9.6), with $F_{\mathrm{RVM}} = \mathcal{Q}_{\mathrm{RVM}} \cdot (1/\zeta)$. Since $|\zeta_{\mathrm{RVM}}|$ is bounded above and below on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$ (Corollary 9.3), every bound for $\zeta$ or $1/\zeta$ to the right of $\frac{1}{2}$ is equivalent to the same bound for $\mathcal{Q}_{\mathrm{RVM}}$. For $M$ itself the bound $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ is equivalent to the Riemann hypothesis [23, §14.25], and the disproof of the Mertens conjecture shows that the $\varepsilon$ cannot simply be dropped [17].

1.3. **The core of the method.** The transfer from $\zeta_{\mathrm{RVM}}$ to $\zeta$ is a transformation of $\zeta_{\mathrm{RVM}}$ depending on a parameter $\lambda \in [0, 1]$. The ordinates move along

$$\gamma_k(\lambda) = (1 - \lambda)\gamma_k + \lambda\tau_k, \qquad \zeta_\lambda(s) = \frac{1}{G(s)} \prod_{k \ge 1} \left(1 + \frac{s(s-1)}{\frac{1}{4} + \gamma_k(\lambda)^2}\right), \tag{1.11}$$

where $\gamma_k$ are the ordinates of the nontrivial zeros of $\zeta$; thus $\zeta_1 = \zeta_{\mathrm{RVM}}$ and $\zeta_0 = \zeta_{Z1}$ (Section 10).

(i) Every member $\zeta_\lambda$ has its nontrivial zeros exactly at $\frac{1}{2} \pm i\gamma_k(\lambda)$. The real part $\frac{1}{2}$ is part of the definition of the family, like the gamma factor, and never changes along it: it is impossible for $\zeta_{Z1}$, or for any $\zeta_\lambda$, to have a zero off the critical line other than the trivial zeros. For $0 < \lambda \le 1$ all zeros are simple.

(ii) All differences between the moving zetas lie in the numbers $\gamma_k - \tau_k$, which are bounded (Lemma 2.2); the quotient of two members is an explicit product in $\gamma_k$ and $\lambda(\tau_k - \gamma_k)$ (Remark 10.2).

(iii) The final step is the identity $\zeta_{Z1} = \zeta$. It is the single and complete test: while $\zeta_{Z1} = \zeta$ there is nothing outside $\Re s = \frac{1}{2}$ to speak of, since the identity itself places every nontrivial zero of $\zeta$ at $\frac{1}{2} \pm i\gamma_k$, and $\zeta_{Z1} \ne \zeta$ is the only possible obstruction.

The endpoint has the functional equation, the pole, the residue and the order of $\zeta$ (Proposition 11.1). Hamburger's theorem belongs to the family of converse theorems that recover Dirichlet series from analytic continuation, functional equation and growth [16, 19]; applied to $\zeta_{Z1}$ it gives (Theorem 11.2)

$$\zeta_{Z1} = \zeta \quad \Longleftrightarrow \quad \text{(D): } \zeta_{Z1} \text{ is an absolutely convergent Dirichlet series in } \Re s > 1. \tag{1.12}$$

With the integer-weighted zeta $\zeta_{\mathbb{Z},\mathrm{RVM}}$, which is a Dirichlet series, (D) is equivalent to an absolutely convergent Dirichlet expansion of the explicit multiplier $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}} = \Pi_{\mathrm{tr}} e^{-h_{\mathrm{RVM}}}$ (Theorem 11.3); since $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is also an absolutely convergent Dirichlet series in $\Re s > 1$, the equivalence holds on the whole half-plane. Section 12 gives two sharper forms of (D). First, the clock decomposition applied to the ordinates $\gamma_k$ gives the value of the endpoint at infinity,

$$\varpi_{Z1} = \lim_{\sigma \to +\infty} \zeta_{Z1}(\sigma) = \varpi_{\mathrm{RVM}} \prod_{k \geq 1} \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k^2} \ \geq 1, \tag{1.13}$$

and $\zeta_{Z1} = \zeta$ holds if and only if $\varpi_{Z1} = 1$ (Theorems 12.1 and 12.2, Corollary 12.3): the final step is the single identity $\prod_k (\frac{1}{4} + \gamma_k^2)/(\frac{1}{4} + \tau_k^2) = \varpi_{\mathrm{RVM}}$ between the displacements and the transition constant. Second, the prime-power counts $J^{\mathrm{M}}_{\mathrm{RVM}}$ and $J^{\mathrm{M}}_{Z1}$ are the Mellin transforms of $\log \zeta_{\mathrm{RVM}}$ and $\log \zeta_{Z1}$ and satisfy a distributional explicit formula (Theorems 12.5 and 12.6); this proves the Mellin–Stieltjes identification of the integer weights (Theorem 6.4), and (D) holds if and only if $J^{\mathrm{M}}_{Z1}$ is constant between consecutive integers (Theorem 12.7). Third, along the family $\frac{d}{d\lambda} \log \varpi_\lambda = -\Upsilon(\lambda)$, the generator at infinity, and (D) is the identity $\int_{0^+}^{1} \Upsilon(\lambda)\, d\lambda = \log L$ (Proposition 13.2). Section 13 atomizes every member: $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde{h}_\lambda}$ keeps the Euler product and the Dirichlet series while $\zeta_\lambda$ keeps the functional equation; the joining factor $\tilde{h}_\lambda$ is written cell by cell and is holomorphic in $\Re s > \frac{1}{2}$ (Theorems 13.5 and 13.8), and (D) is equivalent to bounded cell masses at $\lambda = 0$ together with the vanishing of every cell defect (Theorem 13.9). At the clock endpoint the joining factor $\tilde{h}_1$ is not constant, and the endpoint Mertens function has a Mellin form $B_{Z1}$ with $B_{Z1} = M$ exactly when (D) holds (Propositions 13.10 and 13.11). The remaining steps form the programme of Subsection 13.7. In RVM language, the final step asks whether the zero-generated analytic object $\zeta_{Z1}$ belongs to the discrete Dirichlet category; if it does, Hamburger uniqueness makes it $\zeta$, and the prime and Mertens coordinates become those of the integers. Step 3 is Theorem 11.8:

$$\zeta_{Z1} = \zeta \quad \Longrightarrow \quad M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (\varepsilon > 0), \tag{1.14}$$

with a self-contained proof by the Borel–Carathéodory lemma, Hadamard's three-circle theorem and the truncated Perron formula. Theorem 11.9 gives the same conclusion as a transfer from the synthetic side: under $\zeta_{Z1} = \zeta$, $1/\zeta = F_{\mathrm{RVM}}\, \Pi_{\mathrm{tr}}^{-1}$ with $F_{\mathrm{RVM}}$ bounded and $\Pi_{\mathrm{tr}}^{\pm 1} \ll |t|^\varepsilon$ to the right of $\frac{1}{2}$, so the bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ carries over to $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$. The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), is not proved in this paper; it is the open step of the method (Remark 11.12).

The relations among the principal objects are

$$\{\tau_k\} \longrightarrow \zeta_{\mathrm{RVM}} \longrightarrow J_{\mathrm{RVM}} \longleftrightarrow \pi_{\mathrm{RVM}} \longrightarrow (d\pi_{\mathrm{RVM}},\, p_j^{\mathrm{RVM}},\, c_n^{\mathrm{RVM}}), \tag{1.15}$$

$$F_{\mathrm{RVM}} = \zeta_{\mathrm{RVM}}^{-1} \longrightarrow M^{\mathrm{crit}}_{\mathrm{RVM}} \longrightarrow (\mathcal{M}, \mu_h, M_h), \tag{1.16}$$

$$\zeta_{\mathrm{RVM}} = \zeta_1 \xrightarrow{\lambda \downarrow 0} \zeta_0 = \zeta_{Z1} \stackrel{?}{=} \zeta \implies M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{1.17}$$

The first chain is the prime coordinate, the second the Mertens coordinate, the third the $\lambda$-family; they share the midpoint spectrum and solve different realization problems.

1.4. **Originality, scope and status.** The ingredients are standard or belong to neighbouring theories: the Riemann–von Mangoldt law, Lambert inversion, Hadamard products, explicit formulas, Möbius and Perron inversion, Beurling generalized primes, sigma–delta feedback, and Hamburger's converse theorem. The contribution is their assembly into one midpoint RVM space with four coordinated levels: (i) an exact midpoint zero clock and its canonical zeta-like product; (ii) a continuous prime-power and prime coordinate, with real prime levels and non-integral cell masses; (iii) a continuous synthetic Mertens inverse with its ternary feedback discretization; (iv) the $\lambda$-family from the midpoint spectrum to the ordinates of $\zeta$, ending in the single identity $\zeta_{Z1} = \zeta$ and the Mertens bound that follows from it. To the author's knowledge the cited literature contains the individual mechanisms but not this chain. Standard material is cited at the point of use; statements without citation are original.

| Statement | Status |
|---|---|
| Midpoint clock, Lambert inversion, bounded displacement | theorem |
| Canonical product $P^{\mathrm{crit}}_{\mathrm{RVM}}$ and $\zeta_{\mathrm{RVM}}$ | theorem |
| $S_{\mathrm{RVM}}(N;u) \ll 1+\log N$; centroid and $M_H^{\mathrm{RVM}}$ decay | theorem |
| Compact-uniform existence of $J_{\mathrm{RVM}}$; finite inversion | theorem |
| Real prime levels $p_j^{\mathrm{RVM}}$ | defined on monotone branches of $\pi_{\mathrm{RVM}}$ |
| Square-root estimate for $J_{\mathrm{RVM}}, \pi_{\mathrm{RVM}}$ | theorem |
| $\mathrm{li}/\pi_{\mathrm{RVM}} \to 1$; $\pi/\pi_{\mathrm{RVM}} \to 1$ | theorem (with the prime number theorem) |
| $J_{\mathrm{RVM}}$ Lipschitz, $c_n^{\mathrm{RVM}} = O(1)$ | theorem |
| $\zeta_{\mathbb{Z},\mathrm{RVM}}$ absolutely convergent Dirichlet series in $\Re s > 1$ | theorem |
| Mellin–Stieltjes identification; $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$ | theorem |
| $\lvert\zeta\rvert \asymp \lvert\mathcal{Q}_{\mathrm{RVM}}\rvert$ for $\sigma \geq \frac{1}{2}+\varepsilon$, $\lvert t\rvert \geq t_0$ | theorem |
| Transition theorem $\varpi_{\mathrm{RVM}}(1+w)$, $\varpi_{\mathrm{RVM}}(1+\chi)$ | theorem |
| $C^{-1} \leq \lvert\zeta_{\mathrm{RVM}}\rvert \leq C$ on $\Re s \geq \sigma_0 > 1$; $\zeta_{\mathrm{RVM}} \to \varpi_{\mathrm{RVM}}$ at $+\infty$ | theorem |
| Unsmoothed $M^{\mathrm{crit}}_{\mathrm{RVM}} = L + O_\varepsilon(x^{1/2+\varepsilon})$ | theorem |
| Dirichlet expansion of $\zeta_{\mathrm{RVM}}^{\pm 1}$ | impossible (theorem) |
| Ternary walk $\mu_h$, $M_h$ | definition; comparison object |
| Exact Hadamard product of $2\xi$ | theorem |
| $\lambda$-family, generator, multiplier, simplicity for $\lambda > 0$ | theorem |
| Analytic class of $\zeta_{Z1}$ | theorem |
| $\zeta_{Z1} = \zeta \Leftrightarrow$ (D) $\Leftrightarrow$ RH | theorem |
| (D) $\Leftrightarrow$ Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$ | theorem |
| $\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_k \frac{1/4+\tau_k^2}{1/4+\gamma_k^2}$; (D) $\Leftrightarrow \varpi_{Z1} = 1$ | theorem |
| Mellin prime-power counts; distributional explicit formula | theorem |
| (D) $\Leftrightarrow J^{\mathrm{M}}_{Z1}$ constant between integers | theorem |
| $\frac{d}{d\lambda}\log\varpi_\lambda = -\Upsilon$; (D) $\Leftrightarrow \int_{0^+}^1 \Upsilon = \log L$ | theorem |
| Atomized family; joining factor holomorphic in $\Re s > \frac{1}{2}$ | theorem |
| (D) $\Leftrightarrow$ bounded cell masses and vanishing cell defects at $\lambda = 0$ | theorem |
| $\zeta_{Z1} = \zeta \Rightarrow M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ | theorem (Step 3) |
| $\zeta_{Z1} = \zeta$ | open |

1.5. **Organization.** Section 2 treats the clock, its Lambert inversion and the bounded displacement; Section 3 the canonical product; Section 4 the exponential sums; Sections 5 and 6 the prime coordinate and the integer weights; Sections 7 and 8 the Mertens transform and the transition theorem; Section 9 the exact Hadamard product and the comparison with $\zeta$; Section 10 the $\lambda$-family;

Section 11 the endpoint, Hamburger's theorem and Step 3; Section 12 the value of the endpoint at infinity and the Euler products of $\zeta_{\mathrm{RVM}}$ and $\zeta_{Z1}$; Section 13 the atomized family, Hamburger's hypotheses at the two endpoints and the programme for the final step; Section 14 the numerical illustration; Section 15 the conclusions and open problems.

## 2. The midpoint Riemann–von Mangoldt clock

Let $\rho$ run over the nontrivial zeros of $\zeta$, counted with multiplicity. No nontrivial zero is real, so the positive ordinates can be listed with multiplicity as $0 < \gamma_1 \le \gamma_2 \le \cdots$. Put

$$N_\zeta(T) := \#\{\rho : \ 0 < \Im\rho \le T\} = \#\{k : \gamma_k \le T\}. \tag{2.1}$$

The Riemann–von Mangoldt formula [23, Thms. 9.3–9.4], [24] states that for $T \ge 2$, $T \ne \gamma_k$,

$$N_\zeta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + S(T) + O\Big(\frac{1}{T}\Big), \qquad S(T) := \frac{1}{\pi}\arg\zeta\big(\tfrac12 + iT\big) = O(\log T), \tag{2.2}$$

where $\arg\zeta(\frac12 + iT)$ is obtained by continuous variation of the argument along the segments from 2 to $2 + iT$ and from $2 + iT$ to $\frac12 + iT$, starting from $\arg\zeta(2) = 0$. The $O(1/T)$ term comes from Stirling's formula for $\Gamma$ and is explicit; $S(T)$ is the fluctuating part, with a jump $+m$ at an ordinate of multiplicity $m$. The *midpoint RVM clock* is the smooth part of (2.2):

$$\mathcal{N}_{\mathrm{RVM}}(T) := \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}. \tag{2.3}$$

It is strictly increasing on $[2\pi, \infty)$. The midpoint ordinates $\tau_k$ are the positive solutions of

$$\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac12, \qquad k \ge 1, \tag{2.4}$$

that is, the abscissae where the smooth count crosses the midpoints $k - \frac12$ of the steps of a unit staircase. Thus $\mathcal{N}_{\mathrm{RVM}}$ replaces the staircase $N_\zeta$ by its smooth part, and $\tau_k$ is the point where the $k$th step would be crossed half-way (Figure 1).

The Lambert $W$-function is the multivalued inverse of $v \mapsto ve^v$: $W(z)e^{W(z)} = z$. On $[-1/e, \infty)$ the equation $ve^v = z$ has exactly one solution $v \ge -1$; this defines the principal branch $W_0 : [-1/e, \infty) \to [-1, \infty)$, which is real, increasing, with $W_0(0) = 0$ and $W_0(z) \sim \log z - \log\log z$ as $z \to \infty$ [11, §4.13], [5].

**Proposition 2.1** (Lambert inversion)**.** *For every $k \ge 1$,*

$$\tau_k = \frac{2\pi(k - 11/8)}{W_0\big((k - 11/8)/e\big)}, \tag{2.5}$$

*The quotient is positive also for $k = 1$.*

*Proof.* Put $y = \tau/(2\pi)$. Equation (2.4) reads

$$y(\log y - 1) = k - \frac{11}{8}. \tag{2.6}$$

With $v = \log y - 1$ one has $y = e^{v+1}$ and $ve^v = (k - 11/8)/e$, so $v = W_0((k - 11/8)/e)$ and $y = (k - 11/8)/W_0((k - 11/8)/e)$. For $k = 1$, $(k - 11/8)/e = -3/(8e) \in (-1/e, 0)$, where $W_0$ is real and negative; numerator and denominator are both negative. □

The clock derivative and the resulting growth are

$$\mathcal{N}'_{\mathrm{RVM}}(T) = \frac{1}{2\pi}\log\frac{T}{2\pi}, \tag{2.7}$$

$$\tau_k \asymp \frac{k}{\log k} \qquad (k \ge 2). \tag{2.8}$$

In terms of the clock, (2.2) reads

$$N_\zeta(T) = \mathcal{N}_{\mathrm{RVM}}(T) + S(T) + O(T^{-1}), \qquad S(T) = O(\log T) \qquad (T \geq 2). \tag{2.9}$$

Let

$$N_\zeta^*(T) := \tfrac{1}{2}\big(N_\zeta(T+0) + N_\zeta(T-0)\big) \tag{2.10}$$

be the symmetric count, and define the displacements

$$\eta_k := \tau_k - \gamma_k. \tag{2.11}$$

**Lemma 2.2** (Bounded displacement)**.** *There is a constant $C_\eta$ such that*

$$|\eta_k| \leq C_\eta \qquad (k \geq 1). \tag{2.12}$$

*Proof.* Let $\gamma = \gamma_k$ and let $m = m(\gamma)$ be the number of indices $j$ with $\gamma_j = \gamma$; these indices form a block $j_0 + 1, \ldots, j_0 + m$ containing $k$. Then $N_\zeta(\gamma - 0) = j_0$ and $N_\zeta(\gamma + 0) = j_0 + m$, so $N_\zeta^*(\gamma) = j_0 + m/2$ and

$$\big|N_\zeta^*(\gamma_k) - (k - \tfrac{1}{2})\big| \leq \frac{m(\gamma_k)}{2}. \tag{2.13}$$

By (2.9), $m(\gamma) \leq N_\zeta(\gamma+1) - N_\zeta(\gamma-1) = O(\log\gamma)$, and $N_\zeta^*(\gamma) = \mathcal{N}_{\mathrm{RVM}}(\gamma) + O(\log\gamma)$. With (2.4),

$$\big|\mathcal{N}_{\mathrm{RVM}}(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)\big| = O(\log\gamma_k). \tag{2.14}$$

Both $\gamma_k$ and $\tau_k$ tend to infinity and (2.14) gives $\log\gamma_k \asymp \log\tau_k \asymp \log k$. By the mean-value theorem and (2.7), $\mathcal{N}'_{\mathrm{RVM}}$ is $\gg \log k$ between $\gamma_k$ and $\tau_k$ for large $k$, so $|\gamma_k - \tau_k| = O(1)$. Finitely many initial indices are absorbed into $C_\eta$. □

## 3. The pure critical RVM zeta function

Put

$$q(s) := s(s-1), \tag{3.1}$$

$$P_{\mathrm{RVM}}^{\mathrm{crit}}(s) := \prod_{k\geq 1}\left(1 + \frac{q(s)}{\frac{1}{4} + \tau_k^2}\right), \tag{3.2}$$

$$G(s) := s(s-1)\Gamma(s/2)\pi^{-s/2}, \tag{3.3}$$

and

$$\xi_{\mathrm{RVM}}(s) := \frac{1}{2}P_{\mathrm{RVM}}^{\mathrm{crit}}(s), \qquad \zeta_{\mathrm{RVM}}(s) := \frac{P_{\mathrm{RVM}}^{\mathrm{crit}}(s)}{G(s)}. \tag{3.4}$$

**Theorem 3.1** (Canonical characterization of $\zeta_{\mathrm{RVM}}$)**.** *The product* (3.2) *converges absolutely and locally uniformly in $s \in \mathbb{C}$. It defines an entire function with*

$$P_{\mathrm{RVM}}^{\mathrm{crit}}(1-s) = P_{\mathrm{RVM}}^{\mathrm{crit}}(s), \qquad P_{\mathrm{RVM}}^{\mathrm{crit}}(\bar{s}) = \overline{P_{\mathrm{RVM}}^{\mathrm{crit}}(s)}, \qquad P_{\mathrm{RVM}}^{\mathrm{crit}}(0) = P_{\mathrm{RVM}}^{\mathrm{crit}}(1) = 1. \tag{3.5}$$

*Its zeros are exactly the simple zeros*

$$\rho_k^{\mathrm{RVM}} = \frac{1}{2} + i\tau_k, \qquad \overline{\rho_k^{\mathrm{RVM}}} = \frac{1}{2} - i\tau_k, \qquad k \geq 1. \tag{3.6}$$

*Consequently $\zeta_{\mathrm{RVM}}$ has a simple pole at $s = 1$ with residue 1, the trivial zeros $s = -2, -4, -6, \ldots$, and nontrivial zeros exactly at* (3.6)*; moreover $\xi_{\mathrm{RVM}}(s) = \xi_{\mathrm{RVM}}(1-s)$.*

*Proof.* By (2.8), $\sum_k \tau_k^{-2} < \infty$. On a compact set $K$,

$$\sup_{s\in K}\left|\frac{q(s)}{\frac{1}{4} + \tau_k^2}\right| \ll_K \tau_k^{-2}, \tag{3.7}$$

so the Weierstrass criterion gives normal convergence. Each factor depends on $s$ only through $q(s)$ and equals 1 at $s = 0, 1$, which gives (3.5); real coefficients give the conjugation symmetry. The $k$th factor vanishes exactly when

$$\Big(s - \frac{1}{2}\Big)^2 + \tau_k^2 = 0. \tag{3.8}$$

The clock is strictly increasing on $[2\pi, \infty)$ and $\tau_1 > 2\pi$, so the $\tau_k$ are distinct; the derivative $2s - 1 = \pm 2i\tau_k \neq 0$ of the vanishing factor and the non-vanishing of the other factors make the zeros simple. At $s = 1$, $P^{\mathrm{crit}}_{\mathrm{RVM}}(1) = 1$ and $G(s) = (s-1)(1 + O(s-1))$, so the residue is 1. At $s = -2n$ ($n \geq 1$) the pole of $\Gamma(s/2)$ gives a zero of $1/G$, while $P^{\mathrm{crit}}_{\mathrm{RVM}}(-2n) \neq 0$ because $q(-2n) > 0$. At $s = 0$, $s\Gamma(s/2) \to 2$, so $G(0) \neq 0$ and $s = 0$ is not a zero. □

## 4. RVM exponential sums, centroids, and the height transform

For $u \in \mathbb{R}$ and $N \geq 1$ define

$$S_{\mathrm{RVM}}(N; u) := \sum_{k \leq N} e^{iu\tau_k}, \qquad C^{\mathrm{pts}}_{\mathrm{RVM}}(N; u) := \frac{S_{\mathrm{RVM}}(N; u)}{N}, \tag{4.1}$$

and

$$C^{\mathrm{path}}_{\mathrm{RVM}}(N; u) := \frac{1}{N} \sum_{n \leq N} S_{\mathrm{RVM}}(n; u) = \sum_{k \leq N} \frac{N - k + 1}{N} e^{iu\tau_k}. \tag{4.2}$$

**Theorem 4.1** (Uniform logarithmic exponential-sum bound)**.** *Let $0 < u_1 \leq u_2 < \infty$. Then*

$$\sup_{u \in [u_1, u_2]} |S_{\mathrm{RVM}}(N; u)| \ll_{u_1, u_2} 1 + \log N. \tag{4.3}$$

*Proof.* Let $\mathcal{T} : [-\frac{1}{8}, \infty) \to [2\pi, \infty)$ be the inverse of the clock, so $\tau_k = \mathcal{T}(k - \frac{1}{2})$, and put $\ell_k = \log(\tau_k/2\pi) > 0$. Differentiating $\mathcal{N}_{\mathrm{RVM}}(\mathcal{T}(y)) = y$,

$$\mathcal{T}'(y) = \frac{2\pi}{\log(\mathcal{T}(y)/2\pi)}, \qquad \mathcal{T}''(y) = -\frac{4\pi^2}{\mathcal{T}(y)\log^3(\mathcal{T}(y)/2\pi)}. \tag{4.4}$$

Since $\mathcal{T}'$ decreases and $|\mathcal{T}''|$ decreases,

$$\frac{2\pi}{\ell_{k+1}} \leq \tau_{k+1} - \tau_k \leq \frac{2\pi}{\ell_k}, \tag{4.5}$$

$$|\tau_{k+2} - 2\tau_{k+1} + \tau_k| \leq \frac{4\pi^2}{\tau_k \ell_k^3}. \tag{4.6}$$

From (2.4), $\tau_k \ell_k = 2\pi(k - \frac{11}{8}) + \tau_k$, hence

$$\tau_k \ell_k \asymp k \qquad (k \geq 2). \tag{4.7}$$

Fix $u \in [u_1, u_2]$ and put $\Delta_k = u(\tau_{k+1} - \tau_k)$, $E_k = e^{iu\tau_k}$. Choose $k_0 = k_0(u_2)$ with $\ell_{k_0} \geq 2u_2$; then $0 < \Delta_k \leq \pi$ for $k \geq k_0$. With $b_k := (e^{i\Delta_k} - 1)^{-1}$ one has $E_{k+1} = e^{i\Delta_k} E_k$, hence

$$E_k = b_k(E_{k+1} - E_k). \tag{4.8}$$

Summing (4.8) over $k_0 \leq k \leq N$ and shifting the index in the first part gives the exact identity

$$\sum_{k=k_0}^{N} E_k = b_N E_{N+1} - b_{k_0} E_{k_0} + \sum_{k=k_0+1}^{N} (b_{k-1} - b_k) E_k. \tag{4.9}$$

For $0 < x \leq \pi$, $|e^{ix} - 1| = 2\sin(x/2) \geq 2x/\pi$. Using (4.5) and (4.6),

$$|b_k| \leq \frac{\pi}{2\Delta_k} \leq \frac{\ell_{k+1}}{4u_1}, \qquad |b_{k-1} - b_k| \leq |\Delta_k - \Delta_{k-1}|\,|b_{k-1}||b_k| \leq \frac{\pi^2 u_2}{4u_1^2} \frac{\ell_{k+1}^2}{\tau_{k-1}\ell_{k-1}^3} \ll_{u_1, u_2} \frac{1}{\tau_k \ell_k} \ll \frac{1}{k}, \tag{4.10}$$

the last step by (4.7). Inserting (4.10) into (4.9) gives $|\sum_{k_0}^{N} E_k| \ll \ell_{N+1} + \ell_{k_0+1} + \sum_{k\le N} k^{-1} \ll 1 + \log N$. The indices $k < k_0$ contribute at most $k_0$. All constants depend only on $u_1, u_2$. □

**Corollary 4.2** (Point centroid)**.** *Uniformly for $u \in [u_1, u_2] \subset (0,\infty)$,*

$$C_{\mathrm{RVM}}^{\mathrm{pts}}(N;u) = O_{u_1,u_2}\Big(\frac{1+\log N}{N}\Big) \longrightarrow 0. \tag{4.11}$$

*Theorem 4.1 alone does not imply convergence of the path centroid* (4.2)*.*

*Proof.* The first assertion is (4.3) divided by $N$. For the second, the identity $C_{\mathrm{RVM}}^{\mathrm{path}} = N^{-1}\sum_{n\le N} S_{\mathrm{RVM}}(n;u)$ with $S_{\mathrm{RVM}}(n;u) = O(\log n)$ gives only $O(\log N)$. □

Define the midpoint count

$$n_{\mathrm{RVM}}(H) := \#\{k : \tau_k \le H\}. \tag{4.12}$$

For $x > 0$ and $n_{\mathrm{RVM}}(H) \ge 1$ define

$$M_H^{\mathrm{RVM}}(x) := \frac{1}{n_{\mathrm{RVM}}(H)} \sum_{\tau_k \le H} x^{i\tau_k}, \tag{4.13}$$

and for the ordinates of $\zeta$

$$M_H(x) := \frac{1}{N_\zeta(H)} \sum_{\gamma_k \le H} x^{i\gamma_k}. \tag{4.14}$$

**Corollary 4.3** (Synthetic height-transform decay)**.** *For every fixed $x > 1$, with $N = n_{\mathrm{RVM}}(H)$,*

$$M_H^{\mathrm{RVM}}(x) = O_x\Big(\frac{1+\log N}{N}\Big) \longrightarrow 0 \qquad (H \to \infty). \tag{4.15}$$

*Proof.* Take $u = \log x > 0$ in Theorem 4.1; (4.13) is the point centroid at $N = n_{\mathrm{RVM}}(H)$. □

*Remark* 4.4. Neither (4.13) nor (4.14) is the Mertens function; both are normalized phase averages.

## 5. The RVM explicit formula, $J_{\mathrm{RVM}}$, and $\pi_{\mathrm{RVM}}$

For $w = a + ib$ with $b > 0$ define

$$\mathrm{Ei}(w) := \int_{-\infty}^{a} \frac{e^{v+ib}}{v+ib}\, dv, \qquad \mathrm{Ei}(\bar w) := \overline{\mathrm{Ei}(w)}. \tag{5.1}$$

The path is the horizontal line $\Im = b$, which avoids 0; this is the branch of $\mathrm{li}(x^\rho) = \mathrm{Ei}(\rho \log x)$ used in Riemann's explicit formula [12]. For $x > 1$ put

$$J_{\mathrm{RVM}}(x) := \mathrm{li}(x) - 2\Re \sum_{k\ge 1} \mathrm{Ei}\Big(\big(\tfrac12 + i\tau_k\big)\log x\Big) - \log 2 + \int_x^\infty \frac{dy}{y(y^2-1)\log y}, \tag{5.2}$$

the zero sum being taken in the order of $k$.

**Lemma 5.1** (Asymptotics of Ei off the real axis)**.** *For $w = a + ib$, $b > 0$,*

$$\mathrm{Ei}(w) = \frac{e^w}{w} + \varrho(w), \qquad |\varrho(w)| \le \frac{e^a}{b^2}. \tag{5.3}$$

*Proof.* Integrate (5.1) by parts: $\int_{-\infty}^{a} e^{v+ib}(v+ib)^{-1}dv = e^w/w + \int_{-\infty}^{a} e^{v+ib}(v+ib)^{-2}dv$, the boundary term at $-\infty$ being 0. The last integral is bounded by $\int_{-\infty}^{a} e^v b^{-2} dv = e^a/b^2$. □

**Theorem 5.2** (Compact-uniform existence of the RVM zero sum)**.** *The series $\sum_k \Re\,\mathrm{Ei}\big((\frac12 + i\tau_k)u\big)$ converges uniformly for $u$ in every compact interval $[u_1, u_2] \subset (0,\infty)$. Hence $J_{\mathrm{RVM}}$ is a continuous function on $(1,\infty)$.*

*Proof.* Apply Lemma 5.1 with $w = (\frac{1}{2} + i\tau_k)u$, $a = u/2$, $b = \tau_k u$. Since $\Re\big(e^{iu\tau_k}/(\frac{1}{2} + i\tau_k)\big) = (\frac{1}{2}\cos u\tau_k + \tau_k \sin u\tau_k)/(\frac{1}{4} + \tau_k^2)$,

$$\Re \operatorname{Ei}((\tfrac{1}{2} + i\tau_k)u) = \frac{e^{u/2}}{u}\,\frac{\sin(u\tau_k)}{\tau_k} + O_{u_1,u_2}(\tau_k^{-2}), \tag{5.4}$$

uniformly for $u \in [u_1, u_2]$. The error series converges absolutely and uniformly. For the main series, Abel summation with $S_{\mathrm{RVM}}(K; u) = O_{u_1,u_2}(1 + \log K)$ (Theorem 4.1) and

$$\frac{1}{\tau_K} - \frac{1}{\tau_{K+1}} \le \frac{\tau_{K+1} - \tau_K}{\tau_K^2} \le \frac{2\pi}{\ell_K \tau_K^2} \ll \frac{\log K}{K^2} \tag{5.5}$$

gives a uniformly convergent majorant $\sum_K (\log K)^2 K^{-2}$, while the boundary term $S_{\mathrm{RVM}}(K; u)/\tau_K \to 0$ uniformly. Hence $\sum_k e^{iu\tau_k}/\tau_k$ converges uniformly, and so does its imaginary part. □

Define the support-truncated version

$$J^*_{\mathrm{RVM}}(x) := \begin{cases} J_{\mathrm{RVM}}(x), & x \ge 2, \\ 0, & 0 < x < 2, \end{cases} \tag{5.6}$$

set $\pi_{\mathrm{RVM}}(x) = 0$ for $0 < x < 2$, and for $x \ge 2$

$$\pi_{\mathrm{RVM}}(x) := \sum_{1 \le m \le \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}). \tag{5.7}$$

**Proposition 5.3** (Exact finite prime-power inversion)**.** *For every $x \ge 2$,*

$$J^*_{\mathrm{RVM}}(x) = \sum_{1 \le m \le \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(x^{1/m}). \tag{5.8}$$

*Proof.* All sums are finite because $x^{1/m} < 2$ for $m > \log_2 x$. Substituting (5.7) into the right side of (5.8), the coefficient of $J^*_{\mathrm{RVM}}(x^{1/r})$ is

$$\frac{1}{r} \sum_{d \mid r} \mu(d), \tag{5.9}$$

which is 1 for $r = 1$ and 0 otherwise. □

5.1. **The square-root estimate.** For $u > 0$ put

$$\mathcal{A}_{\mathrm{RVM}}(u) := \sum_{k \ge 1} \frac{e^{iu\tau_k}}{\tau_k}, \tag{5.10}$$

summed in the order of $k$, and write $\ell(t) = \log(t/2\pi)$, so that $\mathcal{N}'_{\mathrm{RVM}}(t) = \ell(t)/2\pi$ and $\ell(\tau_k) = \ell_k$.

**Lemma 5.4** (The weighted clock sum)**.** *For every $u \ge \log 2$ the series* (5.10) *converges, and*

$$|\mathcal{A}_{\mathrm{RVM}}(u)| \le \frac{\log^2(2 + 2u)}{4\pi} + \frac{\log(2 + 2u)}{2\pi} + C_{\mathcal{A}}, \tag{5.11}$$

*with an absolute constant $C_{\mathcal{A}}$.*

*Proof.* Put $Y_j = 2^j \cdot 4\pi(1 + u)$ for $j \ge 0$.

*Head $\tau_k \le Y_0$.* Let $n(t) = \#\{k : \tau_k \le t\}$. By (2.4), $\tau_k \le t$ if and only if $k \le \mathcal{N}_{\mathrm{RVM}}(t) + \frac{1}{2}$, so $n(t) \le \mathcal{N}_{\mathrm{RVM}}(t) + \frac{1}{2} = \frac{t}{2\pi}(\ell(t) - 1) + \frac{11}{8} \le \frac{t\,\ell(t)}{2\pi}$ for $t \ge \tau_1 > \frac{11\pi}{4}$. Stieltjes integration gives

$$\sum_{\tau_k \le Y_0} \frac{1}{\tau_k} = \frac{n(Y_0)}{Y_0} + \int_{\tau_1}^{Y_0} \frac{n(t)}{t^2}\, dt \le \frac{\ell(Y_0)}{2\pi} + \int_{\tau_1}^{Y_0} \frac{\ell(t)}{2\pi t}\, dt \le \frac{\ell(Y_0)}{2\pi} + \frac{\ell(Y_0)^2}{4\pi}, \tag{5.12}$$

and $\ell(Y_0) = \log(2 + 2u)$.

*Blocks.* Let $\mathcal{B}_j = \{k : Y_j < \tau_k \le Y_{j+1}\}$, let $\mathcal{T}$ be the inverse clock of (4.4), and put $\phi(y) = u\,\mathcal{T}(y - \frac{1}{2})/2\pi$, so that $e^{2\pi i\phi(k)} = e^{iu\tau_k}$. By (4.4),

$$\phi''(y) = -\frac{2\pi u}{\mathcal{T}\,\ell(\mathcal{T})^3}, \qquad \mathcal{T} = \mathcal{T}(y - \tfrac{1}{2}). \tag{5.13}$$

On $\mathcal{B}_j$ put $\kappa_j = 2\pi u/(Y_{j+1}\ell(Y_{j+1})^3)$. Since $Y_{j+1} = 2Y_j$ and $\ell(Y_{j+1}) \le 2\ell(Y_j)$ (because $\ell(Y_j) \ge \ell(Y_0) \ge \log 2$), we have $\kappa_j \le -\phi'' \le 16\kappa_j$ there. The number of indices in $\mathcal{B}_j$ is at most $\mathcal{N}_{\mathrm{RVM}}(Y_{j+1}) - \mathcal{N}_{\mathrm{RVM}}(Y_j) + 1 \le Y_j\ell(Y_{j+1})/2\pi + 1$. The second-derivative test for exponential sums [23, Thm. 5.9] gives, for every run $a < k \le b$ of consecutive indices of $\mathcal{B}_j$,

$$\Big|\sum_{a<k\le b} e^{iu\tau_k}\Big| \le A\Big(16\Big(\frac{Y_j\ell(Y_{j+1})}{2\pi} + 1\Big)\kappa_j^{1/2} + \kappa_j^{-1/2}\Big) =: P_j, \tag{5.14}$$

with an absolute constant $A$; a run of length less than 1 has at most one term, and $1 \le \kappa_j^{-1/2}$ because $\kappa_j \le u/(4(1+u)\log^3 2) < 1$. The weights $1/\tau_k$ are positive and decreasing, and $1/\tau_k < 1/Y_j$ on $\mathcal{B}_j$, so Abel's inequality bounds the contribution of any run in $\mathcal{B}_j$ by $P_j/Y_j$:

$$\frac{P_j}{Y_j} \le A\left(\frac{16}{2\pi}\Big(\frac{\pi u}{Y_j\ell(Y_{j+1})}\Big)^{1/2} + \frac{16}{Y_j} + \Big(\frac{\ell(Y_{j+1})^3}{\pi u Y_j}\Big)^{1/2}\right). \tag{5.15}$$

With $Y_j = 2^j \cdot 4\pi(1+u)$ and $\ell(Y_{j+1}) = \log(2^{j+2}(1+u)) \le j + 2 + \log(1+u)$, each of the three terms is at most an absolute constant times $(j+2)^{3/2}2^{-j/2}$, uniformly in $u \ge \log 2$, because $\sup_{u\ge\log 2}(1+\log(1+u))^3/(u(1+u)) < \infty$. Hence $\sum_{j\ge 0} P_j/Y_j \le C_{\mathcal{A}}$ with $C_{\mathcal{A}}$ absolute.

*Convergence and bound.* If $\tau_K \in \mathcal{B}_J$, the terms with $k > K$ form a run of $\mathcal{B}_J$ followed by the full blocks $\mathcal{B}_j$, $j > J$, so $|\sum_{k>K'}^{K''} e^{iu\tau_k}/\tau_k| \le \sum_{j\ge J} P_j/Y_j \to 0$ for $K'' > K' \ge K$, $K \to \infty$. The series converges, and (5.12) together with $\sum_j P_j/Y_j \le C_{\mathcal{A}}$ gives (5.11). □

**Theorem 5.5** (Square-root estimate for $J_{\mathrm{RVM}}$ and $\pi_{\mathrm{RVM}}$)**.** *For $x \ge 8$,*

$$J_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{5.16}$$

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{5.17}$$

*with absolute implied constants. No hypothesis on $\zeta$ is used.*

*Proof.* Let $u = \log x \ge \log 8$ and $\rho_k = \frac{1}{2} + i\tau_k$. Lemma 5.1 with $w = \rho_k u$, $a = u/2$, $b = \tau_k u$ gives

$$\mathrm{Ei}(\rho_k u) = \frac{e^{\rho_k u}}{\rho_k u} + \varrho_k, \qquad |\varrho_k| \le \frac{e^{u/2}}{\tau_k^2 u^2}. \tag{5.18}$$

Moreover $\frac{1}{\rho_k} = -\frac{i}{\tau_k} + r_k$ with $r_k = \frac{i/2}{\tau_k\rho_k}$, so $|r_k| \le \frac{1}{2\tau_k^2}$. Since $\sum_k \tau_k^{-2} < \infty$, Lemma 5.4 shows that the series in (5.2) converges in the order of $k$ and

$$\sum_{k\ge 1}\mathrm{Ei}(\rho_k u) = \frac{e^{u/2}}{u}\Big(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + \sum_k r_k e^{iu\tau_k}\Big) + \sum_k \varrho_k = \frac{e^{u/2}}{u}\big(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + O(1)\big). \tag{5.19}$$

For $u \ge 2$ one has $\log(2+2u) \le \log(2u) + \log 2 \le 3\log u$, so (5.11) gives $\mathcal{A}_{\mathrm{RVM}}(u) = O(\log^2 u)$. The remaining terms $-\log 2 + \int_x^\infty dy/(y(y^2-1)\log y)$ of (5.2) are bounded for $x \ge 2$. With $e^{u/2} = \sqrt{x}$ and $\log u = \log\log x$ this proves (5.16).

For (5.17), (5.7) gives

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = \big(J_{\mathrm{RVM}}(x) - \mathrm{li}(x)\big) + \sum_{2\le m\le \log_2 x}\frac{\mu(m)}{m}\,J^*_{\mathrm{RVM}}(x^{1/m}). \tag{5.20}$$

By (5.16), $\mathrm{li}(y) \ll y/\log y$, and the continuity of $J_{\mathrm{RVM}}$ on $[2,8]$ (Theorem 5.2), $|J^*_{\mathrm{RVM}}(y)| \le C\, y/\log y$ for all $y \ge 2$. The term $m = 2$ is therefore $O(\sqrt{x}/\log x)$, and the terms $3 \le m \le \log_2 x$ add up to at most $C\, x^{1/3} \log_2 x/\log x = O(x^{1/3})$. Both are absorbed in (5.17). □

**Corollary 5.6** (The three limits)**.** *As $x \to \infty$,*

$$\frac{\mathrm{li}(x)}{\pi_{\mathrm{RVM}}(x)} \to 1, \qquad \frac{\pi(x)}{\mathrm{li}(x)} \to 1, \qquad \frac{\pi(x)}{\pi_{\mathrm{RVM}}(x)} \to 1. \tag{5.21}$$

*More precisely, for some absolute $c > 0$,*

$$\pi(x) - \pi_{\mathrm{RVM}}(x) = O\Big(x\, e^{-c\sqrt{\log x}}\Big). \tag{5.22}$$

*Proof.* Since $\mathrm{li}(x) \sim x/\log x$, (5.17) gives $\pi_{\mathrm{RVM}}(x) = \mathrm{li}(x)(1 + O(x^{-1/2}(\log\log x)^2))$; in particular $\pi_{\mathrm{RVM}}(x) > 0$ for large $x$, and the first limit follows. The second is the prime number theorem [15, 8], in the form $\pi(x) = \mathrm{li}(x) + O(x\, e^{-c\sqrt{\log x}})$ [7, Ch. 18]. Subtracting (5.17) gives (5.22), and dividing by $\pi_{\mathrm{RVM}}(x) \sim x/\log x$ gives the third limit. □

The first limit and (5.17) concern only the synthetic system; the second and third compare it with the primes and use the prime number theorem. Figures 8 and 10 show them numerically.

## 6. Integer weights and the continuous-to-atomic problem

Define

$$c_n^{\mathrm{RVM}} := \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1), \qquad n \ge 2, \tag{6.1}$$

and, initially for $\Re s > 1$,

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) := \prod_{n \ge 2} (1 - n^{-s})^{-c_n^{\mathrm{RVM}}}. \tag{6.2}$$

No positivity or 0/1 property is built into (6.1). Define the integer-step prime-power count and the right-continuous extension

$$J_{\mathbb{Z},\mathrm{RVM}}(x) := \sum_{1 \le m \le \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}\big(\lfloor x^{1/m} \rfloor\big), \tag{6.3}$$

$$J^{*,\mathbb{Z}}_{\mathrm{RVM}}(x) := J^*_{\mathrm{RVM}}(\lfloor x \rfloor). \tag{6.4}$$

**Theorem 6.1** (Lipschitz control of $J_{\mathrm{RVM}}$)**.** *There is an absolute constant $C_J$ such that*

$$|J_{\mathrm{RVM}}(x+\theta) - J_{\mathrm{RVM}}(x)| \le C_J\, \theta \qquad (x \ge 2,\ 0 < \theta \le x). \tag{6.5}$$

*Proof.* Put $u_1 = \log x$, $u_2 = \log(x+\theta)$, so $0 < u_2 - u_1 = \log(1 + \theta/x) \le \theta/x$ and $u_2 - u_1 \le \log 2$. The terms li and $\int_x^\infty dy/(y(y^2-1)\log y)$ of (5.2) have derivatives bounded by $1/\log 2$ and $3/\log 2$ on $[2,\infty)$, so they change by $O(\theta)$. It remains to bound $\sum_k D_k$, where, by (5.1) and $\frac{d}{dv}\,\mathrm{Ei}(\rho v) = e^{\rho v}/v$,

$$D_k := \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_2) - \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_1) = \int_{u_1}^{u_2} \frac{e^{v/2} e^{iv\tau_k}}{v}\, dv. \tag{6.6}$$

The series $\sum_k D_k$ converges (Theorem 5.2 and its proof). By Abel's theorem for the increasing exponents $\tau_k$, $\sum_k D_k = \lim_{\varepsilon \downarrow 0} \sum_k e^{-\varepsilon\tau_k} D_k$, and for $\varepsilon > 0$ Fubini gives

$$\sum_k e^{-\varepsilon\tau_k} D_k = \int_{u_1}^{u_2} \frac{e^{v/2}}{v}\, S_\varepsilon(v)\, dv, \qquad S_\varepsilon(v) := \sum_{k \ge 1} g(\tau_k), \quad g(\tau) := e^{(iv-\varepsilon)\tau}. \tag{6.7}$$

*Clock decomposition.* With $\tau_0 = 4\pi$ and the sawtooth count (8.8), Stieltjes integration exactly as in (8.9) ($g(\infty) = 0$, $g' \in L^1$) gives

$$S_\varepsilon(v) = \int_{\tau_0}^\infty g\, d\mathcal{N}_{\mathrm{RVM}} + g(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0) + \int_{\tau_0}^\infty \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}) g'(\tau)\, d\tau. \tag{6.8}$$

The middle term is at most $\frac{1}{2}$. For the first, two integrations by parts with $\mathcal{N}'_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi}\log\frac{\tau}{2\pi}$, $\mathcal{N}''_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi\tau}$ and $v \geq \log 2$ give $|\int_{\tau_0}^{\infty} g\, d\mathcal{N}_{\mathrm{RVM}}| \leq C/v$, uniformly in $\varepsilon \in (0,1]$.

*Harmonics.* The partial sums of $\psi(y+\frac{1}{2}) = -\sum_{m\geq 1}(-1)^m \sin(2\pi m y)/(\pi m)$ are uniformly bounded and converge off a null set, and $g' \in L^1$ for $\varepsilon > 0$, so termwise integration is allowed:

$$\int_{\tau_0}^{\infty} \psi(\mathcal{N}_{\mathrm{RVM}} + \tfrac{1}{2}) g'\, d\tau = -\sum_{m\geq 1} \frac{(-1)^m}{2\pi i\, m}(K_m^+ - K_m^-), \qquad K_m^{\pm} := \int_{\tau_0}^{\infty} (iv - \varepsilon) e^{-\varepsilon\tau} e^{i\Phi_m^{\pm}(\tau)}\, d\tau, \quad (6.9)$$

with $\Phi_m^{\pm}(\tau) = v\tau \pm 2\pi m \mathcal{N}_{\mathrm{RVM}}(\tau)$. The weight $e^{-\varepsilon\tau}$ is positive and decreasing, so by the second mean-value theorem each integral is at most $2(v+1)\sup_{\xi} |\int_a^{\xi} e^{i\Phi} d\tau|$ over the relevant interval $[a,b]$. We use the first- and second-derivative tests [23, Lemmas 4.2 and 4.4]: $|\int_a^b e^{i\Phi}| \leq 4/\mu$ if $\Phi'$ is monotone with $|\Phi'| \geq \mu$, and $\leq 8/\sqrt{\lambda}$ if $|\Phi''| \geq \lambda$.

For $K_m^+$: $(\Phi_m^+)' = v + m\log\frac{\tau}{2\pi} \geq v + m\log 2$ is increasing, so $|K_m^+| \leq 8(v+1)/(v + m\log 2)$.

For $K_m^-$: $(\Phi_m^-)' = v - m\log\frac{\tau}{2\pi}$ is decreasing and vanishes at $\tau_m := 2\pi e^{v/m} > 2\pi$, and $(\Phi_m^-)'' = -m/\tau$. On $[\tau_0, \tau_m/2]$ (if nonempty) $(\Phi_m^-)' \geq m\log 2$; on $[2\tau_m, \infty)$, $(\Phi_m^-)' \leq -m\log 2$; on $[\max(\tau_0, \tau_m/2), 2\tau_m]$, $|(\Phi_m^-)''| \geq m/(2\tau_m)$. Hence

$$|K_m^-| \leq 2(v+1)\Big(\frac{8}{m\log 2} + 8\sqrt{\frac{2\tau_m}{m}}\Big) = 2(v+1)\Big(\frac{8}{m\log 2} + 16\sqrt{\pi}\,\frac{e^{v/(2m)}}{\sqrt{m}}\Big). \quad (6.10)$$

Summing with the weights $1/(2\pi m)$ and using $\sum_m m^{-3/2} e^{v/(2m)} \leq e^{v/2} + \zeta(\frac{3}{2}) e^{v/4}$,

$$|S_\varepsilon(v)| \leq C\,(v+1)\, e^{v/2} \qquad (v \geq \log 2,\ 0 < \varepsilon \leq 1), \quad (6.11)$$

with an absolute constant $C$.

*Conclusion.* Inserting (6.11) into (6.7), for $v \in [u_1, u_2]$ we have $e^{v/2} \leq \sqrt{2x}$ and $(v+1)/v \leq 1 + 1/\log 2$, so

$$\Big|\sum_k e^{-\varepsilon\tau_k} D_k\Big| \leq C'(u_2 - u_1)\, x \leq C'\theta, \quad (6.12)$$

uniformly in $\varepsilon$. Letting $\varepsilon \downarrow 0$ and taking $2\Re$ proves (6.5). □

The $m = 1$ harmonic, with stationary point $\tau_1 = 2\pi x$, produces the unit-periodic ripple of $J_{\mathrm{RVM}}$ (Figure 8); the harmonics $m \geq 2$ have stationary points $2\pi x^{1/m}$ and contribute $O(x^{-1/2+1/(2m)} m^{-3/2})$ to a unit increment.

**Corollary 6.2** (Bounded cell masses)**.**

$$\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y \rfloor) = O(1), \qquad c_n^{\mathrm{RVM}} = O(1). \quad (6.13)$$

*Proof.* By (5.7), $c_n^{\mathrm{RVM}} = \sum_{m \leq \log_2 n} \frac{\mu(m)}{m}(J^*_{\mathrm{RVM}}(n^{1/m}) - J^*_{\mathrm{RVM}}((n-1)^{1/m}))$. The term $m = 1$ is $O(1)$ by (6.5). For $m \geq 2$ and $(n-1)^{1/m} \geq 2$, (6.5) with $\theta = n^{1/m} - (n-1)^{1/m} \leq n^{1/m-1}$ bounds the term by $C_J n^{1/m-1} \leq C_J n^{-1/2}$, and there are at most $\log_2 n$ such terms. The jump of $J^*_{\mathrm{RVM}}$ at 2 enters only for $n = 2^m$, once, with size $|J_{\mathrm{RVM}}(2)|/m$. The same argument applies to $\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y \rfloor)$. □

**Corollary 6.3** (The integer-weighted zeta is a Dirichlet series)**.** *The product* (6.2) *converges absolutely for* $\Re s > 1$ *and*

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{N\geq 1} \frac{a_N^{\mathbb{Z}}}{N^s}, \qquad a_1^{\mathbb{Z}} = 1, \qquad \sum_{N\geq 1} \frac{|a_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma > 1). \quad (6.14)$$

*The inverse is also an absolutely convergent Dirichlet series on the whole half-plane:*

$$\frac{1}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)} = \sum_{N\geq 1} \frac{\bar{a}_N^{\mathbb{Z}}}{N^s}, \qquad \bar{a}_1^{\mathbb{Z}} = 1, \qquad \sum_{N\geq 1} \frac{|\bar{a}_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma > 1). \quad (6.15)$$

*Proof.* By Corollary 6.2,

$$\log \zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{n\ge 2} c_n^{\mathrm{RVM}} \sum_{m\ge 1} \frac{n^{-ms}}{m} = \sum_{N\ge 2} \frac{b_N}{N^s}, \qquad b_N := \sum_{n^m=N} \frac{c_n^{\mathrm{RVM}}}{m}, \tag{6.16}$$

with $\sum_N |b_N| N^{-\sigma} \le C \sum_{n\ge 2} \sum_m n^{-m\sigma}/m < \infty$ for $\sigma > 1$. The exponential of an absolutely convergent Dirichlet series without constant term is an absolutely convergent Dirichlet series with $a_1^{\mathbb{Z}} = 1$, and $\sum_N |a_N^{\mathbb{Z}}| N^{-\sigma} \le \exp(\sum_N |b_N| N^{-\sigma})$. The same applies to $1/\zeta_{\mathbb{Z},\mathrm{RVM}} = \exp(-\sum_N b_N N^{-s})$, which gives (6.15). □

Numerically the cell masses lie in $[-0.38, 1.04]$ for $n \le 1000$ (Experiment 14.6).

**Theorem 6.4** (Mellin–Stieltjes identification)**.** *There is a function $J_{\mathrm{RVM}}^{\mathrm{M}}$ on $(1,\infty)$ with $J_{\mathrm{RVM}}^{\mathrm{M}}(x) x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$ such that*

$$\log \zeta_{\mathrm{RVM}}(s) = s \int_1^\infty J_{\mathrm{RVM}}^{\mathrm{M}}(x)\, x^{-s-1}\, dx \qquad (\Re s > 1), \tag{6.17}$$

*and $J_{\mathrm{RVM}}^{\mathrm{M}} - J_{\mathrm{RVM}}$ is constant on $(1,\infty)$. In particular $J_{\mathrm{RVM}}^{\mathrm{M}} = J_{\mathrm{RVM}}^{*} + C_{\mathrm{RVM}}$ on $[2,\infty)$ for a constant $C_{\mathrm{RVM}}$.*

The proof is given in Section 12 (Theorems 12.5 and 12.6); it uses the half-plane bound of Theorem 8.10.

**Corollary 6.5** (Weighted factorization)**.**

$$J_{\mathbb{Z},\mathrm{RVM}}(x) - J_{\mathrm{RVM}}^{*}(x) = O(\log\log x), \tag{6.18}$$

*and*

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \zeta_{\mathrm{RVM}}(s)\, e^{h_{\mathrm{RVM}}(s)}, \tag{6.19}$$

*where*

$$h_{\mathrm{RVM}}(s) := s \int_1^\infty (J_{\mathbb{Z},\mathrm{RVM}}(x) - J_{\mathrm{RVM}}^{\mathrm{M}}(x)) x^{-s-1}\, dx \tag{6.20}$$

*is holomorphic for $\Re s > 0$. The two functions have the same zeros in $\Re s > 0$.*

*Proof.* By (6.13) replacing $x^{1/m}$ by its integer part costs $O(1/m)$ in the $m$th summand; summing $m^{-1}$ up to $\log_2 x$ gives (6.18). The product (6.2) gives $\log \zeta_{\mathbb{Z},\mathrm{RVM}}(s) = s \int_1^\infty J_{\mathbb{Z},\mathrm{RVM}}(x) x^{-s-1} dx$ for $\Re s > 1$, because $\sum_{n\le y} c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(\lfloor y \rfloor)$. Subtracting (6.17) gives (6.19) on $\Re s > 1$. On $[2,\infty)$, $J_{\mathbb{Z},\mathrm{RVM}} - J_{\mathrm{RVM}}^{\mathrm{M}} = (J_{\mathbb{Z},\mathrm{RVM}} - J_{\mathrm{RVM}}^{*}) - C_{\mathrm{RVM}} = O(\log\log x)$ by (6.18); on $(1,2)$, $J_{\mathbb{Z},\mathrm{RVM}} = 0$ and $J_{\mathrm{RVM}}^{\mathrm{M}}$ is square-integrable. Hence (6.20) converges locally normally for $\Re s > 0$, and exponentiation introduces no zeros. □

**Lemma 6.6** (Exact first-harmonic cell cancellation)**.** *For every integer $n$,*

$$\int_{n-1}^{n} 2\cos(2\pi x)\, dx = 0. \tag{6.21}$$

**Hypothesis 6.7** (Uniform cell remainder)**.**

$$c_n^{\mathrm{RVM}} - \int_{n-1}^{n} \left( \frac{1}{\log x} + 2\cos(2\pi x) \right) dx = o\!\left( \frac{1}{\log n} \right). \tag{6.22}$$

**Proposition 6.8** (Conditional eventual positivity)**.** *Hypothesis 6.7 implies*

$$c_n^{\mathrm{RVM}} \sim \frac{1}{\log n}, \tag{6.23}$$

*hence $0 < c_n^{\mathrm{RVM}} < 1$ for all large $n$.*

*Proof.* The main-density integral is $1/\log n + O(1/(n \log^2 n))$; the first harmonic vanishes by Lemma 6.6. □

## 7. The critical RVM Mertens transform

Define

$$F_{\mathrm{RVM}}(s) = \widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}(s) := \frac{1}{\zeta_{\mathrm{RVM}}(s)} = \frac{G(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{7.1}$$

Formally the associated summatory object is the inverse Mellin transform

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = \frac{1}{2\pi i}\int_{(c)} F_{\mathrm{RVM}}(s)\frac{x^s}{s}\,ds; \tag{7.2}$$

its existence as a symmetric limit, for every $c > \frac{1}{2}$, is proved in Theorem 8.12.

**Proposition 7.1** (Synthetic pole abscissa)**.** *$F_{\mathrm{RVM}}$ is holomorphic in $\Re s > \frac{1}{2}$. Its poles are the simple poles*

$$\frac{1}{2} \pm i\tau_k \quad (k \ge 1) \qquad \textit{and} \qquad s = -2, -4, -6, \dots \tag{7.3}$$

*and $s = 1$ is a simple zero of $F_{\mathrm{RVM}}$.*

*Proof.* This is Theorem 3.1 applied to $F_{\mathrm{RVM}} = G/P^{\mathrm{crit}}_{\mathrm{RVM}}$: the zeros of $P^{\mathrm{crit}}_{\mathrm{RVM}}$ are (3.6), and the poles of $G$ are those of $\Gamma(s/2)$ except $s = 0$, where $s\Gamma(s/2) \to 2$. □

**Proposition 7.2** (Gaussian square-root bound under vertical growth)**.** *Fix $c > 1$ and $0 < \varepsilon < c - \frac{1}{2}$. Assume, uniformly for $\frac{1}{2} + \varepsilon \le \sigma \le c$,*

$$F_{\mathrm{RVM}}(\sigma + it) = \exp(o(t^2)) \qquad (|t| \to \infty). \tag{7.4}$$

*Define*

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) := \frac{1}{2\pi i}\int_{(c)} e^{s^2} F_{\mathrm{RVM}}(s)\frac{x^s}{s}\,ds. \tag{7.5}$$

*Then*

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{7.6}$$

*Proof.* On $s = \sigma + it$, $|e^{s^2}| = e^{\sigma^2 - t^2}$, so (7.4) gives absolute convergence on vertical lines and makes the horizontal integrals tend to 0. The strip $\frac{1}{2} + \varepsilon \le \Re s \le c$ contains no pole of $F_{\mathrm{RVM}}$ (Proposition 7.1) and does not contain $s = 0$, so Cauchy's theorem moves the line to $\Re s = \frac{1}{2} + \varepsilon$, where $|x^s| = x^{1/2+\varepsilon}$. □

Theorem 8.12 below removes both the assumption (7.4) and the smoothing.

## 8. The transition theorem and the unsmoothed synthetic Mertens bound

Let

$$\chi(s) := 2^s\pi^{s-1}\sin\Big(\frac{\pi s}{2}\Big)\Gamma(1-s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \tag{8.1}$$

the factor of the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ [23, §2.1]. Write

$$s = \tfrac{1}{2} + z, \quad z = \sigma' + it, \qquad \tau_* := -iz = t - i\sigma', \qquad w(s) := \exp\{-2\pi i\,\mathcal{N}_{\mathrm{RVM}}(\tau_*)\}. \tag{8.2}$$

Stirling's formula gives, uniformly on compact $\sigma$-sets,

$$|\chi(\sigma + it)| = \Big(\frac{|t|}{2\pi}\Big)^{1/2-\sigma}(1 + O(|t|^{-1})). \tag{8.3}$$

**Theorem 8.1** (Transition evaluation of the midpoint product)**.** *There is a constant $\varpi_{\mathrm{RVM}} > 0$ with the following properties.*

(a) *For every $B > 0$ there are $t_0(B)$ and $C_B$ such that, for $0 < \sigma' \le B$ and $t \ge t_0(B)$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1 + w(s))(1 + \mathcal{E}_0(s)), \qquad |\mathcal{E}_0(s)| \le \frac{C_B}{t}. \tag{8.4}$$

(b) *For every $0<\sigma_1'<\sigma_2'<\frac12$ there are $t_0$ and $C$ such that, for $\sigma_1'\le\sigma'\le\sigma_2'$ and $t\ge t_0$,*

$$\zeta_{\mathrm{RVM}}(s)=\varpi_{\mathrm{RVM}}\big(1+\chi(s)\big)\big(1+\mathcal{E}(s)\big),\qquad |\mathcal{E}(s)|\le\frac{C}{t}. \tag{8.5}$$

*For $t<0$ the same statements hold by $\zeta_{\mathrm{RVM}}(\bar s)=\overline{\zeta_{\mathrm{RVM}}(s)}$.*

The phase identity $w=\chi(s)(1+O(1/t))$ alone does not give (8.4); the product has to be evaluated across the transition region $\tau_k\asymp t$. Log denotes the principal logarithm and $\operatorname{Log}\Gamma$ the branch of $\log\Gamma$ on $\Re v>0$ that is real on $(0,\infty)$. Nothing about $\zeta$ is used.

8.1. **Proof of Theorem 8.1.** Throughout, $t>0$, $0<\sigma'\le B$, $\tau_0:=4\pi$, $t^2\ge 2(\tau_0^2+B^2)$, and implied constants depend only on $B$ (and on $\sigma_1',\sigma_2'$ in part (b)).

*Step 0: reduction to a sum over the clock.* Since $q(s)=z^2-\frac14$, each factor of (3.2) equals $(\tau_k^2+z^2)/(\tau_k^2+\frac14)$, so

$$\frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)}=\prod_{k\ge1}\Big(1+\frac{z^2}{\tau_k^2}\Big)=\exp\Lambda(z),\qquad \Lambda(z):=\sum_{k\ge1}f(\tau_k),\quad f(\tau):=\operatorname{Log}\Big(1+\frac{z^2}{\tau^2}\Big). \tag{8.6}$$

For $\tau>0$, $\Im(1+z^2/\tau^2)=2\sigma' t/\tau^2>0$, so $f$ is smooth on $(0,\infty)$, $f(\tau)=O(|z|^2\tau^{-2})$, the series converges absolutely, and

$$f'(\tau)=-\frac{2z^2}{\tau(\tau^2+z^2)}=-\frac{2}{\tau}+\frac{1}{\tau-iz}+\frac{1}{\tau+iz}. \tag{8.7}$$

*Step 1: the midpoint sawtooth.* The clock increases on $[2\pi,\infty)$, $\mathcal{N}_{\mathrm{RVM}}(\tau_0)=2\log\frac{2}{e}+\frac78\in(0,\frac12)$, and $\tau_1>\tau_0$. With $\psi(y):=y-\lfloor y\rfloor-\frac12$, the count (4.12) satisfies, for $\tau\ge\tau_0$,

$$n_{\mathrm{RVM}}(\tau)=\lfloor\mathcal{N}_{\mathrm{RVM}}(\tau)+\tfrac12\rfloor=\mathcal{N}_{\mathrm{RVM}}(\tau)-\psi\big(\mathcal{N}_{\mathrm{RVM}}(\tau)+\tfrac12\big),\qquad n_{\mathrm{RVM}}(\tau_0)=0. \tag{8.8}$$

Stieltjes integration of $f$ against (8.8), followed by integration by parts in the $\psi$-part ($f(\infty)=0$, $|\psi|\le\frac12$, $\psi(\mathcal{N}_{\mathrm{RVM}}(\tau_0)+\frac12)=\mathcal{N}_{\mathrm{RVM}}(\tau_0)$), gives

$$\begin{aligned}&\Lambda(z)=\Lambda_{\mathrm{sm}}(z)+\Lambda_{\mathrm{saw}}(z),\qquad \Lambda_{\mathrm{sm}}(z):=\int_{\tau_0}^{\infty}f\,d\mathcal{N}_{\mathrm{RVM}}+f(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0),\\ &\Lambda_{\mathrm{saw}}(z):=\int_{\tau_0}^{\infty}\psi\big(\mathcal{N}_{\mathrm{RVM}}(\tau)+\tfrac12\big)f'(\tau)\,d\tau.\end{aligned} \tag{8.9}$$

**Lemma 8.2** (Smooth part)**.** *With $\log G(s):=\operatorname{Log}(z^2-\frac14)+\operatorname{Log}\Gamma(\frac{s}{2})-\frac{s}{2}\log\pi$,*

$$\Lambda_{\mathrm{sm}}(z)-\log G(s)=\nu_3+O(t^{-1}),\qquad \nu_3\in\mathbb{R}\ \textit{independent of } z. \tag{8.10}$$

*Proof.* A second integration by parts ($f\mathcal{N}_{\mathrm{RVM}}\to0$ at $\infty$) cancels the boundary term:

$$\Lambda_{\mathrm{sm}}(z)=-\int_{\tau_0}^{\infty}\mathcal{N}_{\mathrm{RVM}}f'\,d\tau=\int_{\tau_0}^{\infty}\mathcal{N}_{\mathrm{RVM}}(\tau)\Big(\frac{2}{\tau}-\frac{2\tau}{\tau^2+z^2}\Big)d\tau. \tag{8.11}$$

The constant $\frac78$ of the clock contributes $\frac78\operatorname{Log}(1+z^2/\tau_0^2)$, because $\operatorname{Log}\frac{\tau^2}{\tau^2+z^2}$ is an antiderivative of the bracket on $(0,\infty)$ tending to 0 at $\infty$. As $0<\arg z<\frac{\pi}{2}$, $\arg(1+\tau_0^2/z^2)=O(t^{-2})$ and

$$\tfrac78\operatorname{Log}\Big(1+\frac{z^2}{\tau_0^2}\Big)=\tfrac74\operatorname{Log}z-\tfrac74\log\tau_0+O(t^{-2}). \tag{8.12}$$

The term $\frac{\tau}{2\pi}\log\frac{\tau}{2\pi e}$ contributes $\frac1\pi\int_{\tau_0}^{\infty}\log\frac{\tau}{2\pi e}\frac{z^2}{\tau^2+z^2}\,d\tau$. Put

$$\mathcal{I}(z):=\frac1\pi\int_0^{\infty}\log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2+z^2}\,d\tau. \tag{8.13}$$

The integrand is holomorphic in $z$ on $\Re z>0$ (its singularities are at $z=\pm i\tau$), $O(|z|^2\tau^{-2}\log\tau)$ at $\infty$ and $O(|\log\tau|)$ at 0, locally uniformly; so $\mathcal{I}$ is holomorphic on $\Re z>0$. For real $z=x>0$

the substitution $\tau = xv$ and $\int_0^\infty \frac{dv}{1+v^2} = \frac{\pi}{2}$, $\int_0^\infty \frac{\log v}{1+v^2}\,dv = 0$ give $\mathcal{I}(x) = \frac{x}{2}\log\frac{x}{2\pi e}$; by the identity theorem $\mathcal{I}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e}$ on $\Re z > 0$. On $[0,\tau_0]$, $|\tau^2+z^2| \ge t^2/2$, so $\frac{z^2}{\tau^2+z^2} = 1 + O(t^{-2})$ there. Hence

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac{7}{4}\operatorname{Log} z + \nu_1 + O(t^{-2}), \qquad \nu_1 = -\tfrac{7}{4}\log\tau_0 - \tfrac{1}{\pi}\int_0^{\tau_0} \log\tfrac{\tau}{2\pi e}\,d\tau. \tag{8.14}$$

Stirling's formula $\operatorname{Log}\Gamma(v) = (v-\frac12)\operatorname{Log} v - v + \frac12\log 2\pi + O(|v|^{-1})$ [11, §5.11] with $v = \frac{s}{2} = \frac{z+1/2}{2}$ and $\operatorname{Log} v = \operatorname{Log} z - \log 2 + \frac{1}{2z} + O(t^{-2})$ gives

$$\operatorname{Log}\Gamma(\tfrac{s}{2}) = \tfrac{z}{2}\operatorname{Log}\tfrac{z}{2} - \tfrac14\operatorname{Log} z + \tfrac14\log 2 - \tfrac{z}{2} + \tfrac12\log 2\pi + O(t^{-1}), \tag{8.15}$$

and with $\operatorname{Log}(z^2-\frac14) = 2\operatorname{Log} z + O(t^{-2})$, $-\frac{s}{2}\log\pi = -\frac{z}{2}\log\pi - \frac14\log\pi$,

$$\log G(s) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac{7}{4}\operatorname{Log} z + \nu_2 + O(t^{-1}), \qquad \nu_2 = \tfrac14\log 2 + \tfrac12\log 2\pi - \tfrac14\log\pi. \tag{8.16}$$

Subtracting gives (8.10) with $\nu_3 = \nu_1 - \nu_2$. □

**Lemma 8.3** (The clock as a holomorphic phase)**.** *$\mathcal{N}_{\mathrm{RVM}}(\tau) = \frac{\tau}{2\pi}\big(\operatorname{Log}\frac{\tau}{2\pi} - 1\big) + \frac78$ is holomorphic on $\Re\tau > 0$, $\mathcal{N}_{\mathrm{RVM}}(\bar\tau) = \overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$, and for $\Re\tau \ge \tau_0$,*

$$2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \ge (\log 2)\,\Im\tau \quad (\Im\tau \ge 0), \qquad 2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \le (\log 2)\,\Im\tau \quad (\Im\tau \le 0). \tag{8.17}$$

*Proof.* For $\tau = X + iY$, $X \ge \tau_0$, $Y > 0$, put $v = Y/X$. Then

$$2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau) = Y\log\frac{|\tau|}{2\pi e} + X\arg\tau = Y\Big[\log\frac{X}{2\pi e} + \tfrac12\log(1+v^2) + \frac{\arctan v}{v}\Big] \ge Y\big[\log 2 + \varphi(v)\big], \tag{8.18}$$

using $X \ge 4\pi$, where $\varphi(v) = \frac12\log(1+v^2) + \frac{\arctan v}{v} - 1$. Since $\varphi(0^+) = 0$ and $\varphi'(v) = (v - \arctan v)/v^2 \ge 0$, $\varphi \ge 0$. The case $Y < 0$ follows by conjugation. □

In particular $|w(s)| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} \le 2^{-\sigma'} < 1$, and on the vertical rays $|e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0 \pm iy)}| \le 2^{-my}$.

**Lemma 8.4** (Abel-regularized harmonic sum)**.**

$$\Lambda_{\mathrm{saw}}(z) = \operatorname{Log}\big(1 + w(s)\big) + \nu_4 + O(t^{-2}), \qquad \nu_4 \in \mathbb{R} \text{ independent of } z. \tag{8.19}$$

*Proof. Regularization.* For $0 < r < 1$ let

$$\psi_r(y) := -\sum_{m\ge1} r^m \frac{\sin 2\pi m y}{\pi m} = \frac{1}{\pi}\Im\operatorname{Log}(1 - re^{2\pi i y}). \tag{8.20}$$

Since $\Re(1 - re^{2\pi iy}) > 0$, $|\psi_r| \le \frac12$, and $\psi_r(y) \to \psi(y)$ as $r \uparrow 1$ for $y \notin \mathbb{Z}$ (Abel summation of the convergent Fourier series of $\psi$). The set $\{\tau : \mathcal{N}_{\mathrm{RVM}}(\tau) + \frac12 \in \mathbb{Z}\}$ is discrete and $f' \in L^1(\tau_0,\infty)$, so by dominated convergence

$$\Lambda_{\mathrm{saw}}(z) = \lim_{r\uparrow1}\Lambda_r(z), \qquad \Lambda_r(z) := \int_{\tau_0}^\infty \psi_r\big(\mathcal{N}_{\mathrm{RVM}} + \tfrac12\big) f'\,d\tau. \tag{8.21}$$

*Termwise integration.* For fixed $r < 1$ the series for $\psi_r(\mathcal{N}_{\mathrm{RVM}} + \frac12)$, with $\sin 2\pi m(y+\frac12) = (-1)^m \sin 2\pi m y$, is dominated by $\sum_m r^m/m$; by Fubini,

$$\Lambda_r(z) = -\sum_{m\ge1} \frac{(-r)^m}{2\pi i\, m}\big(I_m^+ - I_m^-\big), \qquad I_m^\pm := \int_{\tau_0}^\infty e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau)} f'(\tau)\,d\tau. \tag{8.22}$$

*Contours.* Let $U^\pm = \{\Re\tau \ge \tau_0,\ \pm\Im\tau \ge 0\}$. On $U^\pm$, $|e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}}| \le 1$ by Lemma 8.3, and $|f'(\tau)| \le 4|z|^2|\tau|^{-3}$ for $|\tau| \ge 2|z|$, so the quarter-circle arcs of radius $R$ contribute $O(|z|^2R^{-2}) \to 0$.

The poles of $f'$ are $0$, $iz=-t+i\sigma'\notin U^{\pm}$, and $\tau_*=-iz=t-i\sigma'\in U^-$, the last with residue $1$ (from $\frac{1}{\tau+iz}$). Define the ray integrals

$$R_m^+:=i\int_0^\infty e^{2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0+iy)}f'(\tau_0+iy)\,dy,\qquad R_m^-:=-i\int_0^\infty e^{-2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0-iy)}f'(\tau_0-iy)\,dy. \tag{8.23}$$

On $U^+$ Cauchy's theorem gives $I_m^+=R_m^+$. On $U^-$ the positively oriented boundary of the quarter disc runs down the ray, along the arc, and back along $[\tau_0,\tau_0+R]$ from right to left; letting $R\to\infty$,

$$R_m^- - I_m^- = 2\pi i\, e^{-2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_*)} = 2\pi i\, w^m. \tag{8.24}$$

Therefore, using $|rw|<1$,

$$\Lambda_r(z)=-\sum_{m\ge1}\frac{(-rw)^m}{m}+\Phi_r(z)=\mathrm{Log}(1+rw)+\Phi_r(z),\qquad \Phi_r(z):=-\sum_{m\ge1}\frac{(-r)^m}{2\pi i\,m}(R_m^+-R_m^-). \tag{8.25}$$

*The rays.* Split $f'(\tau)=-\frac{2}{\tau}+g_z(\tau)$, $g_z(\tau)=\frac{2\tau}{\tau^2+z^2}$, and correspondingly $R_m^{\pm}=R_m^{\pm,(0)}+R_m^{\pm,(z)}$, $\Phi_r=\Phi_r^{(0)}+\Phi_r^{(z)}$. The ray integrands carry the factor $2^{-my}$. Since $|2/\tau|\le 2/\tau_0$ on the rays, $|R_m^{\pm,(0)}|\le 2/(\tau_0 m\log 2)$, so the series for $\Phi_r^{(0)}$ is dominated by $\sum_m C/m^2$ uniformly in $r$, and $\Phi_r^{(0)}\to\nu_4:=\Phi_1^{(0)}$, a constant depending only on $\tau_0$. It is real: $\mathcal{N}_{\mathrm{RVM}}(\bar\tau)=\overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$ gives $R_m^{-,(0)}=\overline{R_m^{+,(0)}}$, so each term equals $-(-r)^m\Im R_m^{+,(0)}/(\pi m)$. For $\tau=\tau_0\pm iy$, $\Re(\tau^2+z^2)=\tau_0^2+\sigma'^2-y^2-t^2\le-\frac12(y^2+t^2)$ because $t^2\ge 2(\tau_0^2+B^2)$; hence $|g_z(\tau)|\le 4(\tau_0+y)t^{-2}$ and

$$|R_m^{\pm,(z)}|\le\frac{4}{t^2}\int_0^\infty 2^{-my}(\tau_0+y)\,dy=\frac{4}{t^2}\Big(\frac{\tau_0}{m\log 2}+\frac{1}{(m\log 2)^2}\Big). \tag{8.26}$$

So $|\Phi_r^{(z)}(z)|\le C\,t^{-2}\sum_m m^{-2}$, uniformly in $0<r<1$ and $0<\sigma'\le B$, and $\Phi_r^{(z)}$ converges as $r\uparrow1$ by dominated convergence of the series. Since $\mathrm{Log}(1+rw)\to\mathrm{Log}(1+w)$, letting $r\uparrow1$ in (8.25) and (8.21) gives (8.19). □

*Step 2: assembly.* By (8.6), (8.9) and Lemmas 8.2 and 8.4,

$$\begin{aligned}\zeta_{\mathrm{RVM}}(s)&=\frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)}=P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\exp\big(\Lambda_{\mathrm{sm}}(z)-\log G(s)+\Lambda_{\mathrm{saw}}(z)\big)\\&=P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\,e^{\nu_3+\nu_4}(1+w)(1+O(t^{-1})),\end{aligned} \tag{8.27}$$

uniformly for $0<\sigma'\le B$. This is (8.4) with

$$\varpi_{\mathrm{RVM}}:=P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\,e^{\nu_3+\nu_4}>0, \tag{8.28}$$

which depends only on the clock; its numerical value $0.9619415776\ldots$ plays no role in the proof.

*Step 3: from the clock phase to $\chi$.* Let

$$\vartheta(v):=\frac{1}{2i}\Big(\mathrm{Log}\,\Gamma\big(\tfrac14+\tfrac{iv}{2}\big)-\mathrm{Log}\,\Gamma\big(\tfrac14-\tfrac{iv}{2}\big)\Big)-\frac{v}{2}\log\pi, \tag{8.29}$$

holomorphic on $|\Im v|<\frac12$. The identity $\chi(\frac12+iv)=e^{-2i\vartheta(v)}$ holds for real $v$, hence on the strip, and $s=\frac12+i\tau_*$ with $|\Im\tau_*|=\sigma'<\frac12$. Stirling's formula in the strip gives $\vartheta(v)=\frac{v}{2}\,\mathrm{Log}\,\frac{v}{2\pi}-\frac{v}{2}-\frac{\pi}{8}+O(|v|^{-1})$, i.e.

$$\mathcal{N}_{\mathrm{RVM}}(v)=\frac{\vartheta(v)}{\pi}+1+O(|v|^{-1}). \tag{8.30}$$

Therefore

$$w=e^{-2i\vartheta(\tau_*)-2\pi i+O(1/t)}=\chi(s)(1+O(t^{-1})). \tag{8.31}$$

For $\sigma'\ge\sigma_1'$, (8.3) gives $|\chi(s)|\le\frac12$ when $t\ge t_0(\sigma_1')$, so $1+w=(1+\chi)(1+(w-\chi)/(1+\chi))=(1+\chi)(1+O(t^{-1}))$, which with (8.4) gives (8.5). This completes the proof of Theorem 8.1. □

**Lemma 8.5** (Extension to the critical line)**.** *The function* $\mathcal{E}_0 = \zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1$ *extends holomorphically to* $\{|\sigma'| < \frac{1}{2},\ t > t_0\}$*, and* $|\mathcal{E}_0| \le C/t$ *for* $0 \le \sigma' \le \frac{1}{4}$*.*

*Proof.* By Lemma 8.3, $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) \neq 0$ for $\sigma' \neq 0$, so $1+w$ vanishes in the strip only on $\sigma' = 0$, exactly at $t = \tau_k$; these zeros are simple because $\frac{d}{dt}w = -\log(\tau_*/2\pi)\, w \neq 0$. The zeros of $\zeta_{\mathrm{RVM}}$ in the strip are the same simple points. Hence $\mathcal{E}_0$ has only removable singularities, and the bound extends to $\sigma' = 0$ by continuity. □

**Corollary 8.6** (Derivative at the zeros)**.** *For* $k$ *with* $\tau_k > t_0$*,*

$$\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}}) = \varpi_{\mathrm{RVM}} \log \frac{\tau_k}{2\pi}\Big(1 + O\Big(\frac{1}{\tau_k}\Big)\Big). \tag{8.32}$$

*Proof.* By Lemma 8.5, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ holds holomorphically near $\sigma' = 0$. At $s = \rho_k^{\mathrm{RVM}}$, $1 + w = 0$, so $\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}}) = \varpi_{\mathrm{RVM}} w'(\rho_k^{\mathrm{RVM}})(1 + \mathcal{E}_0(\rho_k^{\mathrm{RVM}}))$. From (8.2), $w'(s) = -2\pi i\, \mathcal{N}'_{\mathrm{RVM}}(\tau_*) \cdot (-i)\, w(s) = -\log(\tau_*/2\pi)\, w(s)$, and at the zero $w = -1$, $\tau_* = \tau_k$. Finally $|\mathcal{E}_0| \le C/\tau_k$. □

*Remark* 8.7 (Structure of the proof). The delicate points are handled as follows: the smooth part is computed with principal branches by the identity theorem (Lemma 8.2); the clock estimate is quantified (Lemma 8.3); the sawtooth series is replaced by its Abel means, termwise integration is justified for $r < 1$, and $r \uparrow 1$ is taken after the contour argument uniformly in $r$ and $z$ (Lemma 8.4); the orientation of the lower contour and the sign of the residue are written out in (8.24); the $z$-dependence of the rays is isolated in $g_z$ and is $O(t^{-2})$, the rest is the explicit real constant $\nu_4$; Stirling is used uniformly for $0 < \sigma' \le B$, and the only place where $\sigma' < \frac{1}{2}$ enters is Step 3.

**Corollary 8.8** (Vertical constant limit)**.** *For every fixed* $\sigma > \frac{1}{2}$*,*

$$\zeta_{\mathrm{RVM}}(\sigma + it) \longrightarrow \varpi_{\mathrm{RVM}}, \qquad F_{\mathrm{RVM}}(\sigma + it) \longrightarrow L := \frac{1}{\varpi_{\mathrm{RVM}}} \qquad (|t| \to \infty). \tag{8.33}$$

*More precisely, with* $\sigma' = \sigma - \frac{1}{2}$*,*

$$|w(\sigma + it)| = O_\sigma(|t|^{-\sigma'}), \tag{8.34}$$

*and, with* $\alpha_c := \min(c - \frac{1}{2}, 1)$ *for* $c > \frac{1}{2}$*,*

$$F_{\mathrm{RVM}}(c + it) - L = O_c(|t|^{-\alpha_c}). \tag{8.35}$$

*Proof.* Expanding the clock at $\tau_* = t - i\sigma'$ gives $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) = -\frac{\sigma'}{2\pi}\log\frac{t}{2\pi} + O_\sigma(t^{-1})$, hence $|w| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} = O_\sigma(t^{-\sigma'})$. Inverting (8.4) for large $t$ gives $F_{\mathrm{RVM}} = L(1 + O(|w|) + O(t^{-1}))$. □

8.2. **Bounds on the whole half-plane.** The constants of Theorem 8.1 depend on the width $B$ of the band. The following estimate holds on the whole half-plane and does not use contours. Put

$$\mathfrak{s}_1(\tau) := \int_{\tau_0}^{\tau} \psi(\mathcal{N}_{\mathrm{RVM}}(v) + \tfrac{1}{2})\, dv \qquad (\tau \ge \tau_0). \tag{8.36}$$

**Lemma 8.9** (Bounded primitive of the midpoint sawtooth)**.** $|\mathfrak{s}_1(\tau)| \le \dfrac{\pi}{4\log 2}$ *for all* $\tau \ge \tau_0$*.*

*Proof.* Substitute $y = \mathcal{N}_{\mathrm{RVM}}(v)$, $dv = \mathcal{T}'(y)\, dy$ with $\mathcal{T}'(y) = 2\pi/\log(\mathcal{T}(y)/2\pi)$ from (4.4). The weight $\mathcal{T}'$ is positive and decreasing, with $\mathcal{T}' \le 2\pi/\log 2$ on $y \ge \mathcal{N}_{\mathrm{RVM}}(\tau_0)$ because $\log(\tau_0/2\pi) = \log 2$. The primitive of $y \mapsto \psi(y + \frac{1}{2})$ is periodic with values in $[-\frac{1}{8}, 0]$, so every integral of $\psi(y + \frac{1}{2})$ over an interval has modulus at most $\frac{1}{8}$. The second mean-value theorem gives $|\mathfrak{s}_1(\tau)| \le \frac{2\pi}{\log 2} \cdot \frac{1}{8}$. □

**Theorem 8.10** (Two-sided bound and limit at $+\infty$)**.** *There is an absolute constant* $R_0$ *such that, for* $s = \frac{1}{2} + z$*,* $z = \sigma' + it$*,* $\sigma' > \frac{1}{2}$ *and* $|z| \ge R_0$*,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\, e^{\theta_{\mathrm{RVM}}(s)}, \qquad |\theta_{\mathrm{RVM}}(s)| \le \frac{\pi^2}{2\log 2} \cdot \frac{1}{\sigma'} + \frac{C}{|z|}. \tag{8.37}$$

*Consequently, for every $\sigma_0 > 1$ there is $C(\sigma_0)$ with*

$$C(\sigma_0)^{-1} \le |\zeta_{\mathrm{RVM}}(s)| \le C(\sigma_0) \qquad (\Re s \ge \sigma_0), \tag{8.38}$$

*and $\zeta_{\mathrm{RVM}}(s) \to \varpi_{\mathrm{RVM}}$ uniformly in $t$ as $\Re s \to +\infty$. Moreover*

$$\nu_4 = -2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau. \tag{8.39}$$

*Proof. Smooth part.* For $\Re z > 0$ the computation in the proof of Lemma 8.2 is exact:

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac{7}{8}\operatorname{Log}\Big(1+\frac{z^2}{\tau_0^2}\Big) - \frac{1}{\pi}\int_0^{\tau_0} \log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2+z^2}\, d\tau. \tag{8.40}$$

For $|z| \ge 2\tau_0$ one has $|\tau^2+z^2| \ge \frac{3}{4}|z|^2$ on $[0,\tau_0]$, and Stirling's formula is uniform in $|\arg(s/2)| < \frac{\pi}{2}$; the expansions (8.14) and (8.16) therefore hold with $O(|z|^{-1})$ in place of $O(t^{-1})$, uniformly on $\Re z > \frac{1}{2}$, and $\Lambda_{\mathrm{sm}}(z) - \log G(s) = \nu_3 + O(|z|^{-1})$.

*Sawtooth part.* Integrating (8.9) by parts with (8.36) ($\mathfrak{s}_1(\tau_0) = 0$, $f'(\tau) = O(|z|^2\tau^{-3})$),

$$\Lambda_{\mathrm{saw}}(z) = -\int_{\tau_0}^{\infty} \mathfrak{s}_1 f''\, d\tau, \qquad f''(\tau) = \frac{2}{\tau^2} - \frac{1}{(\tau - iz)^2} - \frac{1}{(\tau+iz)^2}. \tag{8.41}$$

Since $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$ and $\int_{\mathbb{R}} \frac{d\tau}{(\tau-a)^2+\sigma'^2} = \frac{\pi}{\sigma'}$, Lemma 8.9 gives

$$\Big|\Lambda_{\mathrm{saw}}(z) + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau\Big| \le \frac{\pi}{4\log 2}\cdot\frac{2\pi}{\sigma'}. \tag{8.42}$$

*Assembly.* By (8.27), $\log\zeta_{\mathrm{RVM}}(s) = \log\varpi_\infty + \theta_{\mathrm{RVM}}(s)$ with $\varpi_\infty := P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2})\exp(\nu_3 - 2\int_{\tau_0}^{\infty}\mathfrak{s}_1\tau^{-2}d\tau) > 0$ and $\theta_{\mathrm{RVM}}$ bounded as in (8.37). For fixed $\sigma'$ and $t \to \infty$, Corollary 8.8 gives $|\log\varpi_{\mathrm{RVM}} - \log\varpi_\infty| \le \pi^2/(2\sigma'\log 2)$; letting $\sigma' \to \infty$, $\varpi_\infty = \varpi_{\mathrm{RVM}}$, which is (8.39) by (8.28). On the compact set $\{\Re s \ge \sigma_0,\ |z| \le R_0\}$ the function $\zeta_{\mathrm{RVM}}$ is continuous and has no zeros or poles, which gives (8.38). □

Theorem 8.10 extends the two-sided bound (9.9) of Corollary 9.3 from bands to the whole half-plane $\Re s \ge \sigma_0 > 1$, and identifies $\varpi_{\mathrm{RVM}}$ also as the limit of $\zeta_{\mathrm{RVM}}(\sigma)$ as $\sigma \to +\infty$.

**Theorem 8.11** (No absolutely convergent Dirichlet expansion)**.** *Neither $\zeta_{\mathrm{RVM}}$ nor $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$ is represented by a Dirichlet series $\sum_n a_n n^{-s}$ converging absolutely in some half-plane $\Re s > \sigma_0$.*

*Proof.* Suppose $F_{\mathrm{RVM}}(s) = \sum_{n\ge1} a_n n^{-s}$ absolutely for $\Re s > \sigma_0$. Fix $\sigma > \max(\sigma_0, \frac{1}{2})$ and put $\tilde a_1 = a_1 - L$, $\tilde a_n = a_n$ $(n \ge 2)$, so $F_{\mathrm{RVM}}(\sigma+it) - L = \sum_n \tilde a_n n^{-\sigma-it}$ with $\sum_n |\tilde a_n| n^{-\sigma} < \infty$. Then

$$\frac{1}{2T}\int_{-T}^{T}\big|F_{\mathrm{RVM}}(\sigma+it) - L\big|^2 dt = \sum_{m,n} \tilde a_m \overline{\tilde a_n}(mn)^{-\sigma}\,\frac{1}{2T}\int_{-T}^{T}\Big(\frac{n}{m}\Big)^{it} dt, \tag{8.43}$$

the interchange being justified by absolute convergence of the double series, uniformly in $t$. The inner average is 1 for $m = n$ and is bounded by $\min(1, 1/(T|\log(n/m)|)) \to 0$ for $m \ne n$; each term is bounded by $|\tilde a_m \tilde a_n|(mn)^{-\sigma}$, which is summable, so dominated convergence gives

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\big|F_{\mathrm{RVM}}(\sigma+it) - L\big|^2 dt = \sum_{n\ge1}|\tilde a_n|^2 n^{-2\sigma}. \tag{8.44}$$

By (8.33) the integrand tends to 0 and is bounded on the line (continuous, with a finite limit), so the left side of (8.44) is 0. Hence every $\tilde a_n = 0$, $F_{\mathrm{RVM}} \equiv L$ in the half-plane and, by analytic continuation, everywhere, which contradicts $F_{\mathrm{RVM}}(1) = 0$. The same argument applied to $\zeta_{\mathrm{RVM}} - \varpi_{\mathrm{RVM}}$ forces an absolutely convergent Dirichlet series for $\zeta_{\mathrm{RVM}}$ to be the constant $\varpi_{\mathrm{RVM}}$, which contradicts its pole at $s = 1$. □

**Theorem 8.12** (Unsmoothed synthetic Mertens bound)**.** *For every $x > 1$ and every $c > \frac{1}{2}$ the symmetric Perron integral*

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) := \lim_{U\to\infty} \frac{1}{2\pi i} \int_{c-iU}^{c+iU} F_{\mathrm{RVM}}(s) \frac{x^s}{s}\, ds \tag{8.45}$$

*exists, is independent of $c > \frac{1}{2}$, and equals the absolutely convergent expression*

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + \frac{1}{2\pi i} \int_{(c)} (F_{\mathrm{RVM}}(s) - L) \frac{x^s}{s}\, ds. \tag{8.46}$$

*For every $\varepsilon \in (0, \frac{1}{2})$,*

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon}). \tag{8.47}$$

*Proof.* Fix $c > \frac{1}{2}$. By (8.35) and continuity of $F_{\mathrm{RVM}}$ on $\Re s = c$ (no poles there by Proposition 7.1), the integral in (8.46) converges absolutely. The Perron kernel gives, for $x > 1$ and $c > 0$ [7, Ch. 17],

$$\lim_{U\to\infty} \frac{1}{2\pi i} \int_{c-iU}^{c+iU} \frac{x^s}{s}\, ds = 1. \tag{8.48}$$

Splitting $F_{\mathrm{RVM}} = L + (F_{\mathrm{RVM}} - L)$ proves existence of (8.45) and (8.46). If $\frac{1}{2} < c < c'$, the integrand of (8.45) is holomorphic on the closed strip $c \le \Re s \le c'$ (Proposition 7.1); by Theorem 8.1(a) with $B = c' - \frac{1}{2}$, $F_{\mathrm{RVM}}$ is bounded on the horizontal segments $\Im s = \pm U$, $U \ge t_0(B)$, so these contribute $O(x^{c'}U^{-1}) \to 0$, and Cauchy's theorem gives independence of $c$. Take $c = \frac{1}{2} + \varepsilon$ in (8.46):

$$|M^{\mathrm{crit}}_{\mathrm{RVM}}(x) - L| \le \frac{x^{1/2+\varepsilon}}{2\pi} \int_{-\infty}^{\infty} \frac{|F_{\mathrm{RVM}}(\frac{1}{2} + \varepsilon + it) - L|}{|\frac{1}{2} + \varepsilon + it|}\, dt, \tag{8.49}$$

and the integral is finite by (8.35), the integrand being $O(|t|^{-1-\alpha_c})$. □

*Remark* 8.13 (Scope). Theorem 8.12 concerns $1/\zeta_{\mathrm{RVM}}$. The transfer to $M$ goes through the endpoint of the $\lambda$-family and is Theorem 11.8.

## 9. The exact Hadamard product and the comparison with $\zeta$

Put

$$P_\zeta(s) := G(s)\zeta(s) = 2\xi(s), \qquad \xi(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{9.1}$$

**Theorem 9.1** (Exact Hadamard product)**.** *Let $\rho$ run over the nontrivial zeros of $\zeta$ with $\Im\rho > 0$, with multiplicity, ordered by $\Im\rho$. The product*

$$P_\zeta(s) = \prod_{\Im\rho>0} \Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big) \tag{9.2}$$

*converges absolutely and locally uniformly in $s \in \mathbb{C}$, and the identity holds with no exponential or constant factor.*

*Proof.* $\xi$ is entire of order one with $\xi(0) = \frac{1}{2}$, so Hadamard's factorization [15], [23, §2.12], [7, Ch. 12] gives

$$2\xi(s) = e^{Bs} \prod_\rho \Big(1 - \frac{s}{\rho}\Big) e^{s/\rho}, \tag{9.3}$$

the product running over all nontrivial zeros and converging absolutely because $\sum_\rho |\rho|^{-2} < \infty$. The functional equation $\xi(s) = \xi(1-s)$ makes $\rho \mapsto 1-\rho$ an involution of the zero multiset; no nontrivial zero is real, since $(1-2^{1-\sigma})\zeta(\sigma) = \sum_{n\ge1}(-1)^{n-1}n^{-\sigma} > 0$ for $0 < \sigma < 1$; so this involution exchanges $\Im\rho > 0$ and $\Im\rho < 0$. Grouping $\rho$ with $1-\rho$, which absolute convergence permits,

$$\Big(1 - \frac{s}{\rho}\Big)\Big(1 - \frac{s}{1-\rho}\Big) e^{s/\rho + s/(1-\rho)} = \Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big) e^{s/(\rho(1-\rho))}. \tag{9.4}$$

Since $|\rho(1-\rho)| \geq (\Im\rho)^2$ and $\sum_{\Im\rho>0}(\Im\rho)^{-2} < \infty$, the numbers $1/(\rho(1-\rho))$ are absolutely summable, so the exponentials collect into a single factor:

$$2\xi(s) = e^{B's}\,\Pi_\zeta(s), \qquad \Pi_\zeta(s) := \prod_{\Im\rho>0}\Big(1+\frac{q(s)}{\rho(1-\rho)}\Big), \qquad B' = B + \sum_{\Im\rho>0}\frac{1}{\rho(1-\rho)}. \tag{9.5}$$

$\Pi_\zeta$ depends on $s$ only through $q(s)$, hence $\Pi_\zeta(1-s) = \Pi_\zeta(s)$. From $\xi(1-s) = \xi(s)$ we get $e^{B's} = e^{B'(1-s)}$ on the open set where $\Pi_\zeta \neq 0$, so $B' = 0$. □

Define the exact quotient

$$\mathcal{Q}_{\mathrm{RVM}}(s) := \frac{\zeta(s)}{\zeta_{\mathrm{RVM}}(s)} = \frac{P_\zeta(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{9.6}$$

**Proposition 9.2** (Exact transform relation)**.** *For $\Re s > 1$, and by meromorphic continuation,*

$$F_{\mathrm{RVM}}(s) = \mathcal{Q}_{\mathrm{RVM}}(s)\,\frac{1}{\zeta(s)}. \tag{9.7}$$

*If the inverse Mellin transform $\mathcal{Q}^\vee_{\mathrm{RVM}}$ of $\mathcal{Q}_{\mathrm{RVM}}$ and the inverse defining $M^{\mathrm{crit}}_{\mathrm{RVM}}$ exist in a common distribution space with Mellin convolution, then*

$$M^{\mathrm{crit}}_{\mathrm{RVM}} = \mathcal{Q}^\vee_{\mathrm{RVM}} *_{\mathrm{Mel}} M. \tag{9.8}$$

*Proof.* (9.7) is the identity $1/\zeta_{\mathrm{RVM}} = (\zeta/\zeta_{\mathrm{RVM}})(1/\zeta)$; (9.8) is the Mellin convolution theorem under the stated assumptions. □

**Corollary 9.3** (All growth of $\zeta$ to the right of $\frac12$ lies in $\mathcal{Q}_{\mathrm{RVM}}$)**.** *Let $0 < \varepsilon < c' - \frac12$. There are $t_0$ and $0 < c_- \leq c_+$ such that*

$$c_- \leq |\zeta_{\mathrm{RVM}}(\sigma+it)| \leq c_+ \qquad (\tfrac12+\varepsilon \leq \sigma \leq c',\ |t| \geq t_0), \tag{9.9}$$

*and therefore, in the same region,*

$$c_-|\mathcal{Q}_{\mathrm{RVM}}(s)| \leq |\zeta(s)| \leq c_+|\mathcal{Q}_{\mathrm{RVM}}(s)|, \qquad \frac{1}{c_+|\mathcal{Q}_{\mathrm{RVM}}(s)|} \leq \Big|\frac{1}{\zeta(s)}\Big| \leq \frac{1}{c_-|\mathcal{Q}_{\mathrm{RVM}}(s)|}. \tag{9.10}$$

*Consequently:*

(i) *the Lindelöf hypothesis holds if and only if $\mathcal{Q}_{\mathrm{RVM}}(\sigma+it) = O_{\sigma,\varepsilon}(|t|^\varepsilon)$ for every $\sigma > \frac12$;*
(ii) *every bound for $\zeta$ or $1/\zeta$ on $\frac12+\varepsilon \leq \sigma \leq c'$, $|t| \geq t_0$, is equivalent, up to the constants $c_\pm$, to the same bound for $\mathcal{Q}_{\mathrm{RVM}}$ or $1/\mathcal{Q}_{\mathrm{RVM}}$.*

*Proof.* By Theorem 8.1(a) with $B = c' - \frac12$, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|\mathcal{E}_0| \leq C_B/|t|$ and, by (8.34), $|w| \leq C(|t|/2\pi)^{-\varepsilon}$; for $|t| \geq t_0$ this gives $\frac12 \leq |1+w| \leq 2$ and $\frac12 \leq |1+\mathcal{E}_0| \leq 2$, hence (9.9) with $c_- = \varpi_{\mathrm{RVM}}/4$, $c_+ = 4\varpi_{\mathrm{RVM}}$. Then (9.10) follows from $\zeta = \mathcal{Q}_{\mathrm{RVM}}\zeta_{\mathrm{RVM}}$. For (i): the Lindelöf hypothesis is equivalent to $\zeta(\sigma+it) = O(|t|^\varepsilon)$ for every $\sigma > \frac12$ [23, Ch. XIII], and (9.10) transfers this to $\mathcal{Q}_{\mathrm{RVM}}$. (ii) is (9.10). □

On the prime side define

$$\mathfrak{d}^{\mathrm{RVM}}_n := \mathbf{1}_{\{n \text{ prime}\}} - c^{\mathrm{RVM}}_n. \tag{9.11}$$

Since $\pi_{\mathrm{RVM}}(1) = 0$, for every integer $N \geq 2$

$$\sum_{2\leq n\leq N}\mathfrak{d}^{\mathrm{RVM}}_n = \pi(N) - \pi_{\mathrm{RVM}}(N). \tag{9.12}$$

## 10. The $\lambda$-family from $\zeta_{\mathrm{RVM}}$ to $\zeta_{Z1}$

For $\lambda \in [0,1]$ define the moving ordinates, the family product and the family zeta function

$$\gamma_k(\lambda) := (1-\lambda)\gamma_k + \lambda\tau_k = \gamma_k + \lambda\eta_k, \tag{10.1}$$

$$P_\lambda(s) := \prod_{k\geq 1}\left(1 + \frac{q(s)}{\frac{1}{4}+\gamma_k(\lambda)^2}\right), \tag{10.2}$$

$$\zeta_\lambda(s) := \frac{P_\lambda(s)}{G(s)}, \qquad F_\lambda(s) := \frac{1}{\zeta_\lambda(s)}, \tag{10.3}$$

and the endpoint

$$P_{Z1} := P_0, \qquad \zeta_{Z1} := \zeta_0, \qquad \zeta_1 = \zeta_{\mathrm{RVM}}. \tag{10.4}$$

**Theorem 10.1** (The $\lambda$-family)**.** *The product* (10.2) *converges absolutely and locally uniformly jointly on* $[0,1]\times\mathbb{C}$*. For every* $\lambda$*,* $P_\lambda$ *is entire of order exactly one,* $P_\lambda(1-s) = P_\lambda(s)$*,* $P_\lambda(\bar{s}) = \overline{P_\lambda(s)}$*,* $P_\lambda(0) = P_\lambda(1) = 1$*, and its zeros, with multiplicity, are exactly*

$$\frac{1}{2} \pm i\gamma_k(\lambda), \qquad k \geq 1. \tag{10.5}$$

*For* $0 < \lambda \leq 1$ *the ordinates* $\gamma_k(\lambda)$ *are strictly increasing in* $k$ *and all zeros of* $P_\lambda$ *are simple; at* $\lambda = 0$ *the multiplicities are those of the ordinate list* $(\gamma_k)$*. For every* $j \geq 0$*,* $P_\lambda^{(j)} \to P_{\lambda_0}^{(j)}$ *locally uniformly as* $\lambda \to \lambda_0$*. In particular no member* $\zeta_\lambda$ *of the family,* $\zeta_{Z1} = \zeta_0$ *included, can have a zero off the critical line other than* $-2, -4, \ldots$*.*

*Proof.* By Lemma 2.2, $|\gamma_k(\lambda) - \gamma_k| \leq C_\eta$, and $\gamma_k \asymp k/\log k$; so $\gamma_k(\lambda) \geq \frac{1}{2}\gamma_k$ for $k \geq k_0$, uniformly in $\lambda$. On a compact $K \subset \mathbb{C}$,

$$\sum_{k\geq 1}\sup_{s\in K,\ 0\leq\lambda\leq 1}\left|\frac{q(s)}{\frac{1}{4}+\gamma_k(\lambda)^2}\right| < \infty, \tag{10.6}$$

which gives joint normal convergence and, by Cauchy's formula, convergence of all $s$-derivatives. For $|s| \leq R_0$, uniformly in $\lambda$,

$$\log\max_{|s|\leq R_0}|P_\lambda(s)| \leq \sum_{k\geq 1}\log\Bigl(1 + \frac{R_0^2 + R_0}{\frac{1}{4}+\gamma_k(\lambda)^2}\Bigr) = O(R_0\log R_0), \tag{10.7}$$

so the order is at most one; the zero count is $\asymp T\log T$, so the exponent of convergence is one and the order is exactly one. Symmetries, normalization and divisor follow factor by factor as in Theorem 3.1. For $0 < \lambda \leq 1$,

$$\gamma_{k+1}(\lambda) - \gamma_k(\lambda) = (1-\lambda)(\gamma_{k+1}-\gamma_k) + \lambda(\tau_{k+1}-\tau_k) \geq \lambda(\tau_{k+1}-\tau_k) > 0, \tag{10.8}$$

so distinct factors vanish at distinct points and each zero is simple; at $\lambda = 0$ equal ordinates give equal factors. □

*Remark* 10.2 (All differences lie in $\gamma_k - \tau_k$)*.* For $\lambda, \lambda' \in [0,1]$ and $s$ off the divisors,

$$\frac{\zeta_\lambda(s)}{\zeta_{\lambda'}(s)} = \prod_{k\geq 1}\frac{\bigl(\frac{1}{4}+(\gamma_k+\lambda'\eta_k)^2\bigr)\bigl(\frac{1}{4}+(\gamma_k+\lambda\eta_k)^2+q(s)\bigr)}{\bigl(\frac{1}{4}+(\gamma_k+\lambda\eta_k)^2\bigr)\bigl(\frac{1}{4}+(\gamma_k+\lambda'\eta_k)^2+q(s)\bigr)}. \tag{10.9}$$

The gamma factor cancels; the members differ only through $\eta_k = \tau_k - \gamma_k$, and all of them coincide if every $\eta_k$ vanishes.

**Theorem 10.3** (Infinitesimal generator)**.** *Put* $A_{k,\lambda} := \frac{1}{4} + \gamma_k(\lambda)^2$*. Off the divisor of* $P_\lambda$*,*

$$\mathcal{V}_\lambda(s) := \partial_\lambda \log P_\lambda(s) = -2q(s)\sum_{k\geq 1}\frac{\gamma_k(\lambda)\,\eta_k}{A_{k,\lambda}\bigl(A_{k,\lambda}+q(s)\bigr)}, \tag{10.10}$$

*the series converging locally uniformly on compact sets separated from the divisor. Since $G$ does not depend on $\lambda$,*

$$\partial_\lambda \log \zeta_\lambda = \mathcal{V}_\lambda, \qquad \partial_\lambda \log F_\lambda = -\mathcal{V}_\lambda. \tag{10.11}$$

*Proof.* Since $\partial_\lambda A_{k,\lambda} = 2\gamma_k(\lambda)\eta_k$,

$$\partial_\lambda \log\Big(1 + \frac{q}{A_{k,\lambda}}\Big) = -\frac{2q\,\gamma_k(\lambda)\eta_k}{A_{k,\lambda}(A_{k,\lambda}+q)}. \tag{10.12}$$

On compact sets separated from the divisor the summand is $O_K(|\eta_k|\gamma_k^{-3})$, and $\sum_k \gamma_k^{-3} < \infty$, so termwise differentiation is justified. □

**Proposition 10.4** (Integrated multiplier)**.** *On every component avoiding the moving divisor,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s)\exp\Big(\int_0^\lambda \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big), \tag{10.13}$$

$$F_\lambda(s) = F_{Z1}(s)\exp\Big(-\int_0^\lambda \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big), \tag{10.14}$$

*and the total multiplier is*

$$\Pi_{\mathrm{tr}}(s) := \exp\Big(-\int_0^1 \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big) = \frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}, \qquad \zeta_{Z1} = \zeta_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}. \tag{10.15}$$

*The identities extend meromorphically to $\mathbb{C}$ through the product form.*

*Proof.* (10.13) and (10.14) are the fundamental theorem of calculus applied to (10.11). For (10.15), integrate each mode with $r = \gamma_k(\lambda')$, $dr = \eta_k\,d\lambda'$:

$$\int_0^1 \partial_{\lambda'} \log\Big(1 + \frac{q}{A_{k,\lambda'}}\Big)d\lambda' \equiv \log\frac{1 + q/(\frac14 + \tau_k^2)}{1 + q/(\frac14 + \gamma_k^2)} \pmod{2\pi i}. \tag{10.16}$$

Normal convergence permits summation, and exponentiation removes the branch ambiguity. □

**Corollary 10.5** (Expansion at the endpoint)**.** *Let $K \subset \mathbb{C}$ be compact and disjoint from the zeros of $P_{Z1}$. Uniformly for $s \in K$, as $\lambda \downarrow 0$,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s)\big(1 + \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big), \tag{10.17}$$

$$F_\lambda(s) = F_{Z1}(s)\big(1 - \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big). \tag{10.18}$$

*Proof.* By Theorem 10.1 the zeros of $P_\lambda$ converge to those of $P_{Z1}$, so $K$ avoids the moving divisor for small $\lambda$. The series for $\mathcal{V}_\lambda$ and its $\lambda$-derivative converge normally on $K$ (the differentiated summands are $O_K(\gamma_k^{-3})$), so $\mathcal{V}_\lambda = \mathcal{V}_0 + O_K(\lambda)$, and (10.13) gives $\log(\zeta_\lambda/\zeta_{Z1}) = \lambda\mathcal{V}_0 + O_K(\lambda^2)$. □

**Proposition 10.6** (Residues along the family)**.** *Suppose $\gamma_k$ is a simple ordinate (it occurs once in the list). Put $\rho_{k,\lambda} = \frac12 + i\gamma_k(\lambda)$ and, for $0 < \lambda \le 1$,*

$$\mathfrak{r}_{k,\lambda} := \operatorname*{Res}_{s=\rho_{k,\lambda}} \frac{1}{s\,\zeta_\lambda(s)} = \frac{G(\rho_{k,\lambda})}{\rho_{k,\lambda}P'_\lambda(\rho_{k,\lambda})}. \tag{10.19}$$

*Then*

$$\mathfrak{r}_{k,\lambda} \longrightarrow \frac{G(\frac12 + i\gamma_k)}{(\frac12 + i\gamma_k)P'_{Z1}(\frac12 + i\gamma_k)} \qquad (\lambda \downarrow 0). \tag{10.20}$$

*Proof.* Theorem 10.1 gives convergence of the zero locations and local uniform convergence of $P'_\lambda$; the limit denominator is nonzero because the endpoint zero is simple, and $G$ is continuous. □

*Remark* 10.7 (Multiplicity). For $0 < \lambda \le 1$ every zero of $\zeta_\lambda$ is simple. At $\lambda = 0$ a repeated ordinate of $\zeta$ appears in $\zeta_{Z1}$ with the same multiplicity, and the residues of Proposition 10.6 are replaced by the full principal part. Multiplicity plays no role in Step 3: Theorem 11.8 uses only that $\zeta$ does not vanish in $\Re s > \frac12$.

## 11. The endpoint $\zeta_{Z1}$, Hamburger's theorem, and Step 3

### 11.1. The analytic class of the endpoint.

**Proposition 11.1** (Analytic class of $\zeta_{Z1}$)**.** *The endpoint*

$$P_{Z1}(s) = \prod_{k\ge1}\left(1+\frac{q(s)}{\frac14+\gamma_k^2}\right), \qquad \zeta_{Z1}(s) = \frac{P_{Z1}(s)}{G(s)}, \tag{11.1}$$

*has the following properties.*

(i) *$P_{Z1}$ is entire of order exactly one, $P_{Z1}(1-s)=P_{Z1}(s)$, $P_{Z1}(0)=P_{Z1}(1)=1$;*

(ii) *the nontrivial zeros of $\zeta_{Z1}$ are exactly*

$$\frac12 \pm i\gamma_k, \qquad k\ge1, \tag{11.2}$$

*with the multiplicities of the ordinate list; its other zeros are $s=-2,-4,\dots$. It is impossible for $\zeta_{Z1}$ to vanish at any point off the critical line other than $-2,-4,\dots$: every factor of $P_{Z1}$ vanishes only at $\frac12\pm i\gamma_k$;*

(iii) *$\zeta_{Z1}$ has a simple pole at $s=1$ with residue 1;*

(iv) *$\zeta_{Z1}$ satisfies the functional equation of $\zeta$:*

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta_{Z1}(1-s); \tag{11.3}$$

(v) *$(s-1)\zeta_{Z1}(s)$ is entire of order exactly one.*

*Proof.* (i) and the divisor in (ii) are the case $\lambda=0$ of Theorem 10.1; the trivial zeros and the absence of other zeros come from $1/G$ as in Theorem 3.1. For (iii), $P_{Z1}(1)=1$ and

$$G(s) = (s-1)+O\big((s-1)^2\big) \qquad (s\to1). \tag{11.4}$$

For (iv),

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s) = \frac{P_{Z1}(s)}{q(s)}, \tag{11.5}$$

and both $P_{Z1}$ and $q$ are invariant under $s\mapsto1-s$. For (v),

$$(s-1)\zeta_{Z1}(s) = P_{Z1}(s)\,\frac{\pi^{s/2}}{s\,\Gamma(s/2)}, \tag{11.6}$$

where $s\Gamma(s/2)\to2$ at $s=0$ and $1/\Gamma$ is entire of order one; the product has order at most one, and the zero count $\asymp T\log T$ gives order at least one. □

Define the ratio

$$\mathscr{R}_{Z1}(s) := \frac{\zeta_{Z1}(s)}{\zeta(s)} = \frac{P_{Z1}(s)}{P_\zeta(s)}, \tag{11.7}$$

a meromorphic function; $\mathscr{R}_{Z1}\equiv1$ is the identity $\zeta_{Z1}=\zeta$.

### 11.2. Hamburger's theorem at the endpoint.

**Theorem 11.2** (Hamburger closure)**.** *Consider the condition*

$$\textit{(D)} \qquad \zeta_{Z1}(s) = \sum_{n\ge1}\frac{a_n}{n^s} \qquad \textit{absolutely for } \Re s>1. \tag{11.8}$$

*Then*

$$\text{(D)} \iff \mathscr{R}_{Z1}\equiv1 \iff \zeta_{Z1}=\zeta \iff \text{RH}. \tag{11.9}$$

*Proof.* Assume (D). Then $\zeta_{Z1}(1-s)=\sum_n a_n n^{-(1-s)}$ converges absolutely for $\Re s<0$, $(s-1)\zeta_{Z1}$ is entire of finite order (Proposition 11.1(v)), and (11.3) holds. These are the hypotheses of Hamburger's theorem [16], [23, §2.13], which gives

$$\zeta_{Z1}=C_H\,\zeta \tag{11.10}$$

for a constant $C_H$. Both functions have residue 1 at $s=1$, so $C_H=1$. If $\zeta_{Z1}=\zeta$, the nontrivial zeros of $\zeta$ are those of $\zeta_{Z1}$, namely (11.2), which is RH. If RH holds, every $\rho$ with $\Im\rho>0$ is $\frac12+i\gamma_k$, so $\rho(1-\rho)=\frac14+\gamma_k^2$, and Theorem 9.1 gives

$$P_\zeta(s)=\prod_{k\ge1}\Big(1+\frac{q(s)}{\frac14+\gamma_k^2}\Big)=P_{Z1}(s) \tag{11.11}$$

exactly, hence $\zeta=\zeta_{Z1}$. Finally $\zeta=\sum_n n^{-s}$ converges absolutely for $\Re s>1$, so $\zeta_{Z1}=\zeta$ gives (D). □

11.3. **Condition (D) and the integer-weighted zeta.** By Corollary 6.3, $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s>1$ with first coefficient 1, unconditionally. Define the multiplier

$$\mathcal{K}(s):=\frac{\zeta_{Z1}(s)}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)}. \tag{11.12}$$

**Theorem 11.3** (Reduction of (D) to the multiplier)**.** (i) *If $\mathcal{K}$ has an expansion $\mathcal{K}(s)=\sum_{n\ge1}k_n n^{-s}$ that converges absolutely in $\Re s>1$, then* (11.8) *holds.*

(ii) *If* (11.8) *holds, then $\mathcal{K}(s)=\sum_{n\ge1}k_n n^{-s}$ absolutely on the whole half-plane $\Re s>1$.*

*Thus (D) holds if and only if $\mathcal{K}$ is an absolutely convergent Dirichlet series in $\Re s>1$. Hence, by Theorem 11.2, an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s>1$ gives $\zeta_{Z1}=\zeta$, RH, and, by Theorem 11.8, $M(x)=O_\varepsilon(x^{1/2+\varepsilon})$.*

*Proof.* (i) $\zeta_{Z1}=\zeta_{\mathbb{Z},\mathrm{RVM}}\,\mathcal{K}$ is the Dirichlet convolution of two absolutely convergent series in $\Re s>1$, and $\sum_N|(a^{\mathbb{Z}}*k)_N|N^{-\sigma}\le\sum_N|a^{\mathbb{Z}}_N|N^{-\sigma}\sum_n|k_n|n^{-\sigma}<\infty$. (ii) By (6.15), $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s>1$, and so is its product $\mathcal{K}$ with (11.8). □

By (10.15) and (6.19) (Theorem 6.4),

$$\mathcal{K}(s)=\Pi_{\mathrm{tr}}(s)\,e^{-h_{\mathrm{RVM}}(s)}=\frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}\,e^{-h_{\mathrm{RVM}}(s)}, \tag{11.13}$$

an explicit function of the displacements $\gamma_k-\tau_k$ and of the cell masses $c_n^{\mathrm{RVM}}$. Condition (D) is thereby located in a single object built from the two constructions of this paper: the $\lambda$-family and the integer weights.

*Remark* 11.4 (Continuous-to-atomic meaning of (D))*.* The prime-power count $J^{\mathrm{M}}_{Z1}$ of $\zeta_{Z1}$ (Theorem 12.5) is a continuous object built from the zeros, and (D) makes its measure the atomic measure $\sum_{p,m}m^{-1}\delta_{p^m}$. The final step of the method is therefore the passage of $\zeta_{Z1}$ into the discrete Dirichlet category. Theorem 12.7 states it exactly: (D) holds if and only if $J^{\mathrm{M}}_{Z1}$ is constant between consecutive integers, and Corollary 12.3 reduces it to the single identity (12.12) between the displacements $\gamma_k-\tau_k$ and $\varpi_{\mathrm{RVM}}$.

11.4. **Step 3: the Mertens bound from $\zeta_{Z1}=\zeta$.**

**Lemma 11.5** (Vertical growth of $\zeta$)**.** *For $\frac14\le\sigma\le4$ and $|t|\ge2$,*

$$|\zeta(\sigma+it)|\le15|t|. \tag{11.14}$$

*Proof.* For $\sigma>0$ [23, (2.1.5)],

$$\zeta(s)=\frac{s}{s-1}-s\int_1^\infty\{y\}y^{-s-1}\,dy. \tag{11.15}$$

For $|t| \geq 2$ and $\sigma \leq 4$ one has $|s| \leq 3|t|$, $|s/(s-1)| \leq |s|/|t| \leq 3$, and the integral is at most $1/\sigma \leq 4$; hence $|\zeta(s)| \leq 3 + 12|t| \leq 15|t|$. □

**Lemma 11.6** (Logarithm of $\zeta$ off the critical line)**.** *Assume $\zeta(s) \neq 0$ for $\Re s > \frac{1}{2}$. Let $0 < \delta \leq \frac{1}{4}$ and*

$$\omega(\delta) := \frac{\log(3-2\delta)}{\log(3-\delta)} < 1. \tag{11.16}$$

*Then, with the branch of $\log\zeta$ equal to $\sum_{n\geq 2}\Lambda(n)n^{-s}/\log n$ on $\Re s > 1$,*

$$|\log\zeta(\sigma+it)| \leq \Big(\frac{48}{\delta}\log|t|\Big)^{\omega(\delta)} \qquad (\sigma \geq \tfrac{1}{2}+\delta,\ |t| \geq 4). \tag{11.17}$$

*Consequently, for every $\varepsilon > 0$ there is $t_2 = t_2(\delta,\varepsilon)$ with*

$$\Big|\frac{1}{\zeta(\sigma+it)}\Big| \leq |t|^{\varepsilon} \qquad (\sigma \geq \tfrac{1}{2}+\delta,\ |t| \geq t_2). \tag{11.18}$$

*Proof.* Fix $t$ with $|t| \geq 4$ and put $s_0 = 2+it$, $R = \frac{3}{2} - \frac{\delta}{4}$, $r_3 = \frac{3}{2} - \frac{\delta}{2}$. The closed disc $|s - s_0| \leq R$ lies in $\frac{1}{2} + \frac{\delta}{4} \leq \Re s \leq \frac{7}{2}$, $|\Im s - t| \leq \frac{3}{2}$; it avoids $s = 1$ and, by hypothesis, every zero of $\zeta$. Hence $\log\zeta$ has a holomorphic branch on a neighbourhood of it, equal to the Dirichlet series on its intersection with $\Re s > 1$, which is convex. On the disc $|\Im s| \geq |t|/2 \geq 2$ and $|\Im s| \leq 2|t|$, so Lemma 11.5 gives $\Re\log\zeta = \log|\zeta| \leq \log(30|t|)$ there, while $|\log\zeta(s_0)| \leq \log\zeta(2)$. The Borel–Carathéodory lemma [22, §5.5], applied with radii $r_3 < R$,

$$\max_{|s-s_0|\leq r_3} |\log\zeta(s)| \leq \frac{2r_3}{R-r_3} \max_{|s-s_0|=R} \Re\log\zeta(s) + \frac{R+r_3}{R-r_3}|\log\zeta(s_0)|, \tag{11.19}$$

together with $R - r_3 = \delta/4$, $2r_3 \leq 3$, $R + r_3 \leq 3$ and $\log 30 + \log\zeta(2) < 4 \leq 3\log|t|$, gives

$$\max_{|s-s_0|\leq r_3} |\log\zeta(s)| \leq \frac{12}{\delta}\big(\log(30|t|) + \log\zeta(2)\big) \leq \frac{48}{\delta}\log|t|. \tag{11.20}$$

Now let $\frac{1}{2} + \delta \leq \sigma < \frac{3}{2}$ and set $r_1 = \frac{1}{2}$, $r_2 = 2 - \sigma \in (\frac{1}{2}, \frac{3}{2} - \delta]$. On $|s - s_0| = r_1$ one has $\Re s \geq \frac{3}{2}$, so $|\log\zeta(s)| \leq \log\zeta(\frac{3}{2}) < 1$. Hadamard's three-circle theorem [22, §5.3] gives, with $M_j = \max_{|s-s_0|=r_j}|\log\zeta|$,

$$M_2 \leq M_1^{1-\omega} M_3^{\omega}, \qquad \omega = \frac{\log(r_2/r_1)}{\log(r_3/r_1)} \leq \frac{\log(3-2\delta)}{\log(3-\delta)} = \omega(\delta). \tag{11.21}$$

Since $M_1 < 1$ and $M_3 \leq \frac{48}{\delta}\log|t|$ with $\frac{48}{\delta}\log|t| > 1$, (11.21) gives (11.17) at $s = \sigma + it$, which lies on the circle of radius $r_2$. For $\sigma \geq \frac{3}{2}$, $|\log\zeta| \leq \log\zeta(\frac{3}{2}) < 1$. Finally $|1/\zeta| = \exp(-\Re\log\zeta) \leq \exp|\log\zeta|$, and $(\frac{48}{\delta}\log|t|)^{\omega(\delta)} \leq \varepsilon\log|t|$ for $|t| \geq t_2(\delta,\varepsilon)$ because $\omega(\delta) < 1$. □

**Lemma 11.7** (Perron kernel)**.** *For $y > 0$, $y \neq 1$, $c > 0$, $T > 0$ [7, Ch. 17],*

$$\left|\frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{y^s}{s}\,ds - \mathbf{1}_{\{y>1\}}\right| \leq y^c \min\Big(1, \frac{1}{T|\log y|}\Big). \tag{11.22}$$

**Theorem 11.8** (Step 3)**.** *If $\zeta_{Z1} = \zeta$, then for every $\varepsilon > 0$*

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (x \geq 1). \tag{11.23}$$

*Proof. Non-vanishing.* By Proposition 11.1(ii), $\zeta_{Z1}$ has no zeros in $\Re s > \frac{1}{2}$; since $\zeta = \zeta_{Z1}$, neither has $\zeta$, and Lemma 11.6 applies.

*Truncation.* Let $x = N + \frac{1}{2}$ with $N \geq 3$ an integer, $c = 1 + 1/\log x$ and $T = x$. Applying (11.22) with $y = x/n$ to each term of $\sum_n \mu(n)n^{-s} = 1/\zeta(s)$, which converges absolutely on $\Re s = c$, gives

$$M(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\,ds + O\Big(\sum_{n\geq 1}\Big(\frac{x}{n}\Big)^c \min\Big(1, \frac{1}{T|\log(x/n)|}\Big)\Big). \tag{11.24}$$

For $n \le x/2$ or $n \ge 2x$, $|\log(x/n)| \ge \log 2$, and these terms contribute at most $x^c\zeta(c)/(T\log 2) \le e\,x(1+\log x)/(T\log 2)$. For $x/2 < n < 2x$, $(x/n)^c \le 4$ and $|\log(x/n)| \ge |n-x|/(2x)$; as $|n-x| \in \{j-\frac12 : j \ge 1\}$, each value taken at most twice, these terms contribute at most $8\sum_{1\le j\le 2x}\min(1, 4x/(T(2j-1))) \ll 1 + (x/T)\log x$. Hence

$$M(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\,ds + O(\log x). \tag{11.25}$$

*Contour shift.* Fix $0 < \delta \le \frac14$. The integrand is holomorphic on the closed rectangle with vertices $c \pm iT$, $\frac12 + \delta \pm iT$: $\zeta$ has no zeros there, $s = 1$ is a zero of $1/\zeta$, and $s = 0$ lies outside. By Cauchy's theorem

$$\int_{c-iT}^{c+iT} = \int_{\frac12+\delta-iT}^{\frac12+\delta+iT} + \int_{\frac12+\delta+iT}^{c+iT} - \int_{\frac12+\delta-iT}^{c-iT}. \tag{11.26}$$

For $x \ge t_2(\delta,\varepsilon)$, (11.18) on $|t| = T = x$ bounds the horizontal integrals by

$$\frac{1}{2\pi}\int_{\frac12+\delta}^{c} \frac{x^\sigma T^\varepsilon}{T}\,d\sigma \le x^c T^{\varepsilon-1} = e\,x^\varepsilon. \tag{11.27}$$

On the left side, $1/\zeta$ is continuous on the compact segment $\{\frac12+\delta+it : |t| \le t_2\}$, bounded by a constant $C(\delta,\varepsilon)$, and (11.18) holds for $|t| \ge t_2$; hence

$$\frac{1}{2\pi}\int_{-T}^{T} \frac{x^{1/2+\delta}}{|\frac12+\delta+it|\,|\zeta(\frac12+\delta+it)|}\,dt \le x^{1/2+\delta}\Big(C'(\delta,\varepsilon) + \int_{t_2}^{T} t^{\varepsilon-1}dt\Big) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}. \tag{11.28}$$

*Conclusion.* (11.25)–(11.28) give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$ for $x = N + \frac12$; taking $\delta = \varepsilon = \varepsilon_0/2$ (with $\varepsilon_0 \le \frac12$) gives $O_{\varepsilon_0}(x^{1/2+\varepsilon_0})$. For arbitrary $x \ge 3$, $M(x) = M(\lfloor x\rfloor + \frac12)$ and $\lfloor x\rfloor + \frac12 \le 2x$; for $1 \le x < 3$, $|M(x)| \le 2$. □

**Theorem 11.9** (Transfer of the synthetic Mertens bound to $M$)**.** *Assume $\zeta_{Z1} = \zeta$ and let $0 < \delta \le \frac14$, $\varepsilon > 0$. Then:*

(i) *on $\Re s > \frac12$,*

$$\frac{1}{\zeta(s)} = F_{\mathrm{RVM}}(s)\,\Pi_{\mathrm{tr}}(s)^{-1}, \qquad \Pi_{\mathrm{tr}} = \frac{P_{Z1}}{P^{\mathrm{crit}}_{\mathrm{RVM}}} = \frac{\zeta}{\zeta_{\mathrm{RVM}}}, \tag{11.29}$$

*where $\Pi_{\mathrm{tr}}$ is holomorphic and zero-free in $\Re s > \frac12$ and $\Pi_{\mathrm{tr}}(1) = 1$;*

(ii) *there is $t_3 = t_3(\delta,\varepsilon)$ such that, for $\frac12 + \delta \le \sigma \le 2$ and $|t| \ge t_3$,*

$$|F_{\mathrm{RVM}}(\sigma+it)| \le 2L, \qquad |\Pi_{\mathrm{tr}}(\sigma+it)^{\pm1}| \le |t|^\varepsilon; \tag{11.30}$$

(iii) *the synthetic bound of Theorem 8.12, $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$, carries over to the Mertens function:*

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{11.31}$$

*Proof.* (i) By (10.15), $\zeta_{Z1} = \zeta_{\mathrm{RVM}}\Pi_{\mathrm{tr}}$; with $\zeta_{Z1} = \zeta$ this is (11.29). In $\Re s > \frac12$, $\zeta = \zeta_{Z1}$ has no zeros (Proposition 11.1(ii)) and $\zeta_{\mathrm{RVM}}$ has none (Theorem 3.1); both have a simple pole of residue 1 at $s = 1$, so $\Pi_{\mathrm{tr}}$ is holomorphic there with $\Pi_{\mathrm{tr}}(1) = 1$, and it is zero-free in $\Re s > \frac12$.

(ii) By Theorem 8.1(a) with $B = \frac32$, for $\delta \le \sigma' \le \frac32$ and $|t| \ge t_0$ we have $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|w| \le C|t|^{-\delta} \le \frac12$ and $|\mathcal{E}_0| \le \frac12$ for $|t|$ large; hence $|F_{\mathrm{RVM}}| \le 2L$ and

$$\log\zeta_{\mathrm{RVM}} = \log\varpi_{\mathrm{RVM}} + \mathrm{Log}(1+w) + \mathrm{Log}(1+\mathcal{E}_0) = O(1). \tag{11.32}$$

By Lemma 11.6, which applies because $\zeta = \zeta_{Z1}$ has no zeros in $\Re s > \frac12$, $|\log\zeta(\sigma+it)| \le (\frac{48}{\delta}\log|t|)^{\omega(\delta)}$ with $\omega(\delta) < 1$. Therefore

$$|\log\Pi_{\mathrm{tr}}(\sigma+it)| = |\log\zeta - \log\zeta_{\mathrm{RVM}}| \le \Big(\frac{48}{\delta}\log|t|\Big)^{\omega(\delta)} + O(1) \le \varepsilon\log|t| \tag{11.33}$$

for $|t| \geq t_3(\delta, \varepsilon)$, which gives (11.30).

(iii) The proof of Theorem 8.12 moves the Perron line of $F_{\mathrm{RVM}} x^s/s$ to $\Re s = \frac{1}{2} + \delta$ using only the holomorphy of $F_{\mathrm{RVM}}$ in $\Re s > \frac{1}{2}$ and the vertical control of $F_{\mathrm{RVM}}$. By (11.29) the integrand for $M$ is the same integrand multiplied by $\Pi_{\mathrm{tr}}^{-1}$, which is holomorphic in $\Re s > \frac{1}{2}$ by (i) and at most $|t|^{\varepsilon}$ by (ii). Hence $|1/\zeta(\sigma + it)| \leq 2L|t|^{\varepsilon}$ for $\frac{1}{2} + \delta \leq \sigma \leq 2$, $|t| \geq t_3$, and the truncated Perron formula, contour shift and estimates (11.25)–(11.28) of Theorem 11.8, applied with this bound, give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$. Choosing $\delta = \varepsilon = \varepsilon_0/2$ gives (11.31). □

*Remark* 11.10 (The transfer in Mellin form). Formally, (11.29) reads $M = (\Pi_{\mathrm{tr}}^{-1})^{\vee} *_{\mathrm{Mel}} M_{\mathrm{RVM}}^{\mathrm{crit}}$: the Mertens function is the synthetic Mertens function convolved with the inverse Mellin transform of the inverse transport multiplier. The synthetic factor $F_{\mathrm{RVM}}$ contributes the exponent $\frac{1}{2}$ and is bounded (Theorem 8.1); the multiplier contributes at most $|t|^{\varepsilon}$. All the information of the $\lambda$-family enters through $\Pi_{\mathrm{tr}} = P_{Z1}/P_{\mathrm{RVM}}^{\mathrm{crit}}$, that is, through the displacements $\gamma_k - \tau_k$.

**Corollary 11.11** (Step 3 through (D))**.** *Condition* (11.8) *implies* $M(x) = O_{\varepsilon}(x^{1/2+\varepsilon})$ *for every* $\varepsilon > 0$.

*Proof.* Theorem 11.2 and Theorem 11.8. □

*Remark* 11.12 (Status of Step 3). The implication $\zeta_{Z1} = \zeta \Rightarrow M(x) = O_{\varepsilon}(x^{1/2+\varepsilon})$ is proved completely above, and it requires nothing beyond the identity: no separate statement about the zeros of $\zeta$ enters, because $\zeta = \zeta_{Z1}$ already carries the divisor (11.2). The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), equivalently an absolutely convergent Dirichlet expansion of the multiplier $\mathcal{K}$ of (11.12) in $\Re s > 1$ (Theorem 11.3), is not proved in this paper. It is the open step of the method.

11.5. **Reciprocal endpoint and finite-height Perron integrals.** Put

$$F_{Z1}(s) := \frac{1}{\zeta_{Z1}(s)} = F_{\mathrm{RVM}}(s)\,\Pi_{\mathrm{tr}}(s)^{-1}, \tag{11.34}$$

the second identity by (10.15). For $c > 1$ and finite $U$ define

$$M_{\lambda,U}(x) := \frac{1}{2\pi i} \int_{c-iU}^{c+iU} F_{\lambda}(s) \frac{x^s}{s}\, ds, \qquad M_{Z1,U} := M_{0,U}. \tag{11.35}$$

**Proposition 11.13** (Finite-height Perron integrals along the family)**.** *For fixed* $c > 1$, $U < \infty$ *and* $x > 0$,

$$M_{\lambda,U}(x) \longrightarrow M_{Z1,U}(x) \qquad (\lambda \downarrow 0). \tag{11.36}$$

*Proof.* The segment $\{c + it : |t| \leq U\}$ is compact and disjoint from every divisor (10.5), and $P_{\lambda} \to P_{Z1}$ uniformly on it, so $F_{\lambda} \to F_{Z1}$ uniformly there. □

**Theorem 11.14** (Reciprocal-coefficient criterion)**.** *Assume that in* $\Re s > 1$

$$F_{Z1}(s) = \sum_{n \geq 1} \frac{\mu_{Z1}(n)}{n^s} \tag{11.37}$$

*absolutely, and let* $M_{Z1}(x) = \sum_{n \leq x} \mu_{Z1}(n)$, *so that*

$$\mu_{Z1}(n) = M_{Z1}(n) - M_{Z1}(n-1). \tag{11.38}$$

*Then the following are equivalent: (i)* $M_{Z1}(n) = M(n)$ *for every* $n \geq 1$*; (ii)* $\mu_{Z1} = \mu$*; (iii)* $F_{Z1} = 1/\zeta$ *in* $\Re s > 1$*; (iv)* $\zeta_{Z1} = \zeta$*; (v) RH.*

*Proof.* (i)⇔(ii) by (11.38); (ii)⇔(iii) by uniqueness of absolutely convergent Dirichlet coefficients; (iii)⇔(iv) by analytic continuation, both functions being nonzero in $\Re s > 1$; (iv)⇔(v) by Theorem 11.2. □

*Remark* 11.15 (A route that does not close Step 3). The following approach does not work. One may try to expand both factors of $F_{Z1} = F_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}^{-1}$ in (11.34) as absolutely convergent Dirichlet series and identify their Dirichlet convolution with $\mu$. By Theorem 8.11 the expansion of $F_{\mathrm{RVM}}$ does not exist in any half-plane, so there are no coefficients to convolve. The endpoint criterion concerns $\zeta_{Z1}$ itself, as in Theorem 11.2.

## 12. The endpoint at infinity and the Euler products

This section applies the clock decomposition of Theorem 8.10 to the ordinates $\gamma_k$ of the endpoint, and constructs the prime-power counts of $\zeta_{\mathrm{RVM}}$ and $\zeta_{Z1}$ as Mellin transforms of their logarithms. No hypothesis is used.

12.1. **The value of the endpoint at infinity.** With the zero count $N_\zeta$ of (2.2) put

$$\mathfrak{e}(\tau) := N_\zeta(\tau) - \mathcal{N}_{\mathrm{RVM}}(\tau), \qquad \mathfrak{e}_1(\tau) := \int_{\tau_0}^{\tau} \mathfrak{e}(v)\,dv \qquad (\tau \ge \tau_0 = 4\pi). \tag{12.1}$$

By (2.2), $\mathfrak{e}(\tau) = S(\tau) + O(1/\tau)$, and Littlewood's bound $\int_0^T S(t)\,dt = O(\log T)$ [23, Thm. 9.9] gives an absolute constant $C_{\mathfrak{e}}$ with

$$|\mathfrak{e}_1(\tau)| \le C_{\mathfrak{e}} \log \tau \qquad (\tau \ge \tau_0). \tag{12.2}$$

For the clock the corresponding primitive is $-\mathfrak{s}_1$, bounded by Lemma 8.9.

**Theorem 12.1** (Clock decomposition of the endpoint)**.** *Let $s = \frac12 + z$, $z = \sigma' + it$, $\sigma' > 0$. Then*

$$\log \zeta_{Z1}(s) = \log \varpi_{Z1} - \int_{\tau_0}^{\infty} \mathfrak{e}_1(\tau)\Big(\frac{1}{(\tau - iz)^2} + \frac{1}{(\tau + iz)^2}\Big)d\tau + O\Big(\frac{1}{|z|}\Big) \qquad (|z| \ge R_0), \tag{12.3}$$

*where*

$$\varpi_{Z1} := P_{Z1}(\tfrac12)\exp\Big(\nu_3 + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau)}{\tau^2}\,d\tau\Big) > 0. \tag{12.4}$$

*Consequently:*

(i) *for $\sigma' > 0$ and $|z| \ge R_0$,*

$$|\log \zeta_{Z1}(s) - \log \varpi_{Z1}| \le \frac{2(\pi+2)C_{\mathfrak{e}}}{\sigma'}\Big(1 + \log(2|t| + 2\sigma' + \tau_0)\Big) + \frac{C}{|z|}; \tag{12.5}$$

(ii) *$\zeta_{Z1}(\sigma) \to \varpi_{Z1}$ as $\sigma \to +\infty$;*

(iii) *the value at infinity is the ordered product*

$$\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_{k\ge1} \frac{\frac14 + \tau_k^2}{\frac14 + \gamma_k^2} := \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K} \frac{\frac14 + \tau_k^2}{\frac14 + \gamma_k^2}. \tag{12.6}$$

*Proof.* As in (8.6), $\log P_{Z1}(s) - \log P_{Z1}(\frac12) = \sum_k f(\gamma_k)$ with $f(\tau) = \mathrm{Log}(1 + z^2/\tau^2)$. Since $\gamma_1 > \tau_0$, $N_\zeta(\tau_0) = 0$ and $\mathfrak{e}(\tau_0) = -\mathcal{N}_{\mathrm{RVM}}(\tau_0)$. Stieltjes integration against $N_\zeta = \mathcal{N}_{\mathrm{RVM}} + \mathfrak{e}$ and two integrations by parts ($f\mathfrak{e} \to 0$ and $\mathfrak{e}_1 f' \to 0$ at $\infty$, $\mathfrak{e}_1(\tau_0) = 0$) give

$$\sum_k f(\gamma_k) = \Lambda_{\mathrm{sm}}(z) + \int_{\tau_0}^{\infty} \mathfrak{e}_1 f''\,d\tau, \tag{12.7}$$

with the same $\Lambda_{\mathrm{sm}}$ as in (8.9). The proof of Theorem 8.10 gives $\Lambda_{\mathrm{sm}} - \log G = \nu_3 + O(|z|^{-1})$, and inserting $f''$ from (8.41) gives (12.3).

For (i), $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$. For real $a$, splitting at $X = 2|a| + 2\sigma' + \tau_0$ and using $(\tau - a)^2 \ge \tau^2/4$ for $\tau \ge X$,

$$\int_{\tau_0}^{\infty} \frac{\log \tau}{(\tau - a)^2 + \sigma'^2}\,d\tau \le \frac{\pi}{\sigma'}\log X + 4\,\frac{1 + \log X}{X} \le \frac{\pi+2}{\sigma'}(1 + \log X). \tag{12.8}$$

With (12.2) this gives (12.5). For (ii) take $t = 0$: the right side of (12.5) tends to 0 as $\sigma' \to \infty$.

For (iii), Theorem 8.10 gives $\varpi_{\mathrm{RVM}} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)\exp(\nu_3 - 2\int \mathfrak{s}_1\tau^{-2}d\tau)$, so

$$\frac{\varpi_{Z1}}{\varpi_{\mathrm{RVM}}} = \frac{P_{Z1}(\frac12)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)}\exp\Big(2\int_{\tau_0}^{\infty}\frac{\mathfrak{e}_1(\tau)+\mathfrak{s}_1(\tau)}{\tau^2}\,d\tau\Big). \tag{12.9}$$

Since $\mathfrak{e}+\psi(\mathcal{N}_{\mathrm{RVM}}+\frac12) = N_\zeta - n_{\mathrm{RVM}}$ and $(\mathfrak{e}_1+\mathfrak{s}_1)(X)/X \to 0$,

$$\int_{\tau_0}^{\infty}\frac{\mathfrak{e}_1+\mathfrak{s}_1}{\tau^2}\,d\tau = \lim_{X\to\infty}\int_{\tau_0}^{X}\frac{N_\zeta(\tau)-n_{\mathrm{RVM}}(\tau)}{\tau}\,d\tau = \lim_{K\to\infty}\sum_{k\le K}\log\frac{\tau_k}{\gamma_k}. \tag{12.10}$$

In the last step, $\int_{\tau_0}^{X}(N_\zeta - n_{\mathrm{RVM}})\tau^{-1}d\tau = \sum_{\gamma_k\le X}\log\frac{X}{\gamma_k} - \sum_{\tau_k\le X}\log\frac{X}{\tau_k}$; the indices counted by only one of the two sums are $O(\log X)$ in number and, by Lemma 2.2, have ordinates in $[X-C_\eta, X]$, so they contribute $O(\log X/X)$. Finally $P_{Z1}(\frac12)/P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12) = \prod_k \frac{\gamma_k^2}{\frac14+\gamma_k^2}\cdot\frac{\frac14+\tau_k^2}{\tau_k^2}$, an absolutely convergent product, and multiplying by $\prod_{k\le K}\tau_k^2/\gamma_k^2$ gives (12.6). □

The ordinates enter $\varpi_{Z1}$ only through the displacements $\gamma_k - \tau_k$, in agreement with Remark 10.2.

**Theorem 12.2** (The value at infinity decides the endpoint identity)**.** *$\varpi_{Z1} \ge 1$, and $\varpi_{Z1} = 1$ if and only if $\zeta_{Z1} = \zeta$.*

*Proof.* Write $\zeta_{Z1} = \zeta\,\mathscr{R}_{Z1}$ with $\mathscr{R}_{Z1} = P_{Z1}/P_\zeta$ as in (11.7), and compare the products factor by factor, with the zeros $\rho_k = \beta_k + i\gamma_k$ listed with the ordinates (Theorem 9.1). Factors with $\rho_k(1-\rho_k) = \frac14+\gamma_k^2$ cancel. The remaining factors of $P_\zeta$ come in pairs $\{\rho, 1-\bar\rho\}$ with the same ordinate $\gamma$; with $\delta = \Re\rho - \frac12$ and $W = (s-\frac12)^2+\gamma^2$, the corresponding factor of $\mathscr{R}_{Z1}$ is

$$\frac{W^2}{(W-\delta^2)^2+4\delta^2\gamma^2}\cdot c_\gamma, \qquad c_\gamma = \frac{|\rho(1-\rho)|^2}{(\frac14+\gamma^2)^2} = 1 + \frac{2\delta^2(\gamma^2-\frac14)+\delta^4}{(\frac14+\gamma^2)^2} > 1. \tag{12.11}$$

For real $\sigma > 1$, $W \ge \gamma^2 \ge \gamma_1^2$, and the logarithm of the first factor is at most $14\delta^2/W \le 14\delta^2/\gamma^2$ in modulus; since $\delta \le \frac12$ and $\sum\gamma^{-2} < \infty$, dominated convergence and $\zeta(\sigma)\to 1$ give $\varpi_{Z1} = \lim_{\sigma\to\infty}\zeta_{Z1}(\sigma) = \prod c_\gamma \ge 1$, with equality exactly when there are no such pairs, that is, when $\mathscr{R}_{Z1}\equiv 1$. □

**Corollary 12.3** (The endpoint identity as an identity of the displacements)**.** *The following are equivalent: (D); $\zeta_{Z1} = \zeta$; $\varpi_{Z1} = 1$;*

$$\prod_{k\ge1}\frac{\frac14+\gamma_k^2}{\frac14+\tau_k^2} = \varpi_{\mathrm{RVM}} \tag{12.12}$$

*(ordered product); and $\lim_{\sigma\to+\infty}\Pi_{\mathrm{tr}}(\sigma) = L = 1/\varpi_{\mathrm{RVM}}$.*

*Proof.* Theorems 11.2, 12.1 and 12.2, and $\Pi_{\mathrm{tr}} = \zeta_{Z1}/\zeta_{\mathrm{RVM}} \to \varpi_{Z1}/\varpi_{\mathrm{RVM}}$ by Theorems 8.10 and 12.1(ii). □

**Proposition 12.4** (Boundedness of the endpoint)**.** *Let $\sigma_0 > 1$. Then $|\zeta_{Z1}|$ is bounded above and below on $\Re s \ge \sigma_0$ if and only if*

$$\sup_{\sigma'\ge\sigma_0-\frac12,\ t\ge0}\Big|\Re\int_{\tau_0}^{\infty}\frac{\mathfrak{e}_1(\tau)+\mathfrak{s}_1(\tau)}{(\tau-t+i\sigma')^2}\,d\tau\Big| < \infty. \tag{12.13}$$

*Proof.* $\tau + iz = \tau - t + i\sigma'$. For $t \ge 0$ the term with $(\tau-iz)^{-2}$ in (12.3) is bounded by (12.2), since $|\tau - iz| \ge \tau + t$; the term with $\mathfrak{s}_1$ is bounded by Lemma 8.9. By (12.3), $\log|\zeta_{Z1}|$ is bounded on $\Re s \ge \sigma_0$, $t \ge 0$, exactly when (12.13) holds, and $\zeta_{Z1}(\bar s) = \overline{\zeta_{Z1}(s)}$ covers $t < 0$. □

The function in (12.13) is, after one integration by parts, the conjugate Poisson integral at distance $\sigma'$ of the count difference $N_\zeta - n_{\mathrm{RVM}}$; it depends only on the displacements $\gamma_k - \tau_k$.

12.2. **The Euler products of $\zeta_{\mathrm{RVM}}$ and $\zeta_{Z1}$.**

**Theorem 12.5** (Prime-power counts as Mellin transforms)**.** *Let $X \in \{\mathrm{RVM}, Z1\}$. There is a unique function $J_X^{\mathrm{M}}$ on $(1,\infty)$ with $J_X^{\mathrm{M}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$ and*

$$\log \zeta_X(s) = s \int_1^\infty J_X^{\mathrm{M}}(x)\, x^{-s-1}\, dx \qquad (\Re s > 1), \tag{12.14}$$

*the branch of $\log \zeta_X$ being real on $(1,\infty)$. Equivalently $\zeta_X(s) = \exp\big(\int_{[1,\infty)} x^{-s}\, dJ_X^{\mathrm{M}}(x)\big)$.*

*Proof.* $\zeta_X$ has no zeros or poles in $\Re s > 1$, so $\log \zeta_X$ is holomorphic there. For $c > 1$, Theorem 8.10 ($X = \mathrm{RVM}$) and Theorem 12.1 ($X = Z1$), together with continuity on the compact part, give $|\log \zeta_X(s)| \le C_c(1 + \log(|t| + 2))$ on $\Re s \ge c$. Hence $F_\sigma(t) := \log \zeta_X(\sigma + it)/(\sigma + it)$ lies in $L^2(\mathbb{R})$ with $\|F_\sigma\|_2 \le B_c$ for all $\sigma \ge c$. In the variable $u = \log x$, define $J_X^{\mathrm{M}}(e^u)e^{-\sigma u}$ as the inverse Fourier–Plancherel transform of $F_\sigma$; Cauchy's theorem and the decay of $F$ on horizontal segments show that $J_X^{\mathrm{M}}$ does not depend on $\sigma$. Plancherel gives $\int_{\mathbb{R}} |J_X^{\mathrm{M}}(e^u)|^2 e^{-2\sigma u} du = \|F_\sigma\|_2^2/2\pi \le B_c^2/2\pi$ for all $\sigma \ge c$; letting $\sigma \to \infty$ forces $J_X^{\mathrm{M}}(e^u) = 0$ for almost every $u < 0$. Plancherel again gives (12.14), and uniqueness follows from the injectivity of the Fourier transform. □

**Theorem 12.6** (Distributional explicit formula)**.** *Let $X \in \{\mathrm{RVM}, Z1\}$, with ordinates $\kappa_k = \tau_k$ for $X = \mathrm{RVM}$ and $\kappa_k = \gamma_k$ for $X = Z1$. In $\mathcal{D}'(1,\infty)$,*

$$(\log x)\, \frac{dJ_X^{\mathrm{M}}}{dx} = 1 - \frac{2}{\sqrt{x}}\, \Re \sum_{k \ge 1} x^{i\kappa_k} - \frac{1}{x(x^2-1)}, \tag{12.15}$$

*the series converging in $\mathcal{D}'(1,\infty)$. The explicit-formula count*

$$J_X(x) := \mathrm{li}(x) - 2\Re \sum_{k \ge 1} \mathrm{Ei}\Big(\big(\tfrac{1}{2} + i\kappa_k\big) \log x\Big) - \log 2 + \int_x^\infty \frac{dy}{y(y^2-1)\log y} \tag{12.16}$$

*converges in $\mathcal{D}'(1,\infty)$ ($J_{\mathrm{RVM}}$ is the function (5.2)), and $J_X^{\mathrm{M}} - J_X$ is a constant $C_X$ on $(1,\infty)$.*

*Proof.* With $\rho_k = \frac{1}{2} + i\kappa_k$, the factorization $\zeta_X = P_X/G$, $P_X'/P_X = \sum_k \big(\frac{1}{s-\rho_k} + \frac{1}{s-\bar\rho_k}\big)$, and the digamma expansion $\frac{1}{2}\frac{\Gamma'}{\Gamma}\big(\frac{s}{2}\big) + \frac{1}{s} = -\frac{\gamma_E}{2} + \sum_{n\ge1}\big(\frac{1}{2n} - \frac{1}{s+2n}\big)$ give, for $\Re s > 1$,

$$-\frac{\zeta_X'}{\zeta_X}(s) = \frac{1}{s-1} - \sum_{k\ge1}\Big(\frac{1}{s-\rho_k} + \frac{1}{s-\bar\rho_k}\Big) + \sum_{n\ge1}\Big(\frac{1}{2n} - \frac{1}{s+2n}\Big) - \frac{\gamma_E}{2} - \frac{1}{2}\log\pi. \tag{12.17}$$

Each term is a Mellin–Stieltjes transform $\int_{[1,\infty)} x^{-s} d\mu$: $dx \mapsto \frac{1}{s-1}$, $x^{\rho-1}dx \mapsto \frac{1}{s-\rho}$, $x^{-2n-1}dx \mapsto \frac{1}{s+2n}$, $\delta_1 \mapsto 1$. After multiplication by $x^{-c}$ and the change $u = \log x$, the Fourier transforms of the partial sums over $k$ and over $n$ converge locally uniformly with polynomial bounds, so the series converge in $\mathcal{S}'$, and on $(1,\infty)$ the limit measure is the right side of (12.15) times $dx$. Differentiating (12.14) in $s$ shows that $-\zeta_X'/\zeta_X$ is the Mellin–Stieltjes transform of $(\log x)\, dJ_X^{\mathrm{M}}$; injectivity gives (12.15). The series over $k$ converges in $\mathcal{D}'(1,\infty)$ because $\int \varphi(x) x^{-1/2+i\kappa} dx$ is rapidly decreasing in $\kappa$ for $\varphi \in C_c^\infty(1,\infty)$ and $\kappa_k \asymp k/\log k$. By Lemma 5.1, $\mathrm{Ei}(\rho_k u) = e^{\rho_k u}/(\rho_k u) + \varrho_k$ with $\sum |\varrho_k|$ locally uniformly convergent, so (12.16) converges in $\mathcal{D}'(1,\infty)$ for the same reason; differentiating termwise with $\frac{d}{dx}\mathrm{Ei}(\rho \log x) = x^{\rho-1}/\log x$, $(\log x)J_X'$ equals the right side of (12.15). A distribution on an interval with zero derivative is constant. □

*Proof of Theorem 6.4.* Take $X = \mathrm{RVM}$ in Theorems 12.5 and 12.6. By Theorem 5.2, $J_{\mathrm{RVM}}$ is continuous on $(1,\infty)$, so $J_{\mathrm{RVM}}^{\mathrm{M}} = J_{\mathrm{RVM}} + C_{\mathrm{RVM}}$ pointwise, and $J_{\mathrm{RVM}}^{*} = J_{\mathrm{RVM}}$ on $[2,\infty)$. □

**Theorem 12.7** (Cell form of the endpoint identity)**.** *The following are equivalent.*

(i) *(D), that is, $\zeta_{Z1} = \zeta$.*
(ii) *$J_{Z1}^{\mathrm{M}}$ is constant on every interval $(n, n+1)$, $n \ge 1$, and its jumps $\mathsf{b}_n := v_n - v_{n-1}$, where $v_n$ is its value on $(n, n+1)$ and $v_0 := 0$, satisfy $\sum_{n \le X} |\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$ for every $\varepsilon > 0$.*

(iii) *For every $n \geq 1$, in $\mathcal{D}'(n, n+1)$,*

$$2\,\Re \sum_{k\geq 1} (x^{i\gamma_k} - x^{i\tau_k}) = \sqrt{x}\,\log x\; J'_{\mathrm{RVM}}(x), \tag{12.18}$$

*and the jumps of $J^{\mathrm{M}}_{Z1}$ satisfy the bound in (ii).*

*In this case $J^{\mathrm{M}}_{Z1}$ is Riemann's prime-power count $J(x) = \sum_{p^m \leq x} 1/m$.*

*Proof.* (ii)⇒(i): $|v_n| \leq \sum_{m\leq n} |\mathsf{b}_m| \ll n^{1+\varepsilon}$, so for $\Re s > 1$ summation by parts in (12.14) gives $\log \zeta_{Z1}(s) = \sum_n v_n(n^{-s} - (n+1)^{-s}) = \sum_{n\geq 1} \mathsf{b}_n n^{-s}$, absolutely convergent by partial summation. The exponential of an absolutely convergent Dirichlet series is one, so $\zeta_{Z1}$ satisfies (11.8).

(i)⇒(ii): by Theorem 11.2, $\zeta_{Z1} = \zeta$, and $\log \zeta(s) = s\int_1^\infty J(x) x^{-s-1} dx$ for $\Re s > 1$; by uniqueness in Theorem 12.5, $J^{\mathrm{M}}_{Z1} = J$, which is constant between consecutive integers, with jumps $1/m$ at $p^m$ and $\sum_{n\leq X} |\mathsf{b}_n| = J(X) \ll X$.

(ii)⇔(iii): Theorem 12.6 for $X = Z1$ and $X = \mathrm{RVM}$, with $dJ^{\mathrm{M}}_{\mathrm{RVM}}/dx = J'_{\mathrm{RVM}}$, gives in $\mathcal{D}'(1,\infty)$

$$(\log x)\Big(\frac{dJ^{\mathrm{M}}_{Z1}}{dx} - J'_{\mathrm{RVM}}(x)\Big) = -\frac{2}{\sqrt{x}}\,\Re \sum_{k\geq 1} (x^{i\gamma_k} - x^{i\tau_k}), \tag{12.19}$$

so $dJ^{\mathrm{M}}_{Z1}/dx = 0$ on $(n, n+1)$ is (12.18). □

*Remark* 12.8 (The final step in cell form). Theorem 12.7 is the exact form of the continuous-to-atomic passage of Remark 11.4. Both sides of (12.18) are built from the midpoint construction: the right side from the clock alone ($J'_{\mathrm{RVM}}$ is bounded on $[2,\infty)$ by Theorem 6.1, equal to $1/\log x$ plus the unit-periodic first harmonic and the harmonics with stationary points $2\pi x^{1/m}$, Section 6), the left side from the displacements $\gamma_k - \tau_k$. The final step of the method is the identity (12.18) on every cell, equivalently (12.12). Numerically, the explicit formula (12.16) with the $\gamma_k$ is constant between consecutive integers up to the truncation of the list (Experiment 14.6, Figure 8), and the partial products of (12.12) approach $\varpi_{\mathrm{RVM}}$ (Experiment 14.3).

## 13. The atomized family and the programme for the final step

The continuous family $\zeta_\lambda$ keeps, for every $\lambda$, the functional equation and the zeros on the critical line. Its atomization $\tilde{\zeta}_{\mathbb{Z},\lambda}$, built from the cell masses of the prime coordinate, keeps the Euler product and the Dirichlet series. The two are joined by a factor $e^{\tilde{h}_\lambda}$, and both families have a limit at $\lambda = 0$. Hamburger's theorem needs the functional equation and the Dirichlet series on one and the same function; at $\lambda = 0$ this is the vanishing of the joining factor. This section constructs the atomized family for every $\lambda$, proves its analytic properties, writes the joining factor cell by cell, and lists in Subsection 13.7 the steps that remain, in the order in which they will be followed.

Throughout, $\eta_k = \tau_k - \gamma_k$, $\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k$, $\rho_k(\lambda) = \frac{1}{2} + i\gamma_k(\lambda)$, and $\lambda'$ denotes an integration variable in $[0,1]$.

13.1. **The family at infinity.** Put $N_\lambda(\tau) = \#\{k : \gamma_k(\lambda) \leq \tau\}$, $\mathfrak{e}^{(\lambda)} = N_\lambda - \mathcal{N}_{\mathrm{RVM}}$ and $\mathfrak{e}_1^{(\lambda)}(\tau) = \int_{\tau_0}^\tau \mathfrak{e}^{(\lambda)}$, so that $\mathfrak{e}_1^{(0)} = \mathfrak{e}_1$ and $\mathfrak{e}_1^{(1)} = -\mathfrak{s}_1$.

**Proposition 13.1** (Clock decomposition along the family)**.** *There is an absolute constant $C$ with $|\mathfrak{e}_1^{(\lambda)}(\tau)| \leq C\log\tau$ for $\tau \geq \tau_0$ and all $\lambda \in [0,1]$. Consequently Theorem 12.1 holds for every $\zeta_\lambda$ with $\mathfrak{e}_1$ replaced by $\mathfrak{e}_1^{(\lambda)}$, uniformly in $\lambda$; in particular $\varpi_\lambda := \lim_{\sigma\to+\infty} \zeta_\lambda(\sigma)$ exists and*

$$\varpi_\lambda = \varpi_{\mathrm{RVM}} \prod_{k\geq 1} \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k(\lambda)^2}, \qquad \varpi_1 = \varpi_{\mathrm{RVM}}, \quad \varpi_0 = \varpi_{Z1}. \tag{13.1}$$

*Proof.* $\gamma_k(\lambda)$ lies between $\gamma_k$ and $\tau_k$, so $\gamma_1(\lambda) > \tau_0$. For $X \geq \tau_0$,

$$\int_{\tau_0}^{X} (N_\lambda - n_{\mathrm{RVM}}) = \sum_k \big((X-\gamma_k(\lambda))^+ - (X-\tau_k)^+\big) = (1-\lambda) \sum_{\gamma_k(\lambda),\tau_k \leq X} \eta_k + O(\log X), \quad (13.2)$$

the indices counted by only one of the two sums being $O(\log X)$ in number with ordinates in $[X - C_\eta, X]$ (Lemma 2.2). The case $\lambda = 0$ of (13.2) identifies the last sum with $\mathfrak{e}_1 + \mathfrak{s}_1 + O(\log X) = O(\log X)$, by (12.2) and Lemma 8.9. Since $\mathfrak{e}_1^{(\lambda)} = \int (N_\lambda - n_{\mathrm{RVM}}) - \mathfrak{s}_1$, the bound follows, and the proof of Theorem 12.1 applies verbatim. □

Define the *flow at infinity*

$$\Upsilon(\lambda) := \sum_{k \geq 1} \frac{2\gamma_k(\lambda)\,\eta_k}{\frac{1}{4} + \gamma_k(\lambda)^2} \qquad \text{(summed in the order of } k\text{)}. \quad (13.3)$$

**Proposition 13.2** (The value at infinity along the family)**.** *The series* (13.3) *converges uniformly on* $[0,1]$, $\lambda \mapsto \log \varpi_\lambda$ *is* $C^1$ *on* $[0,1]$, *and*

$$\frac{d}{d\lambda} \log \varpi_\lambda = -\Upsilon(\lambda) = \lim_{\sigma \to +\infty} \mathcal{V}_\lambda(\sigma), \qquad \Upsilon'(\lambda) = \sum_{k \geq 1} \frac{2\eta_k^2 \big(\frac{1}{4} - \gamma_k(\lambda)^2\big)}{\big(\frac{1}{4} + \gamma_k(\lambda)^2\big)^2} < 0, \quad (13.4)$$

*with* $\mathcal{V}_\lambda$ *the generator of Theorem 10.3. Moreover*

$$\int_{0^+}^{1} \Upsilon(\lambda)\, d\lambda = \sum_{k \geq 1} \log \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k^2} = \log L + \log \varpi_{Z1}, \quad (13.5)$$

*so that (D) holds if and only if* $\int_{0^+}^{1} \Upsilon(\lambda)\, d\lambda = \log L$, *and* $\Upsilon(1) < \log L + \log \varpi_{Z1} < \Upsilon(0)$.

*Proof.* Put $g(u) = 2u/(\frac{1}{4} + u^2)$. The series $\Upsilon(1) = \sum_k g(\tau_k)\eta_k$ converges by summation by parts against $N_\zeta - n_{\mathrm{RVM}}$, exactly as in (12.10). Since $|g(\gamma_k(\lambda)) - g(\tau_k)| \leq C|\eta_k|\gamma_k^{-2}$ and $\sum_k \eta_k^2 \gamma_k^{-2} < \infty$, the series $\Upsilon(\lambda) - \Upsilon(1)$ converges absolutely and uniformly; the same holds for the termwise derivative, which is the series in (13.4) (each term is negative because $\gamma_k(\lambda) > \frac{1}{2}$). Differentiating (13.1) termwise gives $\frac{d}{d\lambda} \log \varpi_\lambda = -\Upsilon(\lambda)$, and letting $\sigma \to +\infty$ in the generator (10.10) gives the same series. Integrating from 0 to 1 and using $\varpi_1 = \varpi_{\mathrm{RVM}}$, $\varpi_0 = \varpi_{Z1}$ gives (13.5); the last statement follows from Theorem 12.2 and the monotonicity of $\Upsilon$. □

**Proposition 13.3** (The value at infinity and the verified zeros)**.** *If every zero* $\beta + i\gamma$ *with* $0 < \gamma \leq T_0$ *has* $\beta = \frac{1}{2}$, *then*

$$0 \leq \log \varpi_{Z1} \leq 0.26 \sum_{\gamma > T_0} \frac{1}{\gamma^2}. \quad (13.6)$$

*With* $T_0 = 3 \cdot 10^{12}$ [20], *the right side is* $\approx \frac{0.26}{2\pi T_0}(\log \frac{T_0}{2\pi} + 1) < 4 \cdot 10^{-13}$.

*Proof.* By the proof of Theorem 12.2, $\log \varpi_{Z1} = \sum \log c_\gamma$ over the pairs $\{\rho, 1-\bar{\rho}\}$ with $\rho(1-\rho) \neq \frac{1}{4} + \gamma^2$, and all of them have $\gamma > T_0$. For $\delta \leq \frac{1}{2}$ and $\gamma \geq T_0$, (12.11) gives $\log c_\gamma \leq (2\delta^2\gamma^2 + \delta^4)\gamma^{-4} \leq 0.51\,\gamma^{-2}$; a pair consists of two zeros, which gives (13.6). The numerical value follows from $N_\zeta(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T)$. □

13.2. **Atomization by smooth cells.** Fix $\varphi_n \in C_c^\infty(n-1, n+1)$, $n \geq 2$, with $\sum_{n \geq 2} \varphi_n = 1$ on $[2, \infty)$ and $\varphi_n(m) = \delta_{nm}$ for integers $m$. For $\lambda \in [0,1]$ let $J_\lambda$ be the explicit-formula count (12.16) with the ordinates $\gamma_k(\lambda)$ (a distribution on $(1,\infty)$ by Theorem 12.6, which applies to every $\zeta_\lambda$ by Proposition 13.1), $J_\lambda^* = J_\lambda \mathbf{1}_{[2,\infty)}$, and $\pi_\lambda = \sum_{j \geq 1} \frac{\mu(j)}{j} J_\lambda^*(x^{1/j})$.

**Definition 13.4** (Atomized family)**.** The cell masses, the atomized zeta function and its coefficients are

$$\tilde c_n(\lambda) := \langle d\pi_\lambda, \varphi_n\rangle, \qquad \tilde\zeta_{\mathbb{Z},\lambda}(s) := \prod_{n\ge 2}\bigl(1-n^{-s}\bigr)^{-\tilde c_n(\lambda)} = \sum_{N\ge 1}\frac{\tilde a_N(\lambda)}{N^s}, \tag{13.7}$$

$$\tilde a_N(\lambda) = \sum_{\prod_n n^{r_n}=N}\ \prod_{n:\, r_n\ge 1}\binom{\tilde c_n(\lambda)+r_n-1}{r_n}, \qquad \tilde a_1(\lambda)=1. \tag{13.8}$$

Thus $\tilde a_p(\lambda)=\tilde c_p(\lambda)$ for primes $p$, $\tilde a_4=\tilde c_4+\frac12\tilde c_2(\tilde c_2+1)$, $\tilde a_6=\tilde c_6+\tilde c_2\tilde c_3$. The pairing with a fixed smooth partition defines $\tilde c_n(\lambda)$ for every $\lambda$ without point values of $\pi_\lambda$, and $\lambda\mapsto\tilde c_n(\lambda)$ is continuous. If $d\pi_0=\sum_p\delta_p$, then $\tilde c_n(0)=\mathbf{1}_{\mathrm{prime}}(n)$ and $\tilde a_N(0)=1$ for all $N$. Along the family, in $\mathcal{D}'(1,\infty)$,

$$\frac{\partial J_\lambda}{\partial\lambda}(x) = 2\sum_{k\ge1}\eta_k\,\Im\frac{x^{\rho_k(\lambda)}}{\rho_k(\lambda)}, \qquad \frac{\partial}{\partial\lambda}\Bigl((\log x)\frac{dJ_\lambda}{dx}\Bigr) = \frac{2\log x}{\sqrt{x}}\sum_{k\ge1}\eta_k\sin\bigl(\gamma_k(\lambda)\log x\bigr). \tag{13.9}$$

13.3. **The joining factor cell by cell.** Define the *atomization kernel* and the *cell defects*

$$\mathfrak{a}_s(x) := \sum_{n\ge2}\varphi_n(x)\,n^{-s}-x^{-s}, \qquad \mathfrak{h}_n(\lambda;s) := \bigl\langle d\pi_\lambda,\ \varphi_n(x)\bigl(n^{-s}-x^{-s}\bigr)\bigr\rangle. \tag{13.10}$$

$\mathfrak{a}_s$ vanishes at every integer $\ge 2$, and $\mathfrak{h}_n(\lambda;s)$ measures how far the mass of the cell $n$ is from being concentrated at $n$.

**Theorem 13.5** (The joining factor)**.** *Let $\tilde h_\lambda := \log\tilde\zeta_{\mathbb{Z},\lambda}-\log\zeta_\lambda$ on the half-plane where $\tilde\zeta_{\mathbb{Z},\lambda}$ converges absolutely (Theorem 13.8). Then*

$$\tilde\zeta_{\mathbb{Z},\lambda} = \zeta_\lambda\, e^{\tilde h_\lambda}, \qquad \tilde h_\lambda(s) = -\log\varpi_\lambda + \sum_{m\ge1}\frac1m\sum_{n\ge2}\mathfrak{h}_n(\lambda;ms), \tag{13.11}$$

*and, explicitly,*

$$\tilde h_\lambda(s) = -\log\varpi_\lambda + \sum_{m\ge1}\frac1m\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty \mathfrak{a}_{ms}(y^j)\,\frac{1}{\log y}\Bigl(1-\frac{2}{\sqrt{y}}\Re\sum_{k\ge1}y^{i\gamma_k(\lambda)}-\frac{1}{y(y^2-1)}\Bigr)dy, \tag{13.12}$$

*the series over $k$ converging absolutely after integration. Its derivative along the family is*

$$\partial_\lambda\tilde h_\lambda(s) = \Upsilon(\lambda) + \sum_{m\ge1}\frac1m\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty\mathfrak{a}_{ms}(y^j)\,\frac{2}{\sqrt{y}}\sum_{k\ge1}\eta_k\sin\bigl(\gamma_k(\lambda)\log y\bigr)\,dy. \tag{13.13}$$

*Proof.* By the Euler product, $\log\tilde\zeta_{\mathbb{Z},\lambda}(s)=\sum_{m\ge1}\frac1m\sum_n\tilde c_n(\lambda)n^{-ms}$. By Theorems 12.5 and 12.6, $\log\zeta_\lambda(s)$ equals, up to an additive constant $\kappa_\lambda$, the pairing of $dJ_\lambda$ with $x^{-s}$; finite inversion gives $dJ_\lambda=\sum_m\frac1m\,d[\pi_\lambda(x^{1/m})]$, and $\langle d[\pi_\lambda(x^{1/m})],x^{-s}\rangle=\langle d\pi_\lambda,y^{-ms}\rangle$. Inserting $1=\sum_n\varphi_n$ on $[2,\infty)$ gives $\langle d\pi_\lambda,y^{-ms}\rangle=\sum_n(\tilde c_n(\lambda)n^{-ms}-\mathfrak{h}_n(\lambda;ms))$, hence (13.11) with $-\kappa_\lambda$ in place of $-\log\varpi_\lambda$. Letting $\sigma\to+\infty$, $\log\tilde\zeta_{\mathbb{Z},\lambda}\to0$ (because $\tilde a_1=1$), $\log\zeta_\lambda\to\log\varpi_\lambda$ (Proposition 13.1) and the double series tends to 0 by the bounds of Theorem 13.8; so $\kappa_\lambda=\log\varpi_\lambda$. Atoms of $d\pi_\lambda$ at integers contribute nothing because the test functions vanish there. Formula (13.12) is (13.11) with $d\pi_\lambda$ written by (12.15) and the substitution $x=y^j$; the integrated terms in $k$ are $O(\gamma_k(\lambda)^{-2})$ because $\mathfrak{a}_{ms}(2)=0$. Differentiating in $\lambda$ with (13.9) and (13.4) gives (13.13). □

**Proposition 13.6** (Explicit cell defects)**.** *For every $n\ge2$, $\lambda\in[0,1]$ and $\Re s>1$,*

$$\mathfrak{h}_n(\lambda;s) = \sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty\varphi_n(y^j)\bigl(n^{-s}-y^{-js}\bigr)\,\frac{1}{\log y}\Bigl(1-\frac{2}{\sqrt{y}}\Re\sum_{k\ge1}y^{i\gamma_k(\lambda)}-\frac{1}{y(y^2-1)}\Bigr)dy, \tag{13.14}$$

$$\partial_\lambda\mathfrak{h}_n(\lambda;s) = \sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty\varphi_n(y^j)\bigl(n^{-s}-y^{-js}\bigr)\,\frac{2}{\sqrt{y}}\sum_{k\ge1}\eta_k\sin\bigl(\gamma_k(\lambda)\log y\bigr)\,dy, \tag{13.15}$$

*the sums over $j$ being finite and the sums over $k$ converging absolutely after integration, because the test functions are smooth with compact support. At $\lambda = 1$ the cell defect $\mathfrak{h}_n(1;\cdot)$ is the dispersion of the cell $n$ for $\zeta_{\mathrm{RVM}}$, computed from the clock alone; $d\pi_{\mathrm{RVM}}$ has a bounded density there (Corollary 6.2), so $\mathfrak{h}_n(1;\cdot) \equiv 0$ holds only if this density vanishes on the support of $\varphi_n$.*

*Proof.* Insert (12.15) for $d\pi_\lambda = \sum_j \frac{\mu(j)}{j}\, d[J_\lambda^*(x^{1/j})]$ into the definition (13.10) and substitute $x = y^j$; the atoms of $d\pi_\lambda$ at integers, for instance the jump of $J_\lambda^*$ at 2, contribute nothing because $n^{-s} - x^{-s}$ vanishes at $x = n$ and $\varphi_n$ vanishes at the other integers. Differentiation in $\lambda$ uses (13.9); the factor $\log y$ cancels. For the last statement, $\mathfrak{h}_n(1;\cdot) \equiv 0$ forces $\varphi_n\, d\pi_{\mathrm{RVM}} = \tilde{c}_n(1)\delta_n$ by the injectivity used in Proposition 13.7, and an absolutely continuous measure equals a multiple of $\delta_n$ only if both vanish. □

In the form of the path, the final step requires that in every cell the flow of the displacements cancel exactly the dispersion of the clock, (13.18).

**Proposition 13.7** (The limit $\lambda \to 0$ and the meaning of a vanishing cell)**.** *For every $n \geq 2$ and $\Re s > 1$, $\mathfrak{h}_n(\lambda; s) \to \mathfrak{h}_n(0; s)$ and $\tilde{c}_n(\lambda) \to \tilde{c}_n(0)$ as $\lambda \to 0^+$, and*

$$\mathfrak{h}_n(0; s) = \mathfrak{h}_n(1; s) - \int_{0^+}^{1} \partial_\lambda \mathfrak{h}_n(\lambda; s)\, d\lambda. \tag{13.16}$$

*Moreover $\mathfrak{h}_n(0; s) = 0$ for all $\Re s > 1$ if and only if $\varphi_n\, d\pi_{Z1} = \tilde{c}_n(0)\, \delta_n$.*

*Proof.* The test function $\varphi_n(x)(n^{-s} - x^{-s})$ is smooth with compact support, and the integrated terms over $k$ are dominated uniformly in $\lambda$ because $|\gamma_k(\lambda) - \gamma_k| \leq C_\eta$; dominated convergence gives the limits and (13.16). Since $\mathfrak{h}_n(0; s) = \tilde{c}_n(0)n^{-s} - \langle \varphi_n\, d\pi_{Z1}, x^{-s}\rangle$, its vanishing for all $s$ says that the compactly supported distribution $\varphi_n\, d\pi_{Z1}$ has the Mellin transform of $\tilde{c}_n(0)\delta_n$; the Mellin transform is injective on such distributions. □

13.4. **Analytic continuation of the atomized family.** Let $\mathfrak{L}(x) = \int_1^x (1 - y^{-1})\, dy/\log y$; then $\log \frac{s}{s-1} = \int_1^\infty x^{-s}\, d\mathfrak{L}(x)$ for $\Re s > 1$ (both sides have derivative $\frac{1}{s} - \frac{1}{s-1}$ and tend to 0 as $s \to +\infty$), and $\mathfrak{L}(x) = \mathrm{li}(x) - \log\log x - \gamma_E$.

**Theorem 13.8** (Continuation of the atomized family)**.** *Let $\lambda \in [0, 1]$.*

(i) *For every $c > \frac{1}{2}$, $(\pi_\lambda - \mathfrak{L})\, x^{-c} \in L^2\big((2,\infty), dx/x\big)$.*
(ii) *$\sum_n |\tilde{c}_n(\lambda)| n^{-\sigma} < \infty$ for $\sigma > \frac{3}{2}$; hence $\tilde{\zeta}_{\mathbb{Z},\lambda}$ is an absolutely convergent Dirichlet series in $\Re s > \frac{3}{2}$.*
(iii) *$\tilde{h}_\lambda$ extends holomorphically to $\Re s > \frac{1}{2}$. Hence $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde{h}_\lambda}$ is meromorphic in $\Re s > \frac{1}{2}$, with a single pole at $s = 1$ and no zeros.*
(iv) *For $\lambda = 1$, $\tilde{h}_1$ extends holomorphically to $\Re s > 0$, and the zeros of $\tilde{\zeta}_{\mathbb{Z},1}$ in $\Re s > 0$ are exactly $\frac{1}{2} \pm i\tau_k$.*

*Proof.* (i) By Proposition 13.1, for $c > \frac{1}{2}$ there is $C_c$ with $|\log \zeta_\lambda(s) - \log \frac{s}{s-1}| \leq C_c(1 + \log(|t| + 2))$ on $\Re s \geq c$; the function $\log \zeta_\lambda - \log \frac{s}{s-1}$ is holomorphic on $\Re s > \frac{1}{2}$, where $\zeta_\lambda$ has no zeros, and its pole at 1 is removed. It equals $s \int_1^\infty (J_\lambda^{\mathrm{M}} - \mathfrak{L}) x^{-s-1} dx$ for $\Re s > 1$. The argument of Theorem 12.5, applied on the lines $\Re s = c > \frac{1}{2}$, gives $(J_\lambda^{\mathrm{M}} - \mathfrak{L})x^{-c} \in L^2$. The constant $C_\lambda = J_\lambda^{\mathrm{M}} - J_\lambda$, the truncation at 2, the terms $j \geq 2$ of the Möbius inversion (of size $O(\sqrt{x})$ in mean) and $\mathfrak{L}(x) - \mathfrak{L}(x^{1/j})$ preserve the property for $c > \frac{1}{2}$.

(ii) Put $E = \pi_\lambda - \mathfrak{L}$ and $I_n = \int_{n-1}^{n+1} |E|^2 dx$. Since $\int \varphi_n' = 0$, $\tilde{c}_n(\lambda) = \int \varphi_n\, d\mathfrak{L} - \int E\varphi_n'\, dx$, so $|\tilde{c}_n(\lambda)| \leq C(1 + I_n^{1/2})$. By (i), $\sum_n I_n n^{-2c-1} < \infty$, and Cauchy–Schwarz gives $\sum_n I_n^{1/2} n^{-\sigma} < \infty$ for $\sigma > c + 1$.

(iii) With $\psi_n = \varphi_n(x)(n^{-s} - x^{-s})$ one has $|\psi_n'| \le C|s|n^{-\sigma-1}$ and $\psi_n$ vanishes at the ends of the cell, so

$$|\mathfrak{h}_n(\lambda; s)| \le C|s|\, n^{-\sigma-1}(1 + I_n^{1/2}), \qquad \sum_n I_n^{1/2} n^{-\sigma-1} < \infty \quad (\sigma > c). \tag{13.17}$$

Choosing $c \in (\frac{1}{2}, \sigma)$, the series (13.11) converges locally uniformly on $\Re s > \frac{1}{2}$; the terms $m \ge 2$ are evaluated at $\Re(ms) \ge 2\sigma$. Hence $\tilde{h}_\lambda$ is holomorphic on $\Re s > \frac{1}{2}$, and $\zeta_\lambda$ has no zeros there.

(iv) By Corollary 6.2, $\pi_{\mathrm{RVM}}$ oscillates by $O(1)$ on every unit cell; integrating by parts, $|\mathfrak{h}_n(1; s)| \le C|s|n^{-\sigma-1}$, and the series converges for $\sigma > 0$. The factor $e^{\tilde{h}_1}$ has no zeros, so the zeros of $\tilde{\zeta}_{\mathbb{Z},1}$ in $\Re s > 0$ are those of $\zeta_{\mathrm{RVM}}$. □

**Theorem 13.9** (Closing criterion)**.** *The following are equivalent.*

(i) *(D), that is,* $\zeta_{Z1} = \zeta$.
(ii) $\tilde{c}_n(0) = O(1)$ *and* $\tilde{h}_0 \equiv 0$.
(iii) $\tilde{c}_n(0) = O(1)$ *and, for every* $n \ge 2$ *and every* $\Re s > 1$,

$$\int_{0^+}^{1} \partial_\lambda \mathfrak{h}_n(\lambda; s)\, d\lambda = \mathfrak{h}_n(1; s). \tag{13.18}$$

*In this case* $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, $\tilde{a}_N(0) = 1$ *for all* $N$, *and* $\varpi_{Z1} = 1$.

*Proof.* (ii)⇒(i): $\tilde{h}_0 \equiv 0$ gives $\zeta_{Z1} = \tilde{\zeta}_{\mathbb{Z},0} = \exp(\sum_m \frac{1}{m} \sum_n \tilde{c}_n(0) n^{-ms})$, an absolutely convergent Dirichlet series in $\Re s > 1$ when $\tilde{c}_n(0) = O(1)$. (iii)⇒(ii): by (13.16) every $\mathfrak{h}_n(0; \cdot)$ vanishes, so $\tilde{h}_0 \equiv -\log \varpi_{Z1}$ is constant; then $\zeta_{Z1} = \varpi_{Z1} \tilde{\zeta}_{\mathbb{Z},0}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, Theorem 11.2 gives $\zeta_{Z1} = \zeta$, and comparing the first coefficients, $\varpi_{Z1} = 1$. (i)⇒(iii): if $\zeta_{Z1} = \zeta$, Theorem 12.7 gives $J_{Z1}^{\mathrm{M}} = J$, and $J_{Z1} = J$ by von Mangoldt's explicit formula [12], so $d\pi_{Z1} = \sum_p \delta_p$, $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, every $\mathfrak{h}_n(0; \cdot)$ vanishes, and (13.16) gives (13.18). □

## 13.5. **The two endpoints of the atomized family.**

**Proposition 13.10** (Atomization at the clock endpoint)**.** *At* $\lambda = 1$*:*

(i) $\tilde{c}_n(1) = O(1)$, *and* $\tilde{\zeta}_{\mathbb{Z},1}$ *is an absolutely convergent Dirichlet series in* $\Re s > 1$*;*
(ii) $\tilde{h}_1(\sigma) \to -\log \varpi_{\mathrm{RVM}} = \log L \ne 0$ *as* $\sigma \to +\infty$*;*
(iii) $\tilde{h}_1$ *is not constant: for every* $\sigma > 1$, $\zeta_{\mathrm{RVM}}(\sigma + it) \to \varpi_{\mathrm{RVM}}$ *as* $|t| \to \infty$, *whereas* $\tilde{\zeta}_{\mathbb{Z},1}(\sigma + it)$ *has no limit as* $|t| \to \infty$.

*Proof.* (i) By Corollary 6.2, $\pi_{\mathrm{RVM}}$ oscillates by $O(1)$ on every unit cell, and $\tilde{c}_n(1) = -\int \pi_{\mathrm{RVM}} \varphi_n'\, dx$ with $\int \varphi_n' = 0$. (ii) is (13.11) with $\sigma \to +\infty$, since $\varpi_{\mathrm{RVM}} < 1$. (iii) The first limit is (8.33). If $\tilde{\zeta}_{\mathbb{Z},1}(\sigma + it)$ had a limit $\ell$ as $|t| \to \infty$, the mean-square argument (8.43)–(8.44) applied to $\tilde{\zeta}_{\mathbb{Z},1} - \ell$ on the line $\Re s = \sigma$ would give $\tilde{a}_N(1) = 0$ for $N \ge 2$, so $\tilde{\zeta}_{\mathbb{Z},1} \equiv 1$, which contradicts its pole at $s = 1$ (Theorem 13.8(iii)). If $\tilde{h}_1$ were constant, $\zeta_{\mathrm{RVM}}$ would be a constant multiple of an absolutely convergent Dirichlet series, which Theorem 8.11 excludes. □

The two families thus keep complementary structures and are joined by a zero-free factor:

$$\underbrace{\zeta_\lambda}_{\substack{\text{functional equation}\\ \text{zeros } \frac{1}{2} \pm i\gamma_k(\lambda)}} \cdot\, e^{\tilde{h}_\lambda} \;=\; \underbrace{\tilde{\zeta}_{\mathbb{Z},\lambda}}_{\substack{\text{Dirichlet series}\\ \text{Euler product}}}, \qquad \tilde{h}_1 \text{ not constant}, \qquad (D) \iff \tilde{h}_0 \equiv 0 \text{ and } \tilde{c}_n(0) = O(1). \tag{13.19}$$

At $\lambda = 1$ the joining factor carries the whole difference between the clock and an Euler product; Theorem 13.9 states that this difference disappears at $\lambda = 0$.

The reciprocal of the atomized family is again an Euler product:

$$\frac{1}{\tilde{\zeta}_{\mathbb{Z},\lambda}(s)} = \prod_{n \ge 2} (1 - n^{-s})^{\tilde{c}_n(\lambda)} = \sum_{N \ge 1} \frac{\tilde{\mu}_\lambda(N)}{N^s}, \qquad \tilde{\mu}_\lambda(N) = \sum_{\prod_n n^{r_n} = N} \prod_{n:\, r_n \ge 1} (-1)^{r_n} \binom{\tilde{c}_n(\lambda)}{r_n}, \tag{13.20}$$

absolutely convergent in $\Re s > \frac{3}{2}$ (in $\Re s > 1$ for $\lambda = 1$), because it is the exponential of the absolutely convergent series $-\sum_m \frac{1}{m}\sum_n \tilde{c}_n(\lambda) n^{-ms}$. Thus $\tilde{\mu}_\lambda(p) = -\tilde{c}_p(\lambda)$, and the atomized Mertens functions

$$\tilde{M}_\lambda(x) := \sum_{N \le x} \tilde{\mu}_\lambda(N), \qquad \frac{1}{\tilde{\zeta}_{\mathbb{Z},\lambda}} = e^{-\tilde{h}_\lambda} F_\lambda, \tag{13.21}$$

exist for every $\lambda \in [0,1]$. Under the conditions of Theorem 13.9, $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, so $\tilde{\mu}_0 = \mu$ and $\tilde{M}_0 = M$.

**Proposition 13.11** (Mellin form of the endpoint Mertens function)**.** *There are $c_1 \ge 2$ and a unique measurable $B_{Z1}$ on $(0,\infty)$, vanishing on $(0,1)$, with $x^{-c}B_{Z1} \in L^2((1,\infty), dx/x)$ and*

$$F_{Z1}(s) = s\int_1^\infty B_{Z1}(x)\, x^{-s-1}\, dx = \exp\Big(-\int_{[1,\infty)} x^{-s}\, dJ^{\mathrm{M}}_{Z1}(x)\Big) \qquad (\Re s > c_1), \tag{13.22}$$

*for every $c > c_1$. Moreover $(D)$ holds if and only if $B_{Z1} = M$ almost everywhere; and if $F_{Z1}$ has an absolutely convergent Dirichlet expansion* (11.37)*, then $B_{Z1} = M_{Z1}$.*

*Proof.* Put $A = 2(\pi+2)C_{\mathfrak{e}}$. By (12.5), for $\sigma' \ge \max(4A, R_0)$, $|F_{Z1}(s)| \le C(|t|+\sigma)^{1/4}$; take $c_1 = \max(2, \frac{1}{2}+4A, \frac{1}{2}+R_0)$. Then $\int_{\mathbb{R}} |F_{Z1}(\sigma+it)|^2 |\sigma+it|^{-2} dt \le C'\sigma^{-1/2}$ for $\sigma > c_1$, and the argument of Theorem 12.5 gives $B_{Z1}$, its vanishing on $(0,1)$, its independence of $c$ and its uniqueness; the second equality in (13.22) is Theorem 12.5. If $\zeta_{Z1} = \zeta$, then $F_{Z1}(s) = 1/\zeta(s) = s\int_1^\infty M(x)x^{-s-1}dx$ and uniqueness gives $B_{Z1} = M$. Conversely, $B_{Z1} = M$ gives $F_{Z1} = 1/\zeta$ on $\Re s > c_1$, hence $\zeta_{Z1} = \zeta$ by analytic continuation. Under (11.37), partial summation gives $F_{Z1}(s) = s\int_1^\infty M_{Z1}(x)x^{-s-1}dx$, and uniqueness gives $B_{Z1} = M_{Z1}$. □

13.6. **Hamburger's hypotheses at the two endpoints.** Hamburger's theorem, in the form of [23, §2.13], has four hypotheses: (H0) $f = G/P$ with $G$ entire of finite order and $P$ a polynomial; (H1) $f(s) = \sum_n a_n n^{-s}$ absolutely for $\sigma > 1$, with arbitrary complex $a_n$; (H2) $f(s)\Gamma(\frac{s}{2})\pi^{-s/2} = g(1-s)\Gamma(\frac{1-s}{2})\pi^{-(1-s)/2}$; (H3) $g(1-s) = \sum_n b_n n^{s-1}$ absolutely for $\sigma < -\alpha < 0$. It contains no hypothesis on the location of zeros, and its conclusion is $f = C\zeta$. Absolute convergence for $\sigma > 1$ in (H1) cannot be weakened to some half-plane: Knopp constructed infinitely many linearly independent solutions of (H0), (H2), (H3) that are Dirichlet series absolutely convergent only in some half-plane [18]. The table records the status at the two endpoints and is updated as the programme advances.

| | $\zeta_{Z1}$ (continuous limit) | $\tilde{\zeta}_{\mathbb{Z},0}$ (atomized limit) |
|---|---|---|
| H0 | proved: $(s-1)\zeta_{Z1}$ entire of order 1 (Proposition 11.1) | open: meromorphic in $\Re s > \frac{1}{2}$ only (Theorem 13.8) |
| H1 | open: this is (D) | proved in $\Re s > \frac{3}{2}$; in $\Re s > 1$ if $\tilde{c}_n(0) = O(1)$ (step A6) |
| H2 | proved, with $g = f$ (Proposition 11.1) | open |
| H3 | follows from H1 | open |
| zeros | no hypothesis | no hypothesis |

The two columns meet when $\tilde{h}_0 \equiv 0$: then $\zeta_{Z1} = \tilde{\zeta}_{\mathbb{Z},0}$ has H0–H3, and Hamburger's theorem gives $\zeta_{Z1} = \zeta$ (Theorem 13.9).

| Property | continuous family $\zeta_\lambda$ | atomized family $\tilde{\zeta}_{\mathbb{Z},\lambda}$ |
|---|---|---|
| definition | Hadamard product over $\frac{1}{2} \pm i\gamma_k(\lambda)$ | Euler product over $n \geq 2$ with masses $\tilde{c}_n(\lambda)$ |
| zeros | exactly $\frac{1}{2} \pm i\gamma_k(\lambda)$, impossible elsewhere (Theorem 10.1) | none in $\Re s > \frac{1}{2}$; $\frac{1}{2} \pm i\tau_k$ in $\Re s > 0$ for $\lambda = 1$ (Theorem 13.8) |
| functional equation | proved for every $\lambda$ | open |
| Dirichlet series | not for $\lambda = 1$ (Theorem 8.11); at $\lambda = 0$ this is (D) | proved in $\Re s > \frac{3}{2}$ for every $\lambda$; in $\Re s > 1$ for $\lambda = 1$ |
| value at $+\infty$ | $\varpi_\lambda$ (13.1) | $\tilde{a}_1(\lambda) = 1$ |
| half-plane bounds | $\lambda = 1$: Theorem 8.10; all $\lambda$: $\ll \lvert t\rvert^{A/(\sigma-1/2)}$ | bounded on $\Re s \geq \sigma_0 > \frac{3}{2}$ |
| flow in $\lambda$ | generator $\mathcal{V}_\lambda$; $\Upsilon(\lambda)$ at $+\infty$ | $\partial_\lambda \tilde{c}_n(\lambda)$; joining factor (13.13) |
| prime coordinate | counts $J_\lambda$, $\pi_\lambda$ (distributions, (12.16)) | cell masses $\tilde{c}_n(\lambda) = \langle d\pi_\lambda, \varphi_n \rangle$; $\tilde{c}_n(1) = O(1)$ |
| vertical behaviour | $\lambda = 1$: $\zeta_{\mathrm{RVM}}(\sigma + it) \to \varpi_{\mathrm{RVM}}$ (8.33) | $\lambda = 1$: no limit as $\lvert t\rvert \to \infty$ (Proposition 13.10) |
| reciprocal | $F_\lambda = 1/\zeta_\lambda$; $F_{Z1}$ is a Mellin transform of $B_{Z1}$ (Proposition 13.11) | Dirichlet series with coefficients $\tilde{\mu}_\lambda(N)$ (13.20) |
| Mertens function | $M^{\mathrm{crit}}_{\mathrm{RVM}} = L + O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 8.12); $B_{Z1} = M \iff (D)$ | $\tilde{M}_\lambda$ exists for every $\lambda$; $\tilde{M}_0 = M$ under A6–A7; bound open |
| limit $\lambda \to 0^+$ | $\zeta_{Z1}$ | $\tilde{\zeta}_{\mathbb{Z},0}$ |

13.7. **Programme.** The steps are numbered in the order in which they are followed. The status column is updated with each step.

| Step | Statement | Status |
|---|---|---|
| A1 | Atomization by smooth cells, coefficients $\tilde{a}_N(\lambda)$ | proved (Definition 13.4) |
| A2 | Joining factor and explicit cell defects, (13.11)–(13.15) | proved (Thm. 13.5, Prop. 13.6) |
| A3 | $L^2$ square-root estimate for every $\lambda$; Dirichlet series in $\Re s > \frac{3}{2}$ | proved (Theorem 13.8) |
| A4 | $\tilde{h}_\lambda$ holomorphic in $\Re s > \frac{1}{2}$; in $\Re s > 0$ for $\lambda = 1$ | proved (Theorem 13.8) |
| A5 | Limits $\lambda \to 0^+$ of $\tilde{c}_n$, $\mathfrak{h}_n$, $\varpi_\lambda$; flow at infinity $\Upsilon$ | proved (Propositions 13.2, 13.7) |
| A5′ | Endpoints: $\tilde{h}_1$ not constant; atomized Möbius coefficients $\tilde{\mu}_\lambda$; Mellin form of $M_{Z1}$ | proved (Propositions 13.10, 13.11) |
| A6 | $\tilde{c}_n(0) = O(1)$: bounded oscillation of $\pi_{Z1}$ on unit cells | open |
| A7 | Cell identity (13.18) for every $n \geq 2$ | open |
| A8 | Equivalent scalar form: $\int_{0^+}^1 \Upsilon\, d\lambda = \log L$, i.e. (12.12) | open; follows from A6–A7 |
| A9 | $\zeta_{Z1} = \zeta$ by Hamburger; $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ by Step 3 | proved as implication from A6, A7 (Thms. 13.9, 11.8) |
| A10 | Continuation of $\tilde{h}_\lambda$ to $0 < \Re s \leq \frac{1}{2}$ for $\lambda < 1$ | open; not needed for A9 |

*Remark* 13.12 (Input for A6 and A7). Steps A1–A5 use only two properties of an ordinate list: bounded displacement from the clock and a primitive of the count difference of size $O(\log \tau)$. The list $(\tau_k)$ has both properties, and for it the corresponding function $\zeta_{\mathrm{RVM}}$ is not a Dirichlet series (Theorem 8.11). Steps A6 and A7 must therefore use a property of the ordinates $\gamma_k$ that the clock ordinates do not have; the natural candidate is the arithmetic of $\zeta$ carried by the $\gamma_k$, for instance Landau's formula $\sum_{0<\gamma\leq T} x^\rho = -\frac{T}{2\pi}\Lambda(x) + O(\log T)$, in which the prime powers appear as resonances of the ordinates.

## 14. Numerical illustration with 2 001 052 zeros

This section shows what is being constructed and checks that the theorems are consistent with computation. It uses the ordinates $\gamma_1, \ldots, \gamma_N$ of the first $N = 2\,001\,052$ zeros of $\zeta$ (file `zeross.bin`,

$\gamma_N = 1\,132\,490.6587$) and the midpoint ordinates $\tau_1, \dots, \tau_N$ computed from (2.5) (file `zLamb.bin`, $\tau_N = 1\,132\,490.7226$). The programs are listed in Appendix A and their complete console output in Appendix B; all figures have linear axes. Each item is an *experiment*: it reports a computation and does not replace a proof.

Products over the ordinates are evaluated as follows: the first $10^5$ factors exactly, the factors $10^5 < k \le N$ through power sums of $(\frac14 + o_k^2)^{-1}$, and the factors $k > N$ by the midpoint rule for the clock, which gives the tail $\frac{1}{2\pi}(q\frac{L_U+1}{U} - \frac{q^2}{2}\frac{3L_U+1}{9U^3} + \frac{q^3}{3}\frac{5L_U+1}{25U^5})$ with $\mathcal{N}_{\mathrm{RVM}}(U) = N$, $L_U = \log(U/2\pi)$.

### 14.1. **The clock and the zeros.**

**Experiment 14.1** (Clock, midpoints and displacements)**.** *Object:* $\mathcal{N}_{\mathrm{RVM}}$, $N_\zeta$, $\tau_k$, $\eta_k = \tau_k - \gamma_k$. *Range:* $k \le N$. *Expected:* (2.4) to rounding, and $|\eta_k| \le C_\eta$ (Lemma 2.2). *Observed:* $\max_k |\mathcal{N}_{\mathrm{RVM}}(\tau_k) - (k - \frac12)| = 6.985 \cdot 10^{-10}$; $\max_k |\eta_k| = 0.9937$ (at $k = 34$), mean $|\eta_k| = 0.1512$, root mean square 0.1881, mean $\eta_k = 1.7 \cdot 10^{-7}$. By blocks of $k$, $\max|\eta_k|$ is $0.840, 0.994, 0.915, 0.834, 0.786, 0.763, 0.717$ on $[1, 10]$, $[11, 10^2]$, $[10^2, 10^3]$, $\dots$, $[10^6, N]$.

Figure 1 shows how the clock places the midpoint ordinates. The staircase $N_\zeta$ jumps by one at each $\gamma_k$; the smooth clock $\mathcal{N}_{\mathrm{RVM}}$ crosses the half-integer level $k - \frac12$ at $\tau_k$. Since $N_\zeta - \mathcal{N}_{\mathrm{RVM}} = S(T) + O(1/T)$ (Figure 2), the horizontal distance between the open square $(\gamma_k, k - \frac12)$ and the red point $(\tau_k, k - \frac12)$ is the remainder $S$ converted into ordinate units by the slope $\mathcal{N}'_{\mathrm{RVM}} = \frac{1}{2\pi}\log\frac{T}{2\pi}$. The right panel repeats the picture at height $1.13 \cdot 10^6$, where the mean spacing is 0.519: every $\tau_k$ still lies next to its $\gamma_k$.

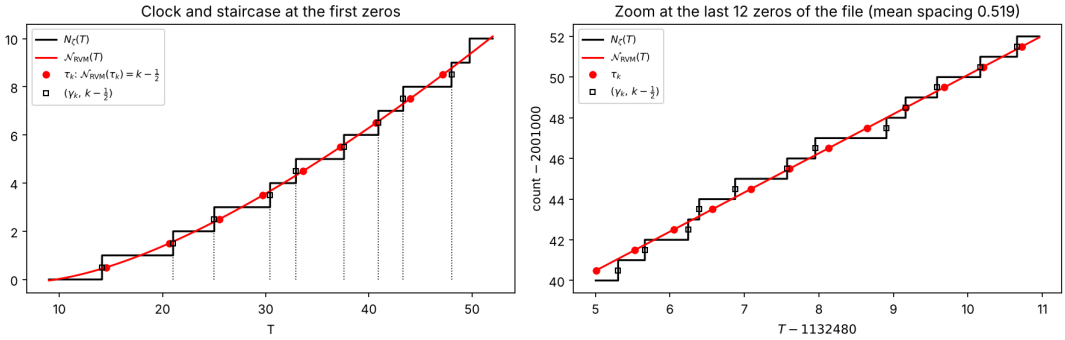



FIGURE 1. The midpoint clock $\mathcal{N}_{\mathrm{RVM}}$ (red), the zero staircase $N_\zeta$ (black), the midpoint ordinates $\tau_k$ (red points) and the points $(\gamma_k, k - \frac12)$ (squares). Left: the first nine zeros. Right: the last twelve zeros of the file.

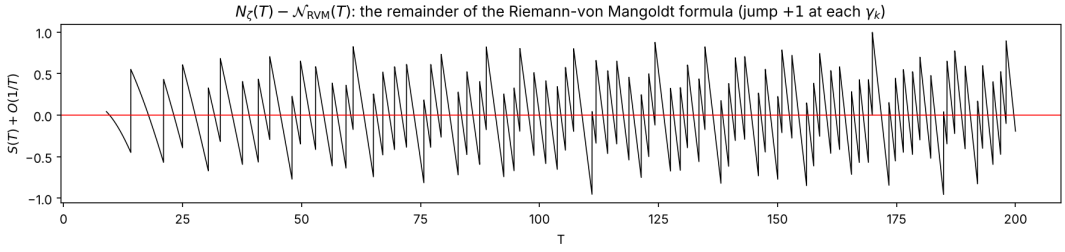



FIGURE 2. The remainder $N_\zeta(T) - \mathcal{N}_{\mathrm{RVM}}(T) = S(T) + O(1/T)$ of (2.2) for $T \le 200$.

14.2. **The $\lambda$-family and the endpoint.** Figure 3 shows the core of the method. The lower left panel draws the moving ordinates $\gamma_k(\lambda)$ in clock units, $\mathcal{N}_{\mathrm{RVM}}(\gamma_k(\lambda)) - (k - \frac{1}{2})$: at $\lambda = 1$ all lines meet at 0 (the $\tau_k$), at $\lambda = 0$ they sit at the ordinates of $\zeta$; for $\lambda > 0$ the ordinates $\gamma_k(\lambda)$ remain strictly increasing in $k$ (Theorem 10.1). The lower right panel draws the real functions $Z_\lambda(t) = e^{i\vartheta(t)}\zeta_\lambda(\frac{1}{2} + it)$ for $\lambda = 0, 0.25, 0.5, 0.75, 1$ together with Hardy's function $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$: the zeros of every $Z_\lambda$ are real, they move continuously with $\lambda$, and the curve for $\lambda = 0$ lies on top of $Z$.

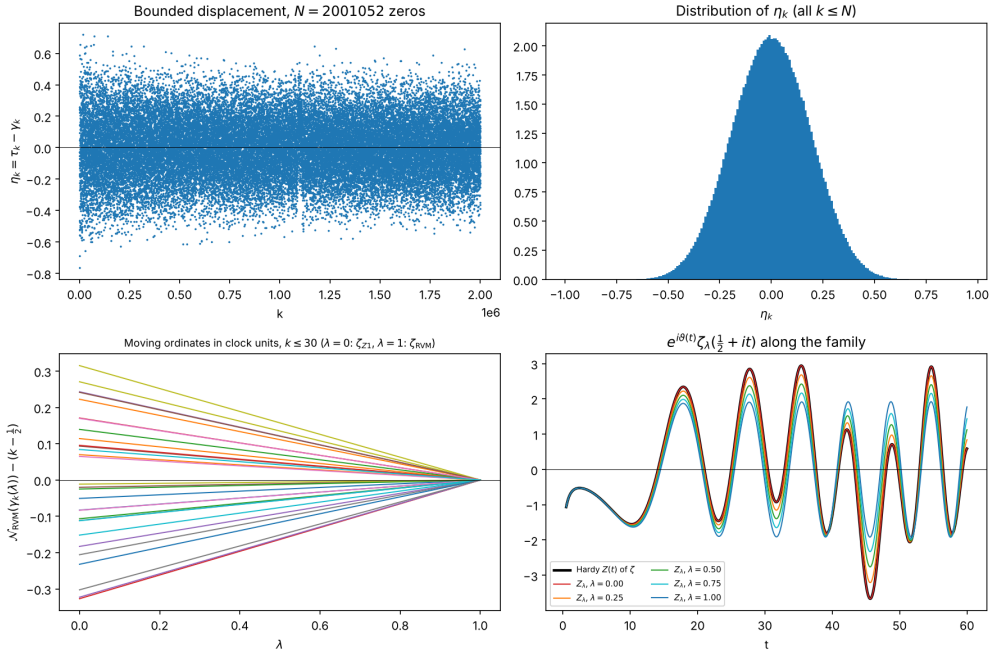



FIGURE 3. Top: $\eta_k$ for every 50th $k \le N$ and its distribution. Bottom: moving ordinates in clock units, $k \le 30$, and the functions $Z_\lambda(t)$ on the critical line.

**Experiment 14.2** (The endpoint $\zeta_{Z1}$ against $\zeta$)**.** *Object:* $\zeta_{Z1} = P_{Z1}/G$ computed from the $N$ ordinates $\gamma_k$, and $\zeta$ computed by mpmath. *Range:* $\sigma \in \{2, 1.25, 0.75\}$, $1 \le t \le 2000$ (499 points per line), and the critical line $0.5 \le t \le 60$. *Expected:* if $\zeta_{Z1} = \zeta$, the ratio is 1 up to the truncation of the product. *Observed:* $\max|\zeta_{Z1}/\zeta - 1| = 8.7 \cdot 10^{-10}$, $1.1 \cdot 10^{-9}$, $2.6 \cdot 10^{-9}$ on the three lines, and $\max|Z_{Z1} - Z|/\max(1, |Z|) = 8.4 \cdot 10^{-10}$ on the critical line (Figure 4).

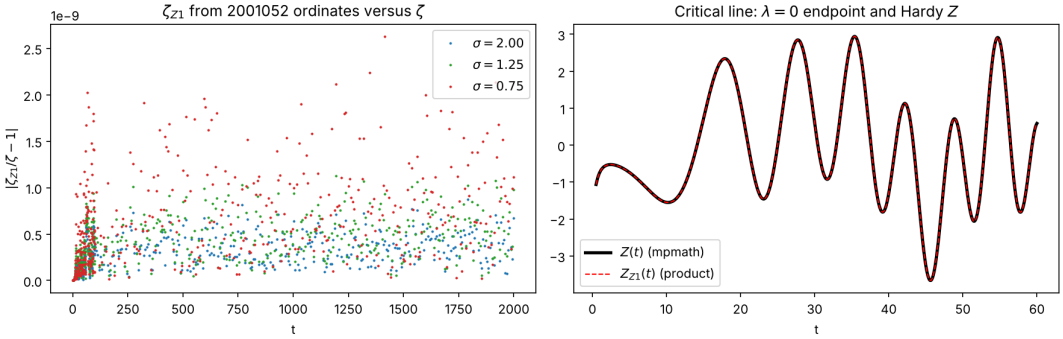



FIGURE 4. Left: $|\zeta_{Z1}/\zeta - 1|$ on three vertical lines. Right: $Z_{Z1}$ and Hardy's $Z$ on the critical line.

Experiment 14.2 is consistent with the identity $\zeta_{Z1} = \zeta$ to nine digits in the range computed; it is a finite computation and not a proof of the identity.

**Experiment 14.3** (The value at infinity: the product identity)**.** *Object:* the ordered partial products $P_K = \prod_{k\le K}(\frac14+\gamma_k^2)/(\frac14+\tau_k^2)$ of (12.12). *Range:* $K \le N$. *Expected:* by Corollary 12.3, $\zeta_{Z1} = \zeta$ is the statement $P_K \to \varpi_{\mathrm{RVM}}$. *Observed:* $P_K = 0.96256,\ 0.96248,\ 0.96229,\ 0.961951,\ 0.961930,\ 0.9619418189,\ 0.9619415390$ for $K = 10, 10^2, \dots, 10^6, N$, against $\varpi_{\mathrm{RVM}} = 0.9619415807$ from Experiment 14.4; for $K \ge 10^6$, $P_K$ stays in $[0.9619389, 0.9619440]$ with mean $0.9619416633$ (Figure 5). The flow at infinity (13.3) is $\Upsilon(\lambda) = 0.0436950,\ 0.0412445,\ 0.0387987,\ 0.0363572,\ 0.0339198$ for $\lambda = 0, \frac14, \frac12, \frac34, 1$, decreasing as in (13.4) with slope close to $-\sum_k 2\eta_k^2/\gamma_k^2 = -0.0098211$, and $\int_0^1 \Upsilon\, d\lambda = \sum_{k\le N} \log\frac{1/4+\tau_k^2}{1/4+\gamma_k^2} = 0.0388016004$ against $\log L = 0.0388015571$.

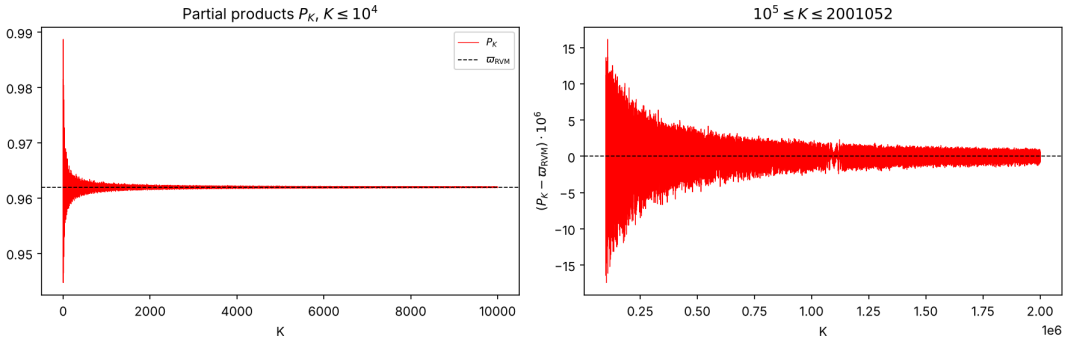



FIGURE 5. The partial products $P_K$ of (12.12) against $\varpi_{\mathrm{RVM}}$ (dashed). Left: $K \le 10^4$. Right: $(P_K - \varpi_{\mathrm{RVM}}) \cdot 10^6$ for $10^5 \le K \le N$. The value of the endpoint at $+\infty$ is $\varpi_{Z1} = \varpi_{\mathrm{RVM}}/\lim_K P_K$ (Theorem 12.1).

### 14.3. The transition theorem and $\zeta_{\mathrm{RVM}}$ against $\zeta$.

**Experiment 14.4** (Transition theorem)**.** *Object:* $\zeta_{\mathrm{RVM}}$, $w(s)$, $\varpi_{\mathrm{RVM}}$, $L$. *Range:* $\sigma' \in \{0.25, 0.5, 1.5\}$, $10 \le t \le 10^4$. *Expected:* Theorem 8.1 and Corollary 8.8: $t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1|$ bounded, $|w| = (t/2\pi)^{-\sigma'}(1 + O(1/t))$. *Observed:* Richardson extrapolation on $\Re s = 2$ gives $\varpi_{\mathrm{RVM}} = 0.9619415807$ and $L = 1.0395641690$; $\max_{t>100} t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1| = 0.0617, 0.0618, 0.0620$ for the three values of $\sigma'$, and $0.0790$ for $\varpi_{\mathrm{RVM}}(1+\chi)$ at $\sigma' = 0.25$; $|w|\,t^{\sigma'} = 1.5832,\ 2.5066,\ 15.66$–$15.75$, i.e. $(2\pi)^{\sigma'}$.

The bottom panels of Figure 6 compare $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on two vertical lines: $\zeta$ carries the Euler product and oscillates with the primes, whereas $\zeta_{\mathrm{RVM}}$ approaches the constant $\varpi_{\mathrm{RVM}}$ on $\sigma = 2$ and $\varpi_{\mathrm{RVM}}|1+\chi|$ on $\sigma = 0.75$. This is the analytic content of Theorem 8.11.

### 14.4. Exponential sums.

**Experiment 14.5** (Exponential sums over the two spectra)**.** *Object:* $S(N; \log x) = \sum_{k\le N} x^{io_k}$ for $o_k = \tau_k$ and $o_k = \gamma_k$. *Range:* $N \le 2\,001\,052$, $x \in \{2, 3, 4, 6, 10\}$. *Expected:* $|S_{\mathrm{RVM}}| \ll 1 + \log N$ (Theorem 4.1); for $\gamma_k$, Landau's term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$. *Observed:* $\max_N |S_{\mathrm{RVM}}(N; \log x)| = 3.57, 3.98, 3.95, 4.53, 9.70$; at $N = 2\,001\,052$, $|\sum x^{i\gamma_k}| = 88\,301,\ 114\,268,\ 62\,435,\ 0.26,\ 2.42$, against Landau's $88\,342,\ 114\,324,\ 62\,467,\ 0,\ 0$.

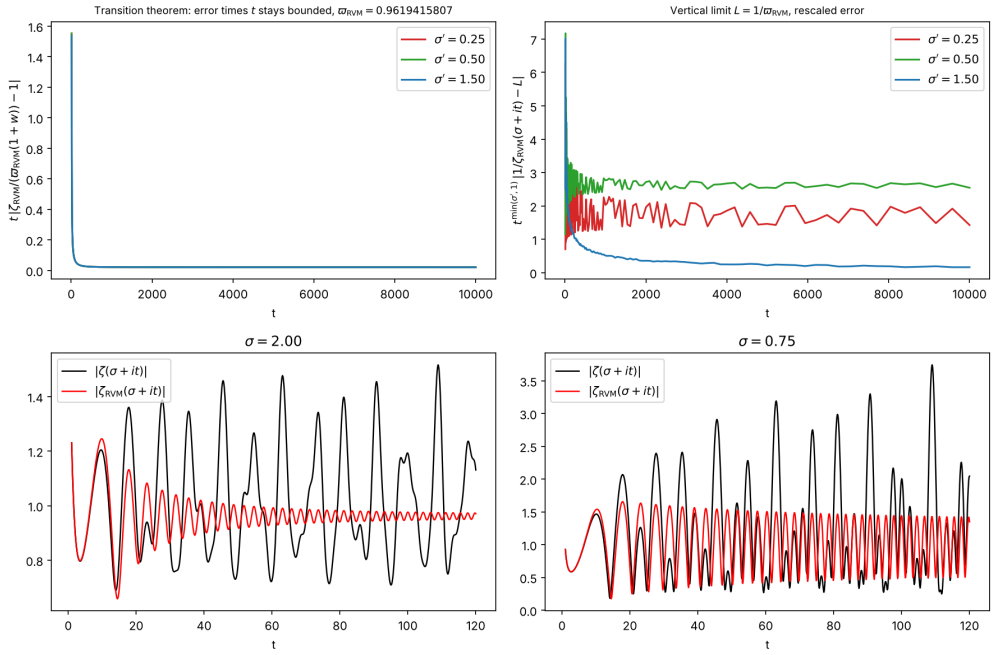



FIGURE 6. Top: the transition error multiplied by $t$, and the rescaled vertical limit $t^{\min(\sigma',1)}|1/\zeta_{\mathrm{RVM}} - L|$. Bottom: $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on $\sigma = 2$ and $\sigma = 0.75$.

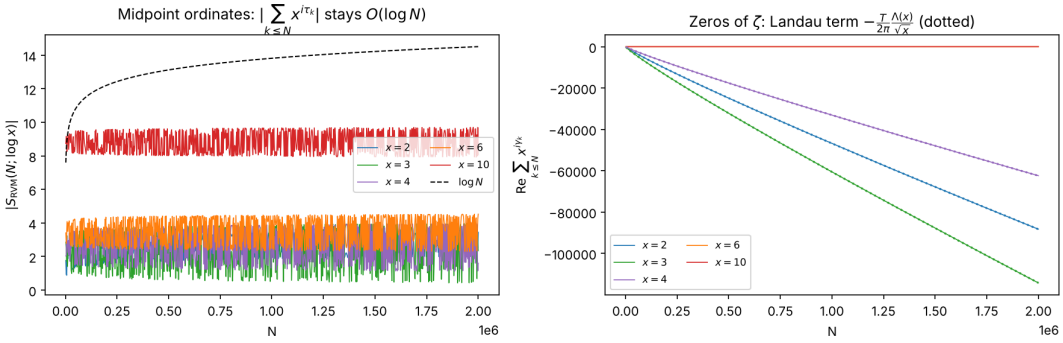



FIGURE 7. The midpoint spectrum carries no prime frequencies (left); the spectrum of $\zeta$ resonates at prime powers (right).

## 14.5. **Prime counts: $\pi$ and $\pi_{\mathrm{RVM}}$.**

**Experiment 14.6** (Prime counts and RVM primes)**.** *Object:* $J_{\mathrm{RVM}}$, $\pi_{\mathrm{RVM}}$, $c_n^{\mathrm{RVM}}$, $p_j^{\mathrm{RVM}}$ against $J$, $\pi$, the primes. *Range:* $2 \le x \le 100$ with all $N$ ordinates of each list (3000 points); $2 \le x \le 1000$ with the first $K = 17\,061$ ordinates and a smooth cut (40 000 points). *Expected:* the explicit formula with the $\gamma_k$ reproduces $\pi(x)$; $\pi_{\mathrm{RVM}}$ is continuous and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 5.5). *Observed:* away from the primes, $|\pi_\gamma - \pi| \le 0.40$ on $[2, 100]$; $\pi_{\mathrm{RVM}}(100) = 26.08$, $\pi(100) = 25$, $\mathrm{li}(100) = 30.13$; $\pi_{\mathrm{RVM}}(1000) = 169.98$, $\pi(1000) = 168$, $\mathrm{li}(1000) = 177.61$. $\pi_{\mathrm{RVM}}$ is not monotone: it carries a ripple of unit scale, with minimum grid increment $-0.25$ on $[2, 100]$. The first crossings of the levels $j = 1, \dots, 8$ are $p_j^{\mathrm{RVM}} = 2.21, 3.16, 5.19, 7.16, 11.14, 14.02, 17.26, 19.18$ against $p_j = 2, 3, 5, 7, 11, 13, 17, 19$.

Figure 8 shows $\pi$ and $\pi_{\mathrm{RVM}}$ in the usual format. The RVM count follows the prime staircase much more closely than li: replacing each $\gamma_k$ by $\tau_k$ moves every zero by less than one unit (Experiment 14.1),

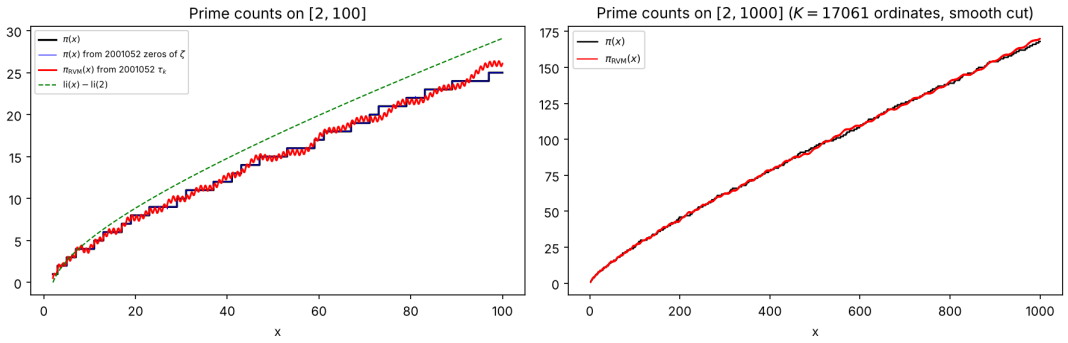



FIGURE 8. $\pi(x)$ (black staircase), the explicit formula with the zeros of $\zeta$ (blue), $\pi_{\mathrm{RVM}}(x)$ (red) and $\mathrm{li}(x) - \mathrm{li}(2)$ (green). The distance between the red and green curves is the quantity bounded in Theorem 5.5; the ratios are in Figure 10.

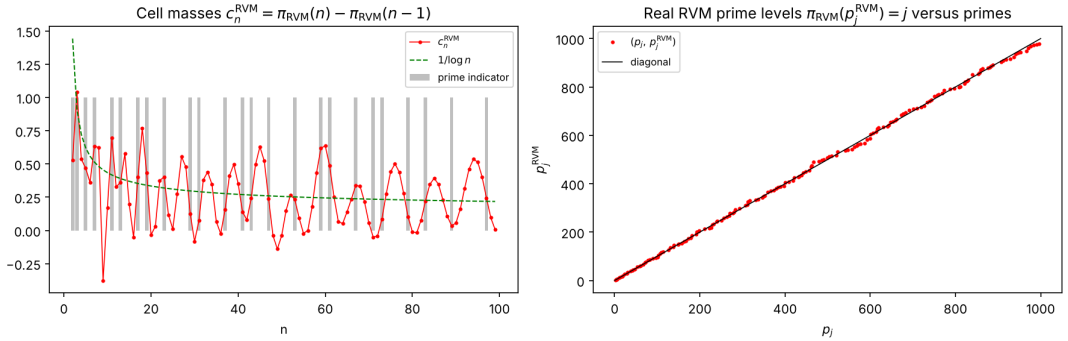



FIGURE 9. Left: cell masses $c_n^{\mathrm{RVM}}$, the prime indicator and $1/\log n$. Right: real RVM prime levels against the primes up to 1000.

and the low ordinates, which shape $\pi(x)$ for small $x$, are moved least in relative terms. The steps are replaced by a continuous ripple, so the cell masses $c_n^{\mathrm{RVM}}$ of Figure 9 are real numbers, occasionally negative, and are not a prime indicator.

**Experiment 14.7** (The three ratios and the square-root scale)**.** *Object:* $\mathrm{li}/\pi_{\mathrm{RVM}}$, $\pi/\mathrm{li}$, $\pi/\pi_{\mathrm{RVM}}$, and $(\pi_{\mathrm{RVM}} - \mathrm{li})/B$, $(\pi - \mathrm{li})/B$ with $B(x) = \sqrt{x}(\log\log x)^2/\log x$. *Range:* the grid of Experiment 14.6 on $[8, 1000]$ ($K = 17\,061$ ordinates). *Expected:* the three ratios tend to 1 (Corollary 5.6) and $(\pi_{\mathrm{RVM}} - \mathrm{li})/B$ stays bounded (Theorem 5.5). *Observed:* at $x = 100, 500, 1000$, $\mathrm{li}/\pi_{\mathrm{RVM}} = 1.155, 1.086, 1.045$, $\pi/\mathrm{li} = 0.830, 0.933, 0.946$, $\pi/\pi_{\mathrm{RVM}} = 0.959, 1.013, 0.988$; $\max|\pi_{\mathrm{RVM}} - \mathrm{li}|/B = 2.72$ on $[8, 1000]$, 1.35 on $[50, 1000]$ and 0.72 on $[500, 1000]$, while $\max|\pi - \mathrm{li}|/B = 2.45$ on $[8, 1000]$ (Figure 10).

On $[2, 1000]$ the two normalized errors in Figure 10 have the same size and sign: both counts lie below li by about $\frac{1}{2}\,\mathrm{li}(\sqrt{x})$, the $m = 2$ term of the Möbius inversion (5.20). This is why $\pi/\pi_{\mathrm{RVM}}$ is much closer to 1 than either ratio with li.

14.6. **The Mertens functions and the ternary division.** The Mertens function is the sum of the ternary sequence $\mu(n) \in \{-1, 0, +1\}$. Its values split into two independent pieces of information:

(a) *the zero set* $\{n : \mu(n) = 0\}$, the non-squarefree integers, of density $1 - 6/\pi^2 = 0.392073$;
(b) *the signs* on the squarefree integers, $\mu(n) = (-1)^{\omega(n)}$, each sign of density $3/\pi^2 = 0.303964$.

The ternary walk $\mu_h$ of (1.9)–(1.10) keeps (a) exactly, $\mu_h(n) = 0 \Leftrightarrow \mu(n) = 0$, and replaces (b): at a squarefree $n$ it steps up when $M_h(n-1)$ lies below the target $\mathcal{M}(n) = M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$ and down

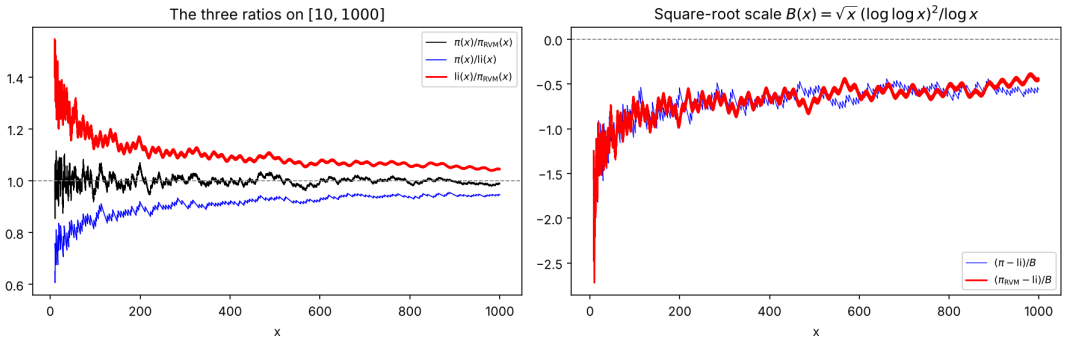



FIGURE 10. Left: the three ratios of Corollary 5.6. Right: $\pi_{\mathrm{RVM}} - \mathrm{li}$ and $\pi - \mathrm{li}$ in units of the scale $B(x)$ of Theorem 5.5. The ratio $\pi/\pi_{\mathrm{RVM}}$ (black) is the closest of the three to 1; the ratios with li approach 1 at the rate $\frac{1}{2}\,\mathrm{li}(\sqrt{x})/\,\mathrm{li}(x)$ of the bias $J - \pi$.

otherwise. Thus

$$M_h(n) = M_h(n-1) + \mu(n)^2\,\varsigma_n, \qquad |M_h(n) - \mathcal{M}(n)| \text{ small}, \tag{14.1}$$

so that $M_h$ is an integer walk with the zero set of $\mu$ that follows $M_{\mathrm{RVM}}^{\mathrm{crit}}$, in the same way as $M$ is the integer walk with the zero set of $\mu$ whose signs are $(-1)^{\omega(n)}$.

| $n$ | $\mu(n)$ | $\mu(n)^2$ | $\mathcal{M}(n)$ | $M_h(n-1)$ | $\varsigma_n$ | $\mu_h(n)$ | $M_h(n)$ | $M(n)$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 0.2925 | 0 | +1 | 1 | 1 | 1 |
| 2 | −1 | 1 | 0.4563 | 1 | −1 | −1 | 0 | 0 |
| 3 | −1 | 1 | −0.6523 | 0 | −1 | −1 | −1 | −1 |
| 4 | 0 | 0 | −0.8720 | −1 | – | 0 | −1 | −1 |
| 5 | −1 | 1 | −1.7255 | −1 | −1 | −1 | −2 | −2 |
| 6 | 1 | 1 | −1.3063 | −2 | +1 | 1 | −1 | −1 |
| 7 | −1 | 1 | −1.5931 | −1 | −1 | −1 | −2 | −2 |
| 8 | 0 | 0 | −2.4029 | −2 | – | 0 | −2 | −2 |
| 9 | 0 | 0 | −1.3292 | −2 | – | 0 | −2 | −2 |
| 10 | 1 | 1 | −1.3280 | −2 | +1 | 1 | −1 | −1 |
| 11 | −1 | 1 | −2.3272 | −1 | −1 | −1 | −2 | −2 |
| 12 | 0 | 0 | −2.2067 | −2 | – | 0 | −2 | −2 |
| 13 | −1 | 1 | −1.8468 | −2 | +1 | 1 | −1 | −3 |
| 14 | 1 | 1 | −1.9324 | −1 | −1 | −1 | −2 | −2 |
| 15 | 1 | 1 | −1.5435 | −2 | +1 | 1 | −1 | −1 |

TABLE 2. The ternary division for $n \le 15$ (the first 40 values are in Appendix B.6). The zero columns of $\mu$ and $\mu_h$ coincide; the signs of $\mu_h$ are chosen by comparing $M_h(n-1)$ with $\mathcal{M}(n)$. For $n \le 12$ the two walks agree; at $n = 13$ the target $-1.85$ lies above $M_h(12) = -2$, so $\mu_h(13) = +1$ while $\mu(13) = -1$.

**Experiment 14.8** (Mertens functions and the ternary walk)**.** *Object:* $M$, $M_{\mathrm{RVM}}^{\mathrm{crit}}$, $M_h$; the explicit formulas. *Range:* $n \le 10^6$; explicit formulas on $[2, 2000]$ with all $N$ ordinates of each list. *Expected:* the explicit formula with the zeros of $\zeta$ reproduces $M$; $M_{\mathrm{RVM}}^{\mathrm{crit}}(x) = O(x^{1/2+\varepsilon})$ (Theorem 8.12); $M_h$ tracks $M_{\mathrm{RVM}}^{\mathrm{crit}}$ with the densities of $\mu$. *Observed:*

- explicit formula for $M$ with the $\gamma_k$: root mean square error 0.0005 on $[2, 2000]$ off the integers;

- $\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}})/\log(\tau_k/2\pi) = 0.96194$ at $k = 1000$, against $\varpi_{\mathrm{RVM}} = 0.96194$ (Corollary 8.6); the values $|\zeta'(\rho_k)|/\log(\gamma_k/2\pi)$ are spread over $(0, 3)$ (Figure 13);
- spectral sums $2\sum|a|^2$: 0.029034 for $\zeta$, 0.027203 for the midpoint system;
- the integer target agrees with the full explicit formula for $M_{\mathrm{RVM}}^{\mathrm{crit}}$ to 0.019 at 100 integers $\leq 10^5$;
- $\max|M_h - \mathcal{M}| = 2.62$ on $[1, 10^6]$ and 1.65 on $[10^4, 10^6]$;
- densities of $(-1, 0, +1)$ for $\mu$: 0.303857, 0.392074, 0.304069;
- densities of $(-1, 0, +1)$ for $\mu_h$: 0.304003, 0.392074, 0.303923;
- $\mu_h(n) = \mu(n)$ at 50.03% of the squarefree $n \leq 10^6$, and $\sum_{n\leq 10^6}\mu(n)\mu_h(n)/\#\{n \leq 10^6 : \mu(n) \neq 0\} = 0.0007$: the sign sequences are uncorrelated although both walks have the same zero set and comparable size;
- $\max_{10\leq n\leq 10^6}|W(n)|/\sqrt{n} = 0.832, 0.754, 0.688$ for $W = M, M_{\mathrm{RVM}}^{\mathrm{crit}}, M_h$;
- energies $\frac{1}{N}\sum_{n\leq N} W(n)^2$ at $N = 10^6$: 14 851, 12 538, 12 538.

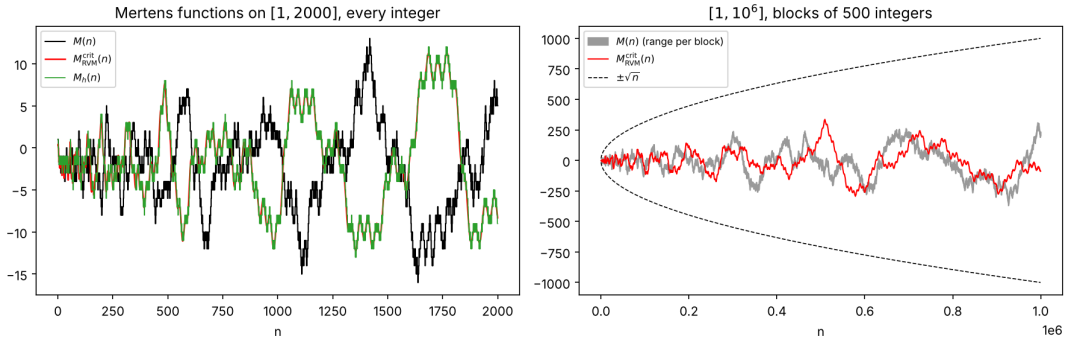



FIGURE 11. Left: $M$ (black), $M_{\mathrm{RVM}}^{\mathrm{crit}}$ (red) and $M_h$ (green) at every integer $n \leq 2000$. Right: range of $M$ and mean of $M_{\mathrm{RVM}}^{\mathrm{crit}}$ in blocks of 500 integers up to $10^6$, with $\pm\sqrt{n}$.

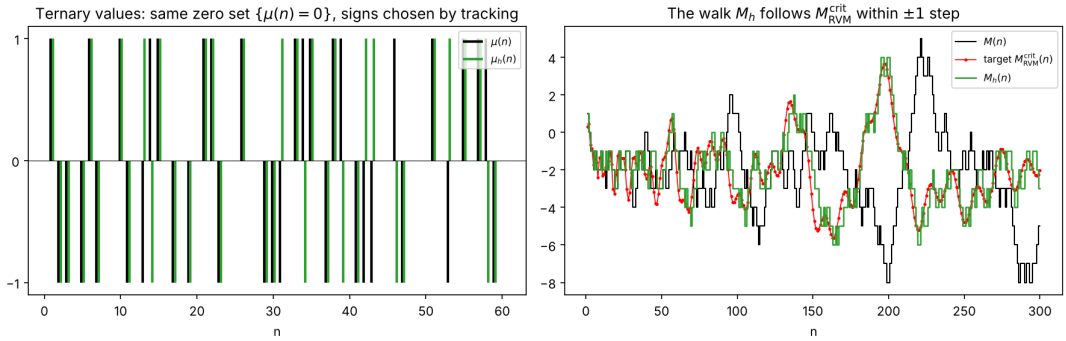



FIGURE 12. Left: $\mu(n)$ and $\mu_h(n)$ for $n \leq 60$: same zeros, different signs. Right: $M$, the target $M_{\mathrm{RVM}}^{\mathrm{crit}}$ and $M_h$ for $n \leq 300$.

Figures 11–13 summarize the Mertens side. $M_{\mathrm{RVM}}^{\mathrm{crit}}$ has the size and the visual texture of $M$, it stays inside $\pm\sqrt{n}$ as Theorem 8.12 requires, and its residues are deterministic; $M_h$ is its ternary integer version with the zero set of $\mu$. The passage from these synthetic objects to $M$ itself is not numerical: it is the identity $\zeta_{Z1} = \zeta$ of Step 3, which Experiment 14.2 is consistent with.

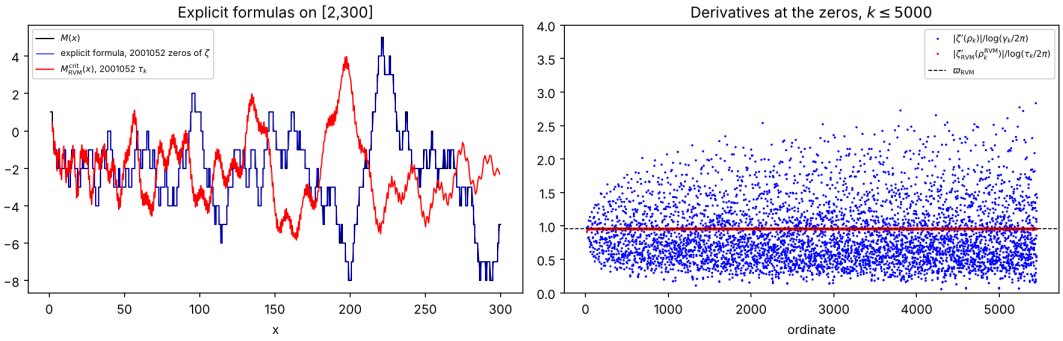



FIGURE 13. Left: $M(x)$, its explicit formula from the zeros of $\zeta$ and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x)$ from the $\tau_k$ on $[2, 300]$. Right: normalized derivatives at the zeros: erratic for $\zeta$, constant $\varpi_{\mathrm{RVM}}$ for $\zeta_{\mathrm{RVM}}$.

## 15. Conclusions and open problems

The following statements are proved:

(1) the midpoint ordinates $\tau_k$ are given by the Lambert formula (2.5) and stay within a bounded distance of the ordinates $\gamma_k$ of $\zeta$ (Lemma 2.2);
(2) $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and symmetric, and $\zeta_{\mathrm{RVM}}$ has exactly the simple zeros $\frac{1}{2} \pm i\tau_k$, the trivial zeros and a residue-one pole at 1 (Theorem 3.1);
(3) $S_{\mathrm{RVM}}(N; u) \ll 1 + \log N$ uniformly on compact positive frequency intervals, so the point centroid and $M^{\mathrm{RVM}}_H(x)$ tend to 0 (Section 4);
(4) $J_{\mathrm{RVM}}$ exists compact-uniformly and is related to $\pi_{\mathrm{RVM}}$ by exact finite Möbius inversion (Section 5);
(5) $J_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$, so $\mathrm{li}\,/\pi_{\mathrm{RVM}} \to 1$, and with the prime number theorem $\pi/\mathrm{li} \to 1$ and $\pi/\pi_{\mathrm{RVM}} \to 1$ (Theorem 5.5, Corollary 5.6);
(6) $J_{\mathrm{RVM}}$ is Lipschitz on $[2, \infty)$, $c^{\mathrm{RVM}}_n = O(1)$, and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Section 6);
(7) $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1 + w)(1 + O(1/|t|))$ on every band to the right of $\frac{1}{2}$ and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1 + \chi)(1 + O(1/|t|))$ on closed substrips of $\frac{1}{2} < \Re s < 1$ (Theorem 8.1);
(8) $1/\zeta_{\mathrm{RVM}} \to L$ on every vertical line to the right of $\frac{1}{2}$, $\zeta^{\pm 1}_{\mathrm{RVM}}$ have no absolutely convergent Dirichlet expansion, and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ (Section 8);
(9) $2\xi$ equals its Hadamard product over $\rho(1-\rho)$ exactly (Theorem 9.1), and $|\zeta| \asymp |\mathcal{Q}_{\mathrm{RVM}}|$ on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$ (Corollary 9.3);
(10) the $\lambda$-family from $\zeta_{\mathrm{RVM}}$ to $\zeta_{Z1}$ converges jointly, has simple zeros for $\lambda > 0$, an explicit generator and total multiplier, and its members differ only through $\gamma_k - \tau_k$ (Section 10);
(11) $\zeta_{Z1}$ has the functional equation, the pole, the residue and the order of $\zeta$, and $\zeta_{Z1} = \zeta \Leftrightarrow$(D)$\Leftrightarrow$RH (Theorem 11.2);
(12) $\zeta_{\mathrm{RVM}}$ is bounded above and below on every half-plane $\Re s \ge \sigma_0 > 1$ and tends to $\varpi_{\mathrm{RVM}}$ as $\Re s \to +\infty$ (Theorem 8.10); $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, so (D) is equivalent to an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s > 1$ (Theorem 11.3);
(13) no member of the $\lambda$-family, $\zeta_{Z1}$ included, can have a zero off the critical line other than the trivial zeros (Theorem 10.1);
(14) the endpoint has the value $\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_k (\frac{1}{4} + \tau_k^2)/(\frac{1}{4} + \gamma_k^2) \ge 1$ at $+\infty$, and $\zeta_{Z1} = \zeta$ if and only if $\varpi_{Z1} = 1$ (Theorems 12.1, 12.2, Corollary 12.3);

(15) the prime-power counts of $\zeta_{\mathrm{RVM}}$ and $\zeta_{Z1}$ are the Mellin transforms of their logarithms and satisfy a distributional explicit formula; this proves the Mellin–Stieltjes identification and the weighted factorization $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$, and (D) holds if and only if the prime-power count of $\zeta_{Z1}$ is constant between consecutive integers (Theorems 12.5, 12.6, 6.4, 12.7);
(16) along the family $\frac{d}{d\lambda}\log\varpi_\lambda = -\Upsilon(\lambda)$, and (D) is the identity $\int_{0^+}^{1} \Upsilon\, d\lambda = \log L$; with the verified zeros, $0 \le \log\varpi_{Z1} < 4\cdot 10^{-13}$ (Propositions 13.2, 13.3);
(17) the atomized family $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde h_\lambda}$ consists of Dirichlet series, the joining factor is explicit cell by cell and holomorphic in $\Re s > \frac12$ (in $\Re s > 0$ for $\lambda = 1$), and (D) is equivalent to bounded cell masses and vanishing cell defects at $\lambda = 0$ (Theorems 13.5, 13.8, 13.9);
(18) at the clock endpoint the joining factor is not constant, $\tilde h_1(\sigma) \to \log L$; the atomized reciprocals have Möbius coefficients $\tilde\mu_\lambda$, and the endpoint Mertens function has a Mellin form $B_{Z1}$ with $B_{Z1} = M$ if and only if (D) (Propositions 13.10, 13.11);
(19) Step 3: $\zeta_{Z1} = \zeta$ implies $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 11.8); equivalently, the synthetic bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ transfers to $M$ through $1/\zeta = F_{\mathrm{RVM}}\Pi_{\mathrm{tr}}^{-1}$ (Theorem 11.9).

The computations of Section 14 with 2 001 052 zeros are consistent with each of these statements.

*Remark* 15.1 (Open problems). (i) Prove the identity $\zeta_{Z1} = \zeta$, equivalently condition (D), equivalently an absolutely convergent Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$, equivalently the product identity (12.12), equivalently the cell identity (12.18) with the jump bound of Theorem 12.7, equivalently steps A6 and A7 of the programme in Subsection 13.7. This is the final step of the method.
(ii) Prove Hypothesis 6.7.
(iii) Determine the monotone branches of $\pi_{\mathrm{RVM}}$, on which the real prime levels $p_j^{\mathrm{RVM}}$ are defined, and explain the observed proximity $p_j^{\mathrm{RVM}} \approx p_j$ for small $j$ (Experiment 14.6).

## Appendix A. Python programs

The programs below produce every number, table and figure of Section 14. They read the two data files `data/zeross.bin` (ordinates $\gamma_k$ of the zeros of $\zeta$) and `data/zLamb.bin` (midpoint ordinates $\tau_k$), both little-endian float64 with 2 001 052 entries, and write their console output, CSV, MAT and binary files to `output/` and the figures to `figures/`. They require Python 3 with NumPy, SciPy, mpmath and Matplotlib, and are run from the directory `code/` in the order `s01`–`s07`.

A.1. **rvm_common.py.**

```
"""Common routines for the midpoint Riemann-von Mangoldt (RVM) zeta system.

Data files (little-endian float64, one ordinate per entry):
    data/zeross.bin  ordinates gamma_k of the nontrivial zeros of zeta (2,001,052 values)
    data/zLamb.bin   midpoint RVM ordinates tau_k, N_RVM(tau_k) = k - 1/2

Notation follows the paper: N_RVM is the clock, tau_k the midpoint ordinates,
gamma_k the ordinates of zeta, eta_k = tau_k - gamma_k, q(s) = s(s-1),
G(s) = s(s-1) Gamma(s/2) pi^(-s/2), zeta_lambda = P_lambda / G with ordinates
gamma_k(lambda) = (1-lambda) gamma_k + lambda tau_k.
"""
import os
import sys
import time
import numpy as np
from scipy import special

ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
DATA = os.path.join(ROOT, "data")
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
```

```
os.makedirs(FIG, exist_ok=True)

TWO_PI = 2.0 * np.pi


class Tee:
    """Write console output both to the screen and to output/<name>_console.txt."""

    def __init__(self, name):
        self.f = open(os.path.join(OUT, name + "_console.txt"), "w")
        self.t0 = time.time()

    def __call__(self, fmt, *args):
        s = fmt % args if args else fmt
        sys.stdout.write(s + "\n")
        sys.stdout.flush()
        self.f.write(s + "\n")
        self.f.flush()

    def close(self):
        self("total run time %.1f s" % (time.time() - self.t0))
        self.f.close()


def read_bin(name):
    return np.fromfile(os.path.join(DATA, name), dtype="<f8")


def load_ordinates():
    gam = read_bin("zeross.bin")
    tau = read_bin("zLamb.bin")
    n = min(gam.size, tau.size)
    return gam[:n], tau[:n]


def clock(T):
    """N_RVM(T) = T/(2 pi) log(T/(2 pi e)) + 7/8 (real or complex T, principal log)."""
    return T / TWO_PI * (np.log(T / TWO_PI) - 1.0) + 0.875


def tail_coefficients(U):
    """Midpoint-rule tail of sum_{k>N} log(1 + q/tau_k^2), N = N_RVM(U):
    returns (c1, c2, c3) with tail(q) = c1 q - c2 q^2 + c3 q^3."""
    LU = np.log(U / TWO_PI)
    c1 = (LU + 1.0) / U / TWO_PI
    c2 = (3.0 * LU + 1.0) / (9.0 * U ** 3) / 2.0 / TWO_PI
    c3 = (5.0 * LU + 1.0) / (25.0 * U ** 5) / 3.0 / TWO_PI
    return c1, c2, c3


def clock_inverse(y):
    """Solve N_RVM(U) = y by Newton iteration."""
    U = TWO_PI * max(y, 2.0) / np.log(max(y, 3.0))
    for _ in range(80):
        U -= (clock(U) - y) / (np.log(U / TWO_PI) / TWO_PI)
    return U


class Product:
    """log P(s) = sum_k log(1 + q(s)/(1/4 + o_k^2)) for an ordinate list o_k.

    Exact summation for k <= K1, power sums for K1 < k <= N, and the clock tail
    for k > N.  Accurate for |q(s)| << o_{K1}^2."""

    def __init__(self, ords, K1=100000, nterms=6):
```

```
        self.o = np.asarray(ords, dtype=float)
        self.N = self.o.size
        self.K1 = min(K1, self.N)
        A = 0.25 + self.o[self.K1:] ** 2
        self.pw = [np.sum(A ** (-j)) for j in range(1, nterms + 1)]
        self.A1 = 0.25 + self.o[: self.K1] ** 2
        self.tail = tail_coefficients(clock_inverse(float(self.N)))

    def logP(self, s):
        s = np.atleast_1d(np.asarray(s, dtype=complex))
        q = s * (s - 1.0)
        out = np.empty(s.shape, dtype=complex)
        for i, qi in enumerate(q):
            v = np.sum(np.log1p(qi / self.A1))
            for j, p in enumerate(self.pw, start=1):
                v += (-1) ** (j + 1) * qi ** j / j * p
            c1, c2, c3 = self.tail
            v += c1 * qi - c2 * qi ** 2 + c3 * qi ** 3
            out[i] = v
        return out


def logG(s):
    s = np.asarray(s, dtype=complex)
    return np.log(s) + np.log(s - 1.0) + special.loggamma(s / 2.0) - s / 2.0 * np.log(np.pi)


def zeta_from_product(prod, s):
    s = np.atleast_1d(np.asarray(s, dtype=complex))
    return np.exp(prod.logP(s) - logG(s))


def chi(s):
    s = np.asarray(s, dtype=complex)
    return np.exp((s - 0.5) * np.log(np.pi) + special.loggamma((1 - s) / 2) - special.loggamma(s / 2))


def w_transition(s):
    """w(s) = exp(-2 pi i N_RVM(tau_*)), tau_* = -i (s - 1/2)."""
    s = np.asarray(s, dtype=complex)
    tstar = -1j * (s - 0.5)
    return np.exp(-2j * np.pi * clock(tstar))


def theta_rs(t):
    t = np.asarray(t, dtype=float)
    return t / 2 * np.log(t / TWO_PI) - t / 2 - np.pi / 8 + 1 / (48 * t) + 7 / (5760 * t ** 3)


def gamma_lambda(gam, tau, lam):
    return (1.0 - lam) * gam + lam * tau


def ei_upper(w):
    """Ei on Im w != 0 along the horizontal line (paper, eq. Ei-def): Ei(w) = -E1(-w)."""
    return -special.exp1(-np.asarray(w, dtype=complex))


def mobius_sieve(X):
    mu = np.ones(X + 1, dtype=np.int8)
    mu[0] = 0
    is_comp = np.zeros(X + 1, dtype=bool)
    for p in range(2, X + 1):
        if not is_comp[p]:
            is_comp[2 * p::p] = True
```

```
            mu[p::p] *= -1
            if p * p <= X:
                mu[p * p::p * p] = 0
    return mu


def primes_upto(X):
    s = np.ones(X + 1, dtype=bool)
    s[:2] = False
    for p in range(2, int(X ** 0.5) + 1):
        if s[p]:
            s[p * p::p] = False
    return np.nonzero(s)[0]


def write_csv(name, header, cols):
    A = np.column_stack(cols)
    np.savetxt(os.path.join(OUT, name), A, delimiter=",", header=",".join(header), comments="", fmt="%.12g")


def savefig(fig, name):
    fig.savefig(os.path.join(FIG, name), dpi=150, bbox_inches="tight")
```

A.2. **s01_clock_and_family.py.**

```
"""Experiment 1: the midpoint clock, the displacements eta_k = tau_k - gamma_k,
the lambda-family zeta_lambda, and the numerical comparison zeta_Z1 versus zeta."""
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, clock, Product, zeta_from_product, gamma_lambda,
                        write_csv, savefig, OUT)
import os

say = Tee("s01_clock_and_family")
gam, tau = load_ordinates()
N = gam.size
k = np.arange(1, N + 1)
say("zeros loaded: N = %d, gamma_1 = %.9f, gamma_N = %.6f, tau_1 = %.9f, tau_N = %.6f",
    N, gam[0], gam[-1], tau[0], tau[-1])

# ---- clock check and displacements
clock_err = np.max(np.abs(clock(tau) - (k - 0.5)))
eta = tau - gam
say("max |N_RVM(tau_k) - (k - 1/2)| = %.3e", clock_err)
say("eta_k = tau_k - gamma_k:  max |eta| = %.6f (k = %d),  mean |eta| = %.6f,  rms = %.6f,  mean eta = %.3e",
    np.max(np.abs(eta)), int(np.argmax(np.abs(eta))) + 1, np.mean(np.abs(eta)),
    np.sqrt(np.mean(eta ** 2)), np.mean(eta))
blocks = [(1, 10), (11, 100), (101, 1000), (1001, 10 ** 4), (10 ** 4 + 1, 10 ** 5), (10 ** 5 + 1, 10 ** 6),
     (10 ** 6 + 1, N)]
rows = []
say("%12s %12s %12s %12s", "k from", "k to", "max|eta|", "rms eta")
for a, b in blocks:
    e = eta[a - 1:b]
    rows.append([a, b, np.max(np.abs(e)), np.sqrt(np.mean(e ** 2))])
    say("%12d %12d %12.6f %12.6f", a, b, rows[-1][2], rows[-1][3])
rows = np.array(rows)
write_csv("s01_eta_blocks.csv", ["k_from", "k_to", "max_abs_eta", "rms_eta"], [rows[:, i] for i in range(4)])
say("first ordinates:")
say("%4s %16s %16s %12s", "k", "gamma_k", "tau_k", "eta_k")
for i in range(12):
    say("%4d %16.9f %16.9f %12.6f", i + 1, gam[i], tau[i], eta[i])
```

```
# ---- zeta_Z1 versus zeta on vertical lines
PZ = Product(gam)
PR = Product(tau)
tline = np.concatenate([np.linspace(1.0, 100.0, 199), np.linspace(101.0, 2000.0, 300)])
mp.mp.dps = 20
cmp_rows = []
for sigma in (2.0, 1.25, 0.75):
    s = sigma + 1j * tline
    zz1 = zeta_from_product(PZ, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    rel = np.abs(zz1 / ze - 1.0)
    cmp_rows.append([sigma, np.max(rel), np.median(rel)])
    say("sigma = %.2f, t in [1,2000] (%d points): max |zeta_Z1/zeta - 1| = %.3e, median = %.3e",
        sigma, tline.size, np.max(rel), np.median(rel))
    write_csv("s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaZ1", "rel_err"],
              [tline, np.abs(ze), np.abs(zz1), rel])
cmp_rows = np.array(cmp_rows)

# ---- Hardy-type functions Z_lambda(t) = exp(i theta(t)) zeta_lambda(1/2+it) on the critical line
def theta_exact(t):
    return np.array([float(mp.siegeltheta(x)) for x in t])

tc = np.linspace(0.5, 60.0, 1200)
th = theta_exact(tc)
lams = [0.0, 0.25, 0.5, 0.75, 1.0]
Zl = {}
for lam in lams:
    P = PZ if lam == 0.0 else (PR if lam == 1.0 else Product(gamma_lambda(gam, tau, lam)))
    Zl[lam] = np.real(np.exp(1j * th) * zeta_from_product(P, 0.5 + 1j * tc))
Zhardy = np.array([float(mp.siegelz(x)) for x in tc])
dZ = np.max(np.abs(Zl[0.0] - Zhardy) / np.maximum(1.0, np.abs(Zhardy)))
say("critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = %.3e", dZ)
write_csv("s01_Z_lambda_critical_line.csv", ["t", "Z_hardy"] + ["Z_lambda_%.2f" % l for l in lams],
          [tc, Zhardy] + [Zl[l] for l in lams])

# ---- figures
dec = np.arange(0, N, 50)
fig, ax = plt.subplots(2, 2, figsize=(12, 8))
ax[0, 0].plot(k[dec], eta[dec], ".", ms=1.5, color="tab:blue")
ax[0, 0].axhline(0, color="k", lw=0.5)
ax[0, 0].set_xlabel("k")
ax[0, 0].set_ylabel(r"$\eta_k=\tau_k-\gamma_k$")
ax[0, 0].set_title(r"Bounded displacement, $N=%d$ zeros" % N)
ax[0, 1].hist(eta, bins=200, density=True, color="tab:blue")
ax[0, 1].set_xlabel(r"$\eta_k$")
ax[0, 1].set_title(r"Distribution of $\eta_k$ (all $k\leq N$)")
lam = np.linspace(0, 1, 101)
for kk in range(30):
    gl = (1 - lam) * gam[kk] + lam * tau[kk]
    ax[1, 0].plot(lam, clock(gl) - (kk + 0.5), "-", lw=1)
ax[1, 0].axhline(0, color="k", lw=0.5)
ax[1, 0].set_xlabel(r"$\lambda$")
ax[1, 0].set_ylabel(r"$\mathcal{N}_{\rm RVM}(\gamma_k(\lambda))-(k-\frac{1}{2})$")
ax[1, 0].set_title(r"Moving ordinates in clock units, $k\leq30$ ($\lambda=0$: $\zeta_{Z1}$, $\lambda=1$: $\
    zeta_{\rm RVM}$)", fontsize=9)
ax[1, 1].plot(tc, Zhardy, "k-", lw=2.2, label=r"Hardy $Z(t)$ of $\zeta$")
cols = ["tab:red", "tab:orange", "tab:green", "tab:cyan", "tab:blue"]
for lam, c in zip(lams, cols):
    ax[1, 1].plot(tc, Zl[lam], "-", color=c, lw=1.0, label=r"$Z_\lambda$, $\lambda=%.2f$" % lam)
ax[1, 1].axhline(0, color="k", lw=0.5)
ax[1, 1].set_xlabel("t")
ax[1, 1].set_title(r"$e^{i\vartheta(t)}\zeta_\lambda(\frac{1}{2}+it)$ along the family")
ax[1, 1].legend(fontsize=7, ncol=2)
fig.tight_layout()
```

```
savefig(fig, "fig01_clock_family.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
for sigma, c in zip((2.0, 1.25, 0.75), ("tab:blue", "tab:green", "tab:red")):
    d = np.loadtxt(os.path.join(OUT, "s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma), delimiter=",", skiprows=1)
    ax[0].plot(d[:, 0], d[:, 3], ".", ms=2, color=c, label=r"$\sigma=%.2f$" % sigma)
ax[0].set_xlabel("t")
ax[0].set_ylabel(r"$|\zeta_{Z1}/\zeta-1|$")
ax[0].set_title(r"$\zeta_{Z1}$ from %d ordinates versus $\zeta$" % N)
ax[0].legend()
ax[1].plot(tc, Zhardy, "k-", lw=2.5, label=r"$Z(t)$ (mpmath)")
ax[1].plot(tc, Z1[0.0], "r--", lw=1.0, label=r"$Z_{Z1}(t)$ (product)")
ax[1].set_xlabel("t")
ax[1].legend()
ax[1].set_title(r"Critical line: $\lambda=0$ endpoint and Hardy $Z$")
fig.tight_layout()
savefig(fig, "fig02_zetaZ1_vs_zeta.png")

sio.savemat(os.path.join(OUT, "s01_results.mat"), {
    "N": N, "clock_err": clock_err, "eta_blocks": rows, "zetaZ1_cmp": cmp_rows,
    "t_critical": tc, "Z_hardy": Zhardy, "Z_lambda": np.array([Z1[l] for l in lams]), "lambdas": np.array(lams
    )})
say.close()

# ---- zoom: how the smooth clock places the midpoint ordinates next to the zeros
say2 = Tee("s01b_clock_zoom")
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
T = np.linspace(9.0, 52.0, 4000)
ax[0].step(T, np.searchsorted(gam, T, side="right"), where="post", color="k", lw=1.5, label=r"$N_\zeta(T)$")
ax[0].plot(T, clock(T), "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
for i in range(9):
    ax[0].plot([gam[i], gam[i]], [0, i + 1], "k:", lw=0.8)
ax[0].plot(tau[:9], np.arange(9) + 0.5, "ro", ms=5, label=r"$\tau_k$: $\mathcal{N}_{\rm RVM}(\tau_k)=k-\frac
    {1}{2}$")
ax[0].plot(gam[:9], np.arange(9) + 0.5, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$")
ax[0].set_xlabel("T")
ax[0].set_title("Clock and staircase at the first zeros")
ax[0].legend(loc="upper left", fontsize=8)
kz = np.arange(N - 12, N)
T3 = np.linspace(gam[kz[0]] - 0.3, gam[kz[-1]] + 0.3, 6000)
ax[1].step(T3 - 1132480, np.searchsorted(gam, T3, side="right") - 2001000, where="post", color="k", lw=1.5,
    label=r"$N_\zeta(T)$")
ax[1].plot(T3 - 1132480, clock(T3) - 2001000, "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
ax[1].plot(tau[kz] - 1132480, kz + 0.5 - 2001000, "ro", ms=5, label=r"$\tau_k$")
ax[1].plot(gam[kz] - 1132480, kz + 0.5 - 2001000, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$
    ")
ax[1].set_xlabel(r"$T-1132480$")
ax[1].set_ylabel(r"count $-\,2001000$")
ax[1].set_title(r"Zoom at the last 12 zeros of the file (mean spacing %.3f)" % (2 * np.pi / np.log(gam[-1] /
    (2 * np.pi))))
ax[1].legend(loc="upper left", fontsize=8)
fig.tight_layout()
savefig(fig, "fig00_clock_zoom.png")
fig, ax = plt.subplots(1, 1, figsize=(12, 3))
T2 = np.linspace(9.0, 200.0, 20000)
ax.plot(T2, np.searchsorted(gam, T2, side="right") - clock(T2), "k-", lw=0.8)
ax.axhline(0, color="r", lw=0.8)
ax.set_xlabel("T")
ax.set_ylabel(r"$S(T)+O(1/T)$")
ax.set_title(r"$N_\zeta(T)-\mathcal{N}_{\rm RVM}(T)$: the remainder of the Riemann-von Mangoldt formula (jump
    $+1$ at each $\gamma_k$)")
fig.tight_layout()
savefig(fig, "fig00b_remainder.png")
say2("zoom table near the top of the file:")
say2("%9s %18s %18s %10s %10s", "k", "gamma_k", "tau_k", "eta_k", "N*-N_RVM")
```

```
for i in kz:
    say2("%9d %18.6f %18.6f %10.6f %10.6f", i + 1, gam[i], tau[i], tau[i] - gam[i],
         np.searchsorted(gam, gam[i], side="right") - 0.5 - clock(gam[i]))
say2.close()
```

A.3. **s02_transition.py.**

```
"""Experiment 2: the transition theorem zeta_RVM = varpi (1 + w)(1 + O(1/t)),
the constant varpi_RVM, the vertical limit of 1/zeta_RVM, and zeta_RVM versus zeta."""
import os
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, zeta_from_product, chi, w_transition,
                        write_csv, savefig, OUT)

say = Tee("s02_transition")
gam, tau = load_ordinates()
PR = Product(tau)

# ---- varpi_RVM by Richardson extrapolation on Re s = 2
Ts = 1.0e4 * np.array([0.25, 0.5, 1.0])
v = np.array([zeta_from_product(PR, 2 + 1j * T)[0] / (1 + w_transition(2 + 1j * T)) for T in Ts])
A = np.column_stack([np.ones(3), 1 / Ts, 1 / Ts ** 2])
c = np.linalg.solve(A.astype(complex), v)
varpi = c[0].real
L = 1.0 / varpi
say("varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000) = %.10f %+.2e i", c[0].real, c[0].imag)
say("L = 1/varpi_RVM = %.10f", L)

# ---- relative error of the transition formula
tt = np.exp(np.linspace(np.log(10.0), np.log(1.0e4), 160))
rows = [tt]
hdr = ["t"]
fits = []
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    zr = zeta_from_product(PR, s)
    err_w = np.abs(zr / (varpi * (1 + w_transition(s))) - 1)
    rows.append(err_w)
    hdr.append("err_w_sigmap%.2f" % sp)
    sel = tt > 100
    slope = np.polyfit(np.log(tt[sel]), np.log(err_w[sel]), 1)[0]
    fits.append([sp, slope, np.max(err_w[sel] * tt[sel])])
    say("sigma' = %.2f: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = %.3f, max t*err = %.4f",
        sp, slope, np.max(err_w[sel] * tt[sel]))
    if sp < 0.5:
        err_c = np.abs(zr / (varpi * (1 + chi(s))) - 1)
        rows.append(err_c)
        hdr.append("err_chi_sigmap%.2f" % sp)
        say("sigma' = %.2f: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = %.4f",
            sp, np.max(err_c[sel] * tt[sel]))
    wmod = np.abs(w_transition(s))
    rows.append(wmod)
    hdr.append("abs_w_sigmap%.2f" % sp)
    Fv = 1 / zr
    rows.append(np.abs(Fv - L))
    hdr.append("abs_F_minus_L_sigmap%.2f" % sp)
    say("          |w| * t^sigma' in [%.4f, %.4f];  |1/zeta_RVM - L| at t = 1e4: %.3e",
        np.min(wmod * tt ** sp), np.max(wmod * tt ** sp), np.abs(Fv[-1] - L))
write_csv("s02_transition_errors.csv", hdr, rows)
```

```
# ---- zeta_RVM versus zeta on two vertical lines and on the critical line
mp.mp.dps = 20
tl = np.linspace(1.0, 120.0, 900)
cmp = {}
for sigma in (2.0, 0.75):
    s = sigma + 1j * tl
    zr = zeta_from_product(PR, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    cmp[sigma] = (zr, ze)
    write_csv("s02_zetaRVM_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaRVM", "
     abs_varpi_1_plus_w"],
              [tl, np.abs(ze), np.abs(zr), np.abs(varpi * (1 + w_transition(s)))])

fig, ax = plt.subplots(2, 2, figsize=(12, 8))
cols = {0.25: "tab:red", 0.5: "tab:green", 1.5: "tab:blue"}
j = 1
for sp in (0.25, 0.5, 1.5):
    ax[0, 0].plot(tt, tt * rows[j], "-", color=cols[sp], label=r"$\sigma'=%.2f$" % sp)
    j += 4 if sp < 0.5 else 3
ax[0, 0].set_xlabel("t")
ax[0, 0].set_ylabel(r"$t\,|\zeta_{\rm RVM}/(\varpi_{\rm RVM}(1+w))-1|$")
ax[0, 0].set_title(r"Transition theorem: error times $t$ stays bounded, $\varpi_{\rm RVM}=%.10f$" % varpi,
     fontsize=9)
ax[0, 0].legend()
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    ax[0, 1].plot(tt, np.abs(1 / zeta_from_product(PR, s) - L) * tt ** min(sp, 1.0), "-", color=cols[sp],
     label=r"$\sigma'=%.2f$" % sp)
ax[0, 1].set_xlabel("t")
ax[0, 1].set_ylabel(r"$t^{\min(\sigma',1)}\,|1/\zeta_{\rm RVM}(\sigma+it)-L|$")
ax[0, 1].set_title(r"Vertical limit $L=1/\varpi_{\rm RVM}$, rescaled error", fontsize=9)
ax[0, 1].legend()
for a, sigma in zip((ax[1, 0], ax[1, 1]), (2.0, 0.75)):
    zr, ze = cmp[sigma]
    a.plot(tl, np.abs(ze), "k-", lw=1.2, label=r"$|\zeta(\sigma+it)|$")
    a.plot(tl, np.abs(zr), "r-", lw=1.2, label=r"$|\zeta_{\rm RVM}(\sigma+it)|$")
    a.set_xlabel("t")
    a.set_title(r"$\sigma=%.2f$" % sigma)
    a.legend()
fig.tight_layout()
savefig(fig, "fig03_transition.png")

sio.savemat(os.path.join(OUT, "s02_results.mat"), {"varpi": varpi, "L": L, "richardson": c,
                                                   "slopes": np.array(fits), "t": tt})
say.close()
```

A.4. **s03_exponential_sums.py.**

```
"""Experiment 3: exponential sums S(N;u) = sum_{k<=N} exp(i u o_k) over the midpoint
ordinates tau_k and over the ordinates gamma_k of zeta, and the height transforms M_H(x)."""
import os
import numpy as np
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, write_csv, savefig, OUT

say = Tee("s03_exponential_sums")
gam, tau = load_ordinates()
N = gam.size
step = 2000
Nk = np.arange(step, N + 1, step)
xs = [2.0, 3.0, 4.0, 6.0, 10.0]
Lam = {2.0: np.log(2), 3.0: np.log(3), 4.0: np.log(2), 6.0: 0.0, 10.0: 0.0}
```

```
res = {}
say("%6s %14s %14s %14s %16s %14s" % ("x", "max|S_RVM|", "max|S_RVM|/log", "|S_zeta(N)|",
                                      "Landau -T/2pi*L/sqrt(x)", "|M_H^RVM(x)|"))
for x in xs:
    u = np.log(x)
    SR = np.cumsum(np.exp(1j * u * tau))[Nk - 1]
    SZ = np.cumsum(np.exp(1j * u * gam))[Nk - 1]
    res[x] = (SR, SZ)
    landau = -gam[-1] / (2 * np.pi) * Lam[x] / np.sqrt(x)
    say("%6.1f %14.4f %14.4f %14.2f %16.2f %14.3e" % (x, np.max(np.abs(SR)), np.max(np.abs(SR) / np.log(Nk)),
                                                      np.abs(SZ[-1]), landau, np.abs(SR[-1]) / N))
    write_csv("s03_expsum_x%g.csv" % x, ["N", "re_S_RVM", "im_S_RVM", "re_S_zeta", "im_S_zeta"],
              [Nk, SR.real, SR.imag, SZ.real, SZ.imag])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
cols = {2.0: "tab:blue", 3.0: "tab:green", 4.0: "tab:purple", 6.0: "tab:orange", 10.0: "tab:red"}
for x in xs:
    ax[0].plot(Nk, np.abs(res[x][0]), "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
ax[0].plot(Nk, np.log(Nk), "k--", lw=1, label=r"$\log N$")
ax[0].set_xlabel("N")
ax[0].set_ylabel(r"$|S_{\rm RVM}(N;\log x)|$")
ax[0].set_title(r"Midpoint ordinates: $|\sum_{k\leq N}x^{i\tau_k}|$ stays $O(\log N)$")
ax[0].legend(fontsize=8, ncol=2)
for x in xs:
    ax[1].plot(Nk, res[x][1].real, "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
T = np.interp(Nk, np.arange(1, N + 1), gam)
for x in (2.0, 3.0, 4.0):
    ax[1].plot(Nk, -T / (2 * np.pi) * Lam[x] / np.sqrt(x), ":", color=cols[x], lw=1.5)
ax[1].set_xlabel("N")
ax[1].set_ylabel(r"$\mathrm{Re}\sum_{k\leq N}x^{i\gamma_k}$")
ax[1].set_title(r"Zeros of $\zeta$: Landau term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$ (dotted)")
ax[1].legend(fontsize=8, ncol=2)
fig.tight_layout()
savefig(fig, "fig04_exponential_sums.png")
sio.savemat(os.path.join(OUT, "s03_results.mat"), {"N": Nk, "x": np.array(xs),
                                                   "S_RVM": np.array([res[x][0] for x in xs]),
                                                   "S_zeta": np.array([res[x][1] for x in xs])})
say.close()
```

A.5. **s04_primes.py.**

```
"""Experiment 4: the prime-power count J_RVM, the synthetic prime count pi_RVM, the cell
masses c_n^RVM and the real RVM prime levels p_j^RVM, compared with J, pi(x) and the primes.

J_o(x) = li(x) - 2 Re sum_k Ei((1/2 + i o_k) log x) - log 2 + int_x^inf dt/(t(t^2-1) log t)
for an ordinate list o_k (o = tau: J_RVM; o = gamma: Riemann's explicit formula for J),
pi_o(x) = sum_{m <= log2 x} mu(m)/m J_o(x^(1/m)).
The zero sums are evaluated on a uniform grid in u = log x by the recurrence
exp(i o_k (u + du)) = exp(i o_k u) exp(i o_k du); Ei is exact (scipy exp1) for k <= 300 and
uses Ei(w) = e^w (1/w + 1/w^2 + 2/w^3 + 6/w^4) for k > 300, where |w| > 400."""
import os
import numpy as np
import scipy.io as sio
from scipy import special, integrate
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, ei_upper, mobius_sieve, primes_upto, write_csv,
                        savefig, clock_inverse, OUT)

say = Tee("s04_primes")
gam, tau = load_ordinates()
KE = 300
```

```
def zero_sum_ei(ords, u, K=None, taper=None):
    """sum_k Ei((1/2 + i o_k) u) on a uniform grid u, k <= K, optional weights."""
    o = ords if K is None else ords[:K]
    wts = np.ones(o.size) if taper is None else taper
    rho = 0.5 + 1j * o
    out = np.zeros(u.size, dtype=complex)
    # exact part
    for i in range(0, KE):
        out += wts[i] * ei_upper(rho[i] * u)
    # asymptotic part by recurrence
    r = rho[KE:]
    b = [wts[KE:] / r ** m for m in (1, 2, 3, 4)]
    du = u[1] - u[0]
    stepf = np.exp(1j * o[KE:] * du)
    for j in range(u.size):
        if j % 200 == 0:
            E = np.exp(1j * o[KE:] * u[j])
        A = [E @ bm for bm in b]
        uj = u[j]
        out[j] += np.exp(uj / 2) * (A[0] / uj + A[1] / uj ** 2 + 2 * A[2] / uj ** 3 + 6 * A[3] / uj ** 4)
        E = E * stepf
    return out


def tail_integral(x):
    return integrate.quad(lambda t: 1.0 / (t * (t * t - 1.0) * np.log(t)), x, np.inf, limit=200)[0]


def J_from(ords, u, K=None, taper=None):
    x = np.exp(u)
    S = zero_sum_ei(ords, u, K, taper)
    ti = np.array([tail_integral(v) for v in x])
    return special.expi(u) - 2 * S.real - np.log(2.0) + ti


def pi_from_J(u, J):
    mu = mobius_sieve(64)
    out = np.zeros(u.size)
    for i, ui in enumerate(u):
        m = 1
        while ui / m >= np.log(2.0) - 1e-12:
            if mu[m] != 0:
                out[i] += mu[m] / m * np.interp(ui / m, u, J)
            m += 1
    return out


def exact_J_pi(x):
    P = primes_upto(int(x.max()) + 1)
    pi = np.searchsorted(P, x, side="right").astype(float)
    J = np.zeros(x.size)
    for m in range(1, 40):
        J += np.searchsorted(P, x ** (1.0 / m) + 1e-12, side="right") / m
    return J, pi


def run(xmax, npts, K, label):
    u = np.linspace(np.log(2.0), np.log(xmax), npts)
    if K is None:
        tg = tr = None
        Kg = Kr = None
    else:
        Kg = Kr = K
        v = np.arange(K) / K
        tg = tr = np.where(v < 0.6, 1.0, np.cos(np.pi / 2 * (v - 0.6) / 0.4) ** 2)
```

```
    JR = J_from(tau, u, Kr, tr)
    JG = J_from(gam, u, Kg, tg)
    PR = pi_from_J(u, JR)
    PG = pi_from_J(u, JG)
    x = np.exp(u)
    Je, Pe = exact_J_pi(x)
    write_csv("s04_prime_counts_%s.csv" % label, ["x", "J_exact", "J_from_zeta_zeros", "J_RVM",
                                                  "pi_exact", "pi_from_zeta_zeros", "pi_RVM", "li"],
              [x, Je, JG, JR, Pe, PG, PR, special.expi(u)])
    return x, Je, JG, JR, Pe, PG, PR


# x <= 100 with all zeros; x <= 1000 with a smooth cut at K zeros
x1, Je1, JG1, JR1, Pe1, PG1, PR1 = run(100.0, 3000, None, "x100_allzeros")
say("x <= 100, all %d zeros of each list, %d grid points:" % (gam.size, x1.size))
say("  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): %.4f",
    np.max(np.abs(PG1 - Pe1)[np.min(np.abs(x1[:, None] - primes_upto(101)[None, :]), axis=1) > 0.2]))
say("  pi_RVM(100) = %.4f, pi(100) = %d, li(100) = %.4f, J_RVM(100) = %.4f, J(100) = %.4f",
    PR1[-1], Pe1[-1], special.expi(np.log(100.0)), JR1[-1], Je1[-1])
dPR = np.diff(PR1)
say("  pi_RVM monotone on [2,100]: %s (min increment on the grid %.3e)", bool(np.all(dPR > 0)), dPR.min())

tauK = 2 * np.pi * 1000.0 * 2.5
K = int(np.searchsorted(tau, tauK))
x2, Je2, JG2, JR2, Pe2, PG2, PR2 = run(1000.0, 40000, K, "x1000_K%d" % K)
say("x <= 1000, smooth cut at K = %d zeros (tau_K = %.1f), %d grid points:" % (K, tau[K - 1], x2.size))
say("  pi_RVM(1000) = %.4f, pi(1000) = %d, li(1000) = %.4f", PR2[-1], Pe2[-1], special.expi(np.log(1000.0)))

# cell masses and real prime levels from the fine x<=1000 run
n = np.arange(2, 1001)
pin = np.interp(n, x2, PR2)
c = np.diff(np.concatenate([[0.0], pin]))
P = primes_upto(1000)
isp = np.isin(n, P).astype(float)
say("cell masses c_n = pi_RVM(n) - pi_RVM(n-1):")
say("%5s %10s %10s %8s" % ("n", "c_n", "1/log n", "prime"))
for i in range(0, 40):
    say("%5d %10.5f %10.5f %8d" % (n[i], c[i], 1 / np.log(n[i]), isp[i]))
say("sum_{n<=1000} c_n = %.4f,  pi(1000) = %d", c.sum(), len(P))
levels = np.arange(1, int(np.floor(PR2.max())) + 1)
pj = np.array([x2[np.argmax(PR2 >= j)] for j in levels])
say("real RVM prime levels p_j^RVM (first crossing of level j) versus p_j:")
say("%5s %12s %8s %10s" % ("j", "p_j^RVM", "p_j", "diff"))
for j in range(min(30, levels.size)):
    say("%5d %12.4f %8d %10.4f" % (levels[j], pj[j], P[j], pj[j] - P[j]))
write_csv("s04_cell_masses.csv", ["n", "c_n_RVM", "inv_log_n", "is_prime"], [n, c, 1 / np.log(n), isp])
m = min(levels.size, P.size)
write_csv("s04_rvm_prime_levels.csv", ["j", "p_j_RVM", "p_j"], [levels[:m], pj[:m], P[:m]])

# ---- figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].step(x1, Pe1, where="post", color="k", lw=1.6, label=r"$\pi(x)$")
ax[0].plot(x1, PG1, "b-", lw=0.6, label=r"$\pi(x)$ from %d zeros of $\zeta$" % gam.size)
ax[0].plot(x1, PR1, "r-", lw=1.4, label=r"$\pi_{\rm RVM}(x)$ from %d $\tau_k$" % tau.size)
ax[0].plot(x1, special.expi(np.log(x1)) - special.expi(np.log(2.0)), "g--", lw=1, label=r"$\mathrm{li}(x)-\
    mathrm{li}(2)$")
ax[0].set_xlabel("x")
ax[0].set_title(r"Prime counts on $[2,100]$")
ax[0].legend(fontsize=7)
ax[1].step(x2, Pe2, where="post", color="k", lw=1.2, label=r"$\pi(x)$")
ax[1].plot(x2, PR2, "r-", lw=1.0, label=r"$\pi_{\rm RVM}(x)$")
ax[1].set_xlabel("x")
ax[1].set_title(r"Prime counts on $[2,1000]$ ($K=%d$ ordinates, smooth cut)" % K)
ax[1].legend(fontsize=8)
fig.tight_layout()
```

```
savefig(fig, "fig05_prime_counts.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
nn = n[:98]
ax[0].bar(nn, isp[:98], width=0.8, color="0.75", label="prime indicator")
ax[0].plot(nn, c[:98], "r.-", lw=0.8, ms=4, label=r"$c_n^{\rm RVM}$")
ax[0].plot(nn, 1 / np.log(nn), "g--", lw=1, label=r"$1/\log n$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Cell masses $c_n^{\rm RVM}=\pi_{\rm RVM}(n)-\pi_{\rm RVM}(n-1)$")
ax[0].legend(fontsize=8)
ax[1].plot(P[:m], pj[:m], "r.", ms=4, label=r"$(p_j,\,p_j^{\rm RVM})$")
ax[1].plot([0, 1000], [0, 1000], "k-", lw=0.8, label="diagonal")
ax[1].set_xlabel(r"$p_j$")
ax[1].set_ylabel(r"$p_j^{\rm RVM}$")
ax[1].set_title(r"Real RVM prime levels $\pi_{\rm RVM}(p_j^{\rm RVM})=j$ versus primes")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig06_rvm_primes.png")

sio.savemat(os.path.join(OUT, "s04_results.mat"), {"x100": x1, "pi100": Pe1, "piZeta100": PG1, "piRVM100": PR1,

                                                   "x1000": x2, "pi1000": Pe2, "piRVM1000": PR2,
                                                   "c_n": c, "p_RVM": pj, "K": K})
say.close()
```

A.6. **s05_mertens.py.**

```
"""Experiment 5: the Mertens function M, the synthetic Mertens function M_RVM^crit and the
ternary walk M_h with values in {-1,0,+1} on the squarefree support.

Explicit formulas (simple zeros, residues a = 1/(rho zeta'(rho))):
  M_zeta(x) = -2 + 2 Re sum_k a_k x^(rho_k) + trivial terms        (zeros of zeta)
  M_RVM(x)  = -2 + 2 Re sum_k b_k x^(1/2 + i tau_k) + trivial terms (b_k = 1/(rho_k zeta_RVM'(rho_k)))
Ternary walk: mu_h(n) = mu(n)^2 * sign, sign = +1 if M_h(n-1) < target(n) else -1."""
import os
import math
import numpy as np
import mpmath as mp
import scipy.io as sio
from scipy.interpolate import PchipInterpolator
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, logG, clock_inverse, tail_coefficients,
                        theta_rs, mobius_sieve, write_csv, savefig, OUT)

say = Tee("s05_mertens")
gam, tau = load_ordinates()
Nz = gam.size
XMAX = 10 ** 6
K0 = 1000

# ------------------------------------------------------------------ RVM residues
c1, c2, c3 = tail_coefficients(clock_inverse(float(Nz)))
t2 = tau ** 2
zpR = np.zeros(K0, dtype=complex)
for k in range(K0):
    tk = tau[k]
    fac = (t2 - tk ** 2) / (t2 + 0.25)
    fac[k] = 1.0
    q = -0.25 - tk ** 2
    lmod = np.sum(np.log(np.abs(fac))) + c1 * q - c2 * q ** 2 + c3 * q ** 3
    rho = 0.5 + 1j * tk
    lP = np.log(2j * tk / (0.25 + tk ** 2)) + lmod
    zpR[k] = (-1) ** k * np.exp(lP - logG(rho))
```

```
ratio = zpR / np.log(tau[:K0] / (2 * np.pi))
PR = Product(tau)
Ts = 1.0e4 * np.array([0.25, 0.5, 1.0])
from rvm_common import zeta_from_product, w_transition
v = np.array([zeta_from_product(PR, 2 + 1j * T)[0] / (1 + w_transition(2 + 1j * T)) for T in Ts])
varpi = np.linalg.solve(np.column_stack([np.ones(3), 1 / Ts, 1 / Ts ** 2]).astype(complex), v)[0].real
sel = np.arange(K0 // 2, K0)
cfit = np.mean((ratio[sel] / varpi - 1) * tau[sel])
say("zeta_RVM'(rho_k) exact for k <= %d; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = %d: %.10f %+.2e i",
    K0, K0, ratio[-1].real, ratio[-1].imag)
say("varpi_RVM = %.10f; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = %.4f %+.4f i",
    varpi, cfit.real, cfit.imag)
zR = varpi * np.log(tau / (2 * np.pi)) * (1 + cfit / tau)
zR[:K0] = zpR
bR = 1.0 / ((0.5 + 1j * tau) * zR)

# ------------------------------------------------------------- residues of zeta
def rs_Z(t):
    a = np.sqrt(t / (2 * np.pi))
    Nn = np.floor(a).astype(int)
    p = a - Nn
    nmax = Nn.max()
    n = np.arange(1, nmax + 1)
    ph = theta_rs(t)[:, None] - t[:, None] * np.log(n)[None, :]
    mask = n[None, :] <= Nn[:, None]
    Z = 2 * np.sum(np.cos(ph) * mask / np.sqrt(n)[None, :], axis=1)
    cc = np.cos(2 * np.pi * p)
    C0 = np.cos(2 * np.pi * (p ** 2 - p - 1.0 / 16)) / cc
    return Z + (-1.0) ** (Nn - 1) * (t / (2 * np.pi)) ** (-0.25) * C0


mp.mp.dps = 20
KM = 1000
zG = np.zeros(Nz, dtype=complex)
for k in range(KM + 50):
    zG[k] = complex(mp.zeta(mp.mpc(0.5, gam[k]), 1, 1))
h = 1e-5
for i0 in range(KM, Nz, 4000):
    i1 = min(Nz, i0 + 4000)
    t = gam[i0:i1]
    Zp = (rs_Z(t + h) - rs_Z(t - h)) / (2 * h)
    val = -1j * np.exp(-1j * theta_rs(t)) * Zp
    if i0 == KM:
        chk = np.max(np.abs(val[:50] / zG[KM:KM + 50] - 1))
        say("Riemann-Siegel zeta'(rho) versus mpmath at k = %d..%d: max relative difference %.2e",
            KM + 1, KM + 50, chk)
        zG[KM + 50:i1] = val[50:]
    else:
        zG[i0:i1] = val
aG = 1.0 / ((0.5 + 1j * gam) * zG)
S2G = 2 * np.sum(np.abs(aG) ** 2)
S2R = 2 * np.sum(np.abs(bR) ** 2)
say("median |zeta'(rho)| = %.4f;  spectral sums 2 sum|a|^2: zeta %.6f, RVM %.6f", np.median(np.abs(zG)), S2G,
    S2R)


def trivial_zeta(x, n=6):
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        v += (-1) ** (k - 1) * (2 * np.pi / x) ** (2 * k) / (math.factorial(2 * k) * k * float(mp.zeta(2 * k +
     1)))
    return v


def trivial_rvm(x, n=6):
```

```
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        s = -2.0 * k
        lP = np.real(PR.logP(s)[0])
        v -= x ** (-2 * k) * (2 * k + 1) * np.pi ** k * 2 * (-1) ** k / (math.factorial(k) * np.exp(lP))
    return v


def explicit_uniform(ords, wts, u):
    """2 Re sqrt(x) sum_k w_k exp(i o_k u) on a uniform u grid (recurrence)."""
    out = np.zeros(u.size)
    st = np.exp(1j * ords * (u[1] - u[0]))
    for j in range(u.size):
        if j % 200 == 0:
            E = np.exp(1j * ords * u[j])
        out[j] = 2 * np.exp(u[j] / 2) * np.real(E @ wts)
        E = E * st
    return out


# ---------------------------------------------------------------- exact M by sieve
mu = mobius_sieve(XMAX)
M = np.cumsum(mu.astype(np.int64))
sqf = mu != 0
say("M(10^6) = %d;  squarefree density Q(10^6)/10^6 = %.6f (6/pi^2 = %.6f)", M[XMAX], sqf[1:].mean(), 6 / np.
    pi ** 2)

# ---------------------------------------------------------------- explicit formulas on [2, 2000]
u = np.linspace(np.log(2.0), np.log(2000.0), 2500)
xg = np.exp(u)
Mz = -2 + explicit_uniform(gam, aG, u) + trivial_zeta(xg)
Mc = -2 + explicit_uniform(tau, bR, u) + trivial_rvm(xg)
Mfl = M[np.floor(xg).astype(int)]
off = np.abs(xg - np.round(xg)) > 0.05
say("explicit formula with %d zeros of zeta on [2,2000]: rms(M_zeta - M) = %.4f off integers (|x-n|>0.05)",
    Nz, np.sqrt(np.mean((Mz - Mfl)[off] ** 2)))
write_csv("s05_explicit_formulas.csv", ["x", "M_floor", "M_zeta_explicit", "M_RVM_explicit"], [xg, Mfl, Mz, Mc
    ])

# ---------------------------------------------------------------- integer target for M_RVM^crit
def target_grid(X, div=24):
    x = [1.0]
    while x[-1] < X:
        x.append(min(X, x[-1] + max(1.0, np.floor(np.sqrt(x[-1]) / div))))
    return np.array(x)


def tapered(x, ords, a, C1=2.0, C2=4.0):
    S = np.zeros(x.size, dtype=complex)
    lx = np.log(x)
    i0 = 0
    while i0 < x.size:
        i1 = min(x.size, i0 + 200)
        K = max(1, int(np.searchsorted(ords, C2 * 2 * np.pi * np.sqrt(x[i1 - 1]))))
        uu = ords[None, :K] / (2 * np.pi * np.sqrt(x[i0:i1, None]))
        w = np.where(uu <= C1, 1.0, np.where(uu >= C2, 0.0, np.cos(np.pi / 2 * (uu - C1) / (C2 - C1)) ** 2))
        E = np.exp(1j * lx[i0:i1, None] * ords[None, :K])
        S[i0:i1] = np.sqrt(x[i0:i1]) * np.sum(E * w * a[None, :K], axis=1)
        i0 = i1
    return S


xT = target_grid(float(XMAX))
TT = -2 + 2 * np.real(tapered(xT, tau, bR)) + trivial_rvm(xT)
nn = np.arange(1, XMAX + 1)
```

```
T = np.concatenate([[0.0], PchipInterpolator(xT, TT)(nn)])
nchk = np.arange(1000, 100001, 997)
full = -2 + 2 * np.real(np.array([np.sqrt(n) * np.sum(np.exp(1j * np.log(n) * tau) * bR) for n in nchk])) +
    trivial_rvm(nchk.astype(float))
say("target on %d grid points; at %d integers <= 1e5: max |M_RVM(n) (all zeros) - target(n)| = %.4f",
    xT.size, nchk.size, np.max(np.abs(full - T[nchk])))

# ------------------------------------------------------------------ ternary walk
mu_h = np.zeros(XMAX + 1, dtype=np.int8)
Mh = np.zeros(XMAX + 1)
sgn = np.zeros(XMAX + 1, dtype=np.int8)
cur = 0.0
for n in range(1, XMAX + 1):
    if sqf[n]:
        sgn[n] = 1 if cur < T[n] else -1
        cur += sgn[n]
        mu_h[n] = sgn[n]
    Mh[n] = cur
err = Mh[1:] - T[1:]
say("ternary walk: max |M_h - target| on [1,1e6] = %.4f, on [1e4,1e6] = %.4f", np.max(np.abs(err)), np.max(np.
    abs(err[9999:])))
dens = lambda a: [np.mean(a[1:] == -1), np.mean(a[1:] == 0), np.mean(a[1:] == 1)]
say("densities of (-1, 0, +1):  mu   %.6f %.6f %.6f", *dens(mu))
say("                           mu_h %.6f %.6f %.6f   (3/pi^2 = %.6f, 1 - 6/pi^2 = %.6f)", *dens(mu_h), 3 / np.
    pi ** 2, 1 - 6 / np.pi ** 2)
say("sign agreement mu_h = mu on squarefrees: %.6f;  sum mu mu_h / Q = %.6f",
    np.mean(mu_h[sqf] == mu[sqf]), np.sum(mu_h.astype(int) * mu.astype(int)) / sqf[1:].sum())
say("max |M(n)|/sqrt(n) on [10,1e6] = %.4f;  max |M_RVM(n)|/sqrt(n) = %.4f;  max |M_h(n)|/sqrt(n) = %.4f",
    np.max(np.abs(M[10:]) / np.sqrt(nn[9:])), np.max(np.abs(T[10:]) / np.sqrt(nn[9:])), np.max(np.abs(Mh[10:])
     / np.sqrt(nn[9:])))
say("\nNUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)")
say("%4s %4s %6s %10s %8s %6s %6s %6s %6s" % ("n", "mu", "mu^2", "target", "M_h(n-1)", "sign", "mu_h", "M_h",
    "M"))
rows = []
for n in range(1, 41):
    say("%4d %4d %6d %10.4f %8d %6s %6d %6d %6d" % (n, mu[n], abs(mu[n]), T[n], Mh[n - 1],
                                                    ("+1" if sgn[n] > 0 else ("-1" if sgn[n] < 0 else "-")),
                                                    mu_h[n], Mh[n], M[n]))
    rows.append([n, mu[n], abs(mu[n]), T[n], Mh[n - 1], sgn[n], mu_h[n], Mh[n], M[n]])
rows = np.array(rows, dtype=float)
write_csv("s05_ternary_example.csv", ["n", "mu", "mu2", "target", "Mh_prev", "sign", "mu_h", "Mh", "M"],
          [rows[:, i] for i in range(rows.shape[1])])

Ns = 10 ** np.arange(3, 7)
say("\nENERGY E_N = (1/N) sum_{n<=N} W(n)^2")
say("%10s %14s %14s %14s" % ("N", "E_N(M)", "E_N(M_RVM)", "E_N(M_h)"))
EN = []
for Nn in Ns:
    EN.append([Nn, np.mean(M[1:Nn + 1].astype(float) ** 2), np.mean(T[1:Nn + 1] ** 2), np.mean(Mh[1:Nn + 1] **
      2)])
    say("%10d %14.4f %14.4f %14.4f" % tuple(EN[-1]))
EN = np.array(EN)
write_csv("s05_energy.csv", ["N", "E_M", "E_MRVM", "E_Mh"], [EN[:, i] for i in range(4)])
dec = np.arange(1, XMAX + 1, 25)
write_csv("s05_walks_decimated.csv", ["n", "M", "M_RVM_target", "M_h"], [dec, M[dec], T[dec], Mh[dec]])
mu_h[1:].astype(np.int8).tofile(os.path.join(OUT, "mu_h_int8_1e6.bin"))

# ------------------------------------------------------------------ figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 2001)
ax[0].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[0].plot(w, T[w], "r-", lw=1.2, label=r"$M_{\rm RVM}^{\rm crit}(n)$")
ax[0].step(w, Mh[w], where="post", color="tab:green", lw=0.8, label=r"$M_h(n)$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Mertens functions on $[1,2000]$, every integer")
```

```
ax[0].legend(fontsize=8)
nb = 2000
bw = XMAX // nb
xb = (np.arange(nb) + 0.5) * bw
Mb = M[1:].reshape(nb, bw)
Tb = T[1:].reshape(nb, bw)
ax[1].fill_between(xb, Mb.min(1), Mb.max(1), color="0.6", label=r"$M(n)$ (range per block)")
ax[1].plot(xb, Tb.mean(1), "r-", lw=1.0, label=r"$M_{\rm RVM}^{\rm crit}(n)$")
ax[1].plot(xb, np.sqrt(xb), "k--", lw=0.8, label=r"$\pm\sqrt{n}$")
ax[1].plot(xb, -np.sqrt(xb), "k--", lw=0.8)
ax[1].set_xlabel("n")
ax[1].set_title(r"$[1,10^6]$, blocks of %d integers" % bw)
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig07_mertens.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 61)
ax[0].vlines(w - 0.15, 0, mu[w], color="k", lw=2, label=r"$\mu(n)$")
ax[0].vlines(w + 0.15, 0, mu_h[w], color="tab:green", lw=2, label=r"$\mu_h(n)$")
ax[0].axhline(0, color="k", lw=0.5)
ax[0].set_xlabel("n")
ax[0].set_yticks([-1, 0, 1])
ax[0].set_title(r"Ternary values: same zero set $\{\mu(n)=0\}$, signs chosen by tracking")
ax[0].legend(fontsize=8)
w = np.arange(1, 301)
ax[1].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[1].plot(w, T[w], "r.-", lw=0.8, ms=3, label=r"target $M_{\rm RVM}^{\rm crit}(n)$")
ax[1].step(w, Mh[w], where="post", color="tab:green", lw=1.2, label=r"$M_h(n)$")
ax[1].set_xlabel("n")
ax[1].set_title(r"The walk $M_h$ follows $M_{\rm RVM}^{\rm crit}$ within $\pm1$ step")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig08_ternary_walk.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
sel = xg <= 300
ax[0].step(np.arange(1, 301), M[1:301], where="post", color="k", lw=1.0, label=r"$M(x)$")
ax[0].plot(xg[sel], Mz[sel], "b-", lw=0.7, label=r"explicit formula, %d zeros of $\zeta$" % Nz)
ax[0].plot(xg[sel], Mc[sel], "r-", lw=1.0, label=r"$M_{\rm RVM}^{\rm crit}(x)$, %d $\tau_k$" % Nz)
ax[0].set_xlabel("x")
ax[0].set_title("Explicit formulas on [2,300]")
ax[0].legend(fontsize=7)
kk = np.arange(5000)
ax[1].plot(gam[kk], np.abs(zG[kk]) / np.log(gam[kk] / (2 * np.pi)), "b.", ms=1.5, label=r"$|\zeta'(\rho_k)|/\
    log(\gamma_k/2\pi)$")
ax[1].plot(tau[kk], np.abs(zR[kk]) / np.log(tau[kk] / (2 * np.pi)), "r.", ms=1.5, label=r"$|\zeta_{\rm RVM}'(\
    rho^{\rm RVM}_k)|/\log(\tau_k/2\pi)$")
ax[1].axhline(varpi, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
ax[1].set_xlabel("ordinate")
ax[1].set_ylim(0, 4)
ax[1].set_title(r"Derivatives at the zeros, $k\leq5000$")
ax[1].legend(fontsize=7)
fig.tight_layout()
savefig(fig, "fig09_explicit_weights.png")

sio.savemat(os.path.join(OUT, "s05_results.mat"), {"varpi": varpi, "cfit": cfit, "S2G": S2G, "S2R": S2R,
                                                   "energy": EN, "ternary_example": rows,
                                                   "x_explicit": xg, "M_zeta_explicit": Mz, "M_RVM_explicit":
    Mc})
say.close()
```

A.7. **s06_prime_ratios.py.**

```
"""Experiment 6: the three ratios li/pi_RVM, pi/li, pi/pi_RVM and the square-root estimate
```

```
pi_RVM(x) - li(x) = O(sqrt(x) (log log x)^2 / log x), from the prime counts written by s04_primes.py.

Input: output/s04_prime_counts_x1000_K17061.csv (columns x, pi_exact, pi_RVM, li, ...).
B(x) = sqrt(x) (log log x)^2 / log x is the scale of the square-root estimate."""
import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, savefig, write_csv, OUT

say = Tee("s06_prime_ratios")
D = np.genfromtxt(os.path.join(OUT, "s04_prime_counts_x1000_K17061.csv"), delimiter=",", names=True)
x, pe, pr, li = D["x"], D["pi_exact"], D["pi_RVM"], D["li"]
say("grid: %d points on [%.0f, %.0f]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)"
    % (x.size, x[0], x[-1]))

sel = x >= 10
r1, r2, r3 = li[sel] / pr[sel], pe[sel] / li[sel], pe[sel] / pr[sel]
xs = x[sel]
say("")
say("ratios at selected x")
say("%8s %12s %12s %12s %12s %12s %12s" % ("x", "pi", "pi_RVM", "li", "li/pi_RVM", "pi/li", "pi/pi_RVM"))
for X in [10, 20, 50, 100, 200, 500, 1000]:
    i = int(np.argmin(np.abs(x - X)))
    say("%8.1f %12.4f %12.4f %12.4f %12.6f %12.6f %12.6f"
        % (x[i], pe[i], pr[i], li[i], li[i] / pr[i], pe[i] / li[i], pe[i] / pr[i]))

s8 = x >= 8
B = np.sqrt(x[s8]) * np.log(np.log(x[s8])) ** 2 / np.log(x[s8])
e_rvm = (pr[s8] - li[s8]) / B
e_pi = (pe[s8] - li[s8]) / B
say("")
say("square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]")
say("max |pi_RVM - li| / B = %.4f   (at x = %.2f)" % (np.max(np.abs(e_rvm)), x[s8][np.argmax(np.abs(e_rvm))]))
say("max |pi - li| / B     = %.4f   (at x = %.2f)" % (np.max(np.abs(e_pi)), x[s8][np.argmax(np.abs(e_pi))]))
for X in [100, 1000]:
    m = x[s8] >= X / 2
    say("max |pi_RVM - li| / B on [%d, 1000] = %.4f" % (X // 2, np.max(np.abs(e_rvm[m]))))

# CSV files: every 10th grid point
write_csv("s06_prime_ratios.csv", ["x", "li_over_pi_RVM", "pi_over_li", "pi_over_pi_RVM"],
          [xs[::10], r1[::10], r2[::10], r3[::10]])
write_csv("s06_square_root_scale.csv", ["x", "pi_RVM_minus_li_over_B", "pi_minus_li_over_B"],
          [x[s8][::10], e_rvm[::10], e_pi[::10]])

# ---- figure (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].plot(xs, r3, "k-", lw=0.8, label=r"$\pi(x)/\pi_{\rm RVM}(x)$")
ax[0].plot(xs, r2, "b-", lw=0.8, label=r"$\pi(x)/\mathrm{li}(x)$")
ax[0].plot(xs, r1, "r-", lw=1.4, label=r"$\mathrm{li}(x)/\pi_{\rm RVM}(x)$")
ax[0].axhline(1.0, color="0.5", lw=0.8, ls="--")
ax[0].set_xlabel("x")
ax[0].set_title(r"The three ratios on $[10,1000]$")
ax[0].legend(fontsize=8)
ax[1].plot(x[s8], e_pi, "b-", lw=0.6, label=r"$(\pi-\mathrm{li})/B$")
ax[1].plot(x[s8], e_rvm, "r-", lw=1.4, label=r"$(\pi_{\rm RVM}-\mathrm{li})/B$")
ax[1].axhline(0.0, color="0.5", lw=0.8, ls="--")
ax[1].set_xlabel("x")
ax[1].set_title(r"Square-root scale $B(x)=\sqrt{x}\,(\log\log x)^2/\log x$")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig10_prime_ratios.png")
say.close()
```

A.8. **s07_product_identity.py.**

```
"""Experiment 7: the value at infinity of the endpoint.

varpi_Z1 = lim_{sigma -> +inf} zeta_Z1(sigma) = varpi_RVM * prod_k (1/4 + tau_k^2) / (1/4 + gamma_k^2),
so zeta_Z1 = zeta holds exactly when the ordered partial products
P_K = prod_{k <= K} (1/4 + gamma_k^2) / (1/4 + tau_k^2)
converge to varpi_RVM.  The partial products are computed from data/zeross.bin (gamma_k) and
data/zLamb.bin (tau_k) and compared with varpi_RVM from s02_transition.py."""
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, savefig, write_csv

say = Tee("s07_product_identity")
gam, tau = load_ordinates()
VARPI = 0.9619415807          # Richardson value of varpi_RVM (s02_transition.py)
N = gam.size
logP = np.cumsum(np.log1p(0.25 / gam ** 2) + 2 * np.log(gam) - np.log1p(0.25 / tau ** 2) - 2 * np.log(tau))
P = np.exp(logP)
say("N = %d ordinates; varpi_RVM = %.10f" % (N, VARPI))
say("%10s %16s %14s" % ("K", "P_K", "P_K - varpi"))
for K in [10, 100, 1000, 10 ** 4, 10 ** 5, 10 ** 6, N]:
    say("%10d %16.10f %14.3e" % (K, P[K - 1], P[K - 1] - VARPI))
for K0 in [10 ** 4, 10 ** 5, 10 ** 6]:
    seg = P[K0 - 1:]
    say("K >= %7d: min P_K = %.10f, max P_K = %.10f, mean = %.10f" % (K0, seg.min(), seg.max(), seg.mean()))
# the flow along the family: Upsilon(lam) = sum_k 2 gamma_k(lam) eta_k / (1/4 + gamma_k(lam)^2),
# eta_k = tau_k - gamma_k, gamma_k(lam) = (1 - lam) gamma_k + lam tau_k; d/dlam log varpi_lam = -Upsilon(lam)
eta = tau - gam
say("")
say("flow along the lambda-family")
for lam in [0.0, 0.25, 0.5, 0.75, 1.0]:
    gl = (1 - lam) * gam + lam * tau
    say("Upsilon(%.2f) = %.10f" % (lam, np.sum(2 * gl * eta / (0.25 + gl ** 2))))
say("sum_k 2 eta_k^2 / gamma_k^2 = %.10f" % np.sum(2 * eta ** 2 / gam ** 2))
total = np.sum(np.log((0.25 + tau ** 2) / (0.25 + gam ** 2)))
say("integral_0^1 Upsilon = sum_k log((1/4+tau_k^2)/(1/4+gamma_k^2)) = %.10f" % total)
say("log L = log(1/varpi_RVM)                                  = %.10f" % np.log(1 / VARPI))
step = 50
idx = np.arange(step - 1, N, step)
write_csv("s07_partial_products.csv", ["K", "P_K"], [idx + 1, P[idx]])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
k1 = np.arange(1, 10 ** 4 + 1)
ax[0].plot(k1, P[:10 ** 4], "r-", lw=0.6, label=r"$P_K$")
ax[0].axhline(VARPI, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
ax[0].set_xlabel("K")
ax[0].set_title(r"Partial products $P_K$, $K\leq 10^4$")
ax[0].legend(fontsize=8)
sel = idx >= 10 ** 5 - 1
ax[1].plot(idx[sel] + 1, (P[idx[sel]] - VARPI) * 1e6, "r-", lw=0.6)
ax[1].axhline(0.0, color="k", lw=0.8, ls="--")
ax[1].set_xlabel("K")
ax[1].set_ylabel(r"$(P_K-\varpi_{\rm RVM})\cdot 10^{6}$")
ax[1].set_title(r"$10^5\leq K\leq %d$" % N)
fig.tight_layout()
savefig(fig, "fig11_product_identity.png")
say.close()
```

## Appendix B. Console output of the programs

The complete console output of each program, as written to `output/*_console.txt`.

B.1. **s01_clock_and_family.**

```
zeros loaded: N = 2001052, gamma_1 = 14.134725142, gamma_N = 1132490.658714, tau_1 = 14.521346953, tau_N =
    1132490.722588
max |N_RVM(tau_k) - (k - 1/2)| = 6.985e-10
eta_k = tau_k - gamma_k:  max |eta| = 0.993711 (k = 34),  mean |eta| = 0.151209,  rms = 0.188066,  mean eta =
    1.731e-07
     k from         k to     max|eta|      rms eta
          1           10     0.840022     0.545768
         11          100     0.993711     0.453166
        101         1000     0.915450     0.345177
       1001        10000     0.833881     0.274177
      10001       100000     0.785835     0.226352
     100001      1000000     0.762816     0.192411
    1000001      2001052     0.717359     0.178900
first ordinates:
   k          gamma_k            tau_k        eta_k
   1     14.134725142     14.521346953     0.386622
   2     21.022039639     20.655740356    -0.366299
   3     25.010857580     25.492675432     0.481818
   4     30.424876126     29.739411632    -0.685464
   5     32.935061588     33.624531889     0.689470
   6     37.586178159     37.257370087    -0.328808
   7     40.918719012     40.700551604    -0.218167
   8     43.327073281     43.994013953     0.666941
   9     48.005150881     47.165128427    -0.840022
  10     49.773832478     50.233653391     0.459821
  11     52.970321478     53.214422874     0.244101
  12     56.446247697     56.118921948    -0.327326
sigma = 2.00, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 8.706e-10, median = 2.685e-10
sigma = 1.25, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 1.128e-09, median = 3.693e-10
sigma = 0.75, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 2.629e-09, median = 5.179e-10
critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = 8.370e-10
total run time 30.6 s
```

B.2. **s01b_clock_zoom.**

```
zoom table near the top of the file:
        k            gamma_k              tau_k      eta_k   N*-N_RVM
  2001041     1132485.303923     1132485.011569  -0.292355  -0.563104
  2001042     1132485.662638     1132485.530752  -0.131886  -0.254025
  2001043     1132486.244063     1132486.049936  -0.194127  -0.373908
  2001044     1132486.392256     1132486.569120   0.176863   0.340656
  2001045     1132486.874881     1132487.088303   0.213422   0.411072
  2001046     1132487.571868     1132487.607487   0.035619   0.068605
  2001047     1132487.950694     1132488.126670   0.175977   0.338949
  2001048     1132488.905328     1132488.645854  -0.259474  -0.499774
  2001049     1132489.162780     1132489.165037   0.002257   0.004348
  2001050     1132489.585215     1132489.684221   0.099006   0.190696
  2001051     1132490.165305     1132490.203404   0.038099   0.073383
  2001052     1132490.658714     1132490.722588   0.063873   0.123026
total run time 0.6 s
```

B.3. **s02_transition.**

```
varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000) = 0.9619415807 -1.28e-10 i
L = 1/varpi_RVM = 1.0395641690
sigma' = 0.25: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0617
sigma' = 0.25: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = 0.0790
          |w| * t^sigma' in [1.5832, 1.5832];  |1/zeta_RVM - L| at t = 1e4: 1.425e-01
sigma' = 0.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0618
          |w| * t^sigma' in [2.5061, 2.5066];  |1/zeta_RVM - L| at t = 1e4: 2.543e-02
sigma' = 1.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.182, max t*err = 0.0620
          |w| * t^sigma' in [15.6619, 15.7496];  |1/zeta_RVM - L| at t = 1e4: 1.598e-05
total run time 11.5 s
```

B.4. **s03_exponential_sums.**

```
     x     max|S_RVM| max|S_RVM|/log    |S_zeta(N)| Landau -T/2pi*L/sqrt(x)   |M_HˆRVM(x)|
   2.0         3.5745         0.2695       88300.77        -88341.59      1.722e-06
   3.0         3.9750         0.3759      114267.92       -114324.31      1.175e-06
   4.0         3.9523         0.3732       62435.29        -62466.94      8.004e-07
   6.0         4.5264         0.4547           0.26            -0.00      2.262e-06
  10.0         9.6989         1.1495           2.42            -0.00      4.031e-06
total run time 1.7 s
```

B.5. **s04_primes.**

```
x <= 100, all 2001052 zeros of each list, 3000 grid points:
  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): 0.4035
  pi_RVM(100) = 26.0782, pi(100) = 25, li(100) = 30.1261, J_RVM(100) = 29.5889, J(100) = 28.5333
  pi_RVM monotone on [2,100]: False (min increment on the grid -2.528e-01)
x <= 1000, smooth cut at K = 17061 zeros (tau_K = 15707.6), 40000 grid points:
  pi_RVM(1000) = 169.9750, pi(1000) = 168, li(1000) = 177.6097
cell masses c_n = pi_RVM(n) - pi_RVM(n-1):
    n        c_n    1/log n    prime
    2    0.52944    1.44270        1
    3    1.04188    0.91024        1
    4    0.53593    0.72135        0
    5    0.46739    0.62133        1
    6    0.36089    0.55811        0
    7    0.63254    0.51390        1
    8    0.62393    0.48090        0
    9   -0.37669    0.45512        0
   10    0.17151    0.43429        0
   11    0.69779    0.41703        1
   12    0.32703    0.40243        0
   13    0.36227    0.38987        1
   14    0.57662    0.37892        0
   15    0.19671    0.36927        0
   16   -0.05284    0.36067        0
   17    0.39958    0.35296        1
   18    0.76865    0.34598        0
   19    0.43500    0.33962        1
   20   -0.03342    0.33381        0
   21    0.03009    0.32846        0
   22    0.37466    0.32352        0
   23    0.40277    0.31893        1
   24    0.11768    0.31466        0
   25    0.01006    0.31067        0
   26    0.27350    0.30693        0
   27    0.55508    0.30341        0
   28    0.47627    0.30010        0
   29    0.12577    0.29697        1
   30   -0.08367    0.29401        0
   31    0.07329    0.29121        1
   32    0.38040    0.28854        0
   33    0.43927    0.28600        0
   34    0.34619    0.28358        0
   35    0.06419    0.28127        0
   36   -0.02475    0.27906        0
   37    0.15448    0.27694        1
   38    0.40992    0.27491        0
   39    0.49628    0.27296        0
   40    0.35226    0.27109        0
   41    0.14030    0.26928        1
sum_{n<=1000} c_n = 169.9750,  pi(1000) = 168
real RVM prime levels p_jˆRVM (first crossing of level j) versus p_j:
    j      p_jˆRVM      p_j       diff
    1       2.2139        2     0.2139
    2       3.1550        3     0.1550
    3       5.1887        5     0.1887
    4       7.1593        7     0.1593
```

```
    5      11.1406       11     0.1406
    6      14.0208       13     1.0208
    7      17.2552       17     0.2552
    8      19.1781       19     0.1781
    9      26.0536       23     3.0536
   10      28.0447       29    -0.9553
   11      33.0707       31     2.0707
   12      38.0931       37     1.0931
   13      40.2282       41    -0.7718
   14      44.1861       43     1.1861
   15      46.1221       47    -0.8779
   16      57.1779       53     4.1779
   17      59.1201       59     0.1201
   18      61.0809       61     0.0809
   19      67.0384       67     0.0384
   20      74.0931       71     3.0931
   21      76.1235       73     3.1235
   22      83.0822       79     4.0822
   23      86.0513       83     3.0513
   24      92.0825       89     3.0825
   25      94.0781       97    -2.9219
   26      96.1319      101    -4.8681
   27     103.2219      103     0.2219
   28     112.1163      107     5.1163
   29     115.0277      109     6.0277
   30     118.1431      113     5.1431
total run time 107.7 s
```

B.6. **s05_mertens.**

```
zeta_RVM'(rho_k) exact for k <= 1000; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = 1000: 0.9619388429 -1.41e-05 i
varpi_RVM = 0.9619415807; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = -0.0052 -0.0208 i
Riemann-Siegel zeta'(rho) versus mpmath at k = 1001..1050: max relative difference 1.19e-05
median |zeta'(rho)| = 7.8899;  spectral sums 2 sum|a|^2: zeta 0.029034, RVM 0.027203
M(10^6) = 212;  squarefree density Q(10^6)/10^6 = 0.607926 (6/pi^2 = 0.607927)
explicit formula with 2001052 zeros of zeta on [2,2000]: rms(M_zeta - M) = 0.0005 off integers (|x-n|>0.05)
target on 49896 grid points; at 100 integers <= 1e5: max |M_RVM(n) (all zeros) - target(n)| = 0.0191
ternary walk: max |M_h - target| on [1,1e6] = 2.6155, on [1e4,1e6] = 1.6539
densities of (-1, 0, +1):  mu   0.303857 0.392074 0.304069
                           mu_h 0.304003 0.392074 0.303923   (3/pi^2 = 0.303964, 1 - 6/pi^2 = 0.392073)
sign agreement mu_h = mu on squarefrees: 0.500344;  sum mu mu_h / Q = 0.000688
max |M(n)|/sqrt(n) on [10,1e6] = 0.8321;  max |M_RVM(n)|/sqrt(n) = 0.7541;  max |M_h(n)|/sqrt(n) = 0.6882

NUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)
   n   mu   mu^2     target M_h(n-1)   sign   mu_h    M_h      M
   1    1      1     0.2925        0     +1      1      1      1
   2   -1      1     0.4563        1     -1     -1      0      0
   3   -1      1    -0.6523        0     -1     -1     -1     -1
   4    0      0    -0.8720       -1      -      0     -1     -1
   5   -1      1    -1.7255       -1     -1     -1     -2     -2
   6    1      1    -1.3063       -2     +1      1     -1     -1
   7   -1      1    -1.5931       -1     -1     -1     -2     -2
   8    0      0    -2.4029       -2      -      0     -2     -2
   9    0      0    -1.3292       -2      -      0     -2     -2
  10    1      1    -1.3280       -2     +1      1     -1     -1
  11   -1      1    -2.3272       -1     -1     -1     -2     -2
  12    0      0    -2.2067       -2      -      0     -2     -2
  13   -1      1    -1.8468       -2     +1      1     -1     -3
  14    1      1    -1.9324       -1     -1     -1     -2     -2
  15    1      1    -1.5435       -2     +1      1     -1     -1
  16    0      0    -1.0414       -1      -      0     -1     -1
  17   -1      1    -1.6970       -1     -1     -1     -2     -2
  18    0      0    -3.0095       -2      -      0     -2     -2
  19   -1      1    -3.2869       -2     -1     -1     -3     -3
  20    0      0    -2.2604       -3      -      0     -3     -3
```

```
  21    1      1    -1.2515       -3     +1      1     -2     -2
  22    1      1    -1.1120       -2     +1      1     -1     -1
  23   -1      1    -1.3607       -1     -1     -1     -2     -2
  24    0      0    -1.3669       -2      -      0     -2     -2
  25    0      0    -1.3810       -2      -      0     -2     -2
  26    1      1    -1.9805       -2     +1      1     -1     -1
  27    0      0    -3.0141       -1      -      0     -1     -1
  28    0      0    -3.6155       -1      -      0     -1     -1
  29   -1      1    -3.2066       -1     -1     -1     -2     -2
  30   -1      1    -2.1599       -2     -1     -1     -3     -3
  31   -1      1    -1.3649       -3     +1      1     -2     -4
  32    0      0    -1.2973       -2      -      0     -2     -4
  33    1      1    -1.6435       -2     +1      1     -1     -3
  34    1      1    -1.7783       -1     -1     -1     -2     -2
  35    1      1    -1.4711       -2     +1      1     -1     -1
  36    0      0    -1.0723       -1      -      0     -1     -1
  37   -1      1    -1.0774       -1     -1     -1     -2     -2
  38    1      1    -1.5897       -2     +1      1     -1     -1
  39    1      1    -2.2227       -1     -1     -1     -2      0
  40    0      0    -2.4881       -2      -      0     -2      0

ENERGY E_N = (1/N) sum_{n<=N} W(n)^2
         N         E_N(M)     E_N(M_RVM)       E_N(M_h)
      1000        14.6400        18.6280        18.5960
     10000       163.6747       195.6965       196.0683
    100000      1622.3189      1308.3041      1308.8186
   1000000     14851.0857     12537.9008     12538.1174
total run time 132.6 s
```

B.7. **s06_prime_ratios.**

```
grid: 40000 points on [2, 1000]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)

ratios at selected x
       x           pi       pi_RVM           li    li/pi_RVM        pi/li    pi/pi_RVM
    10.0       4.0000       3.9876       6.1657     1.546218     0.648747     1.003105
    20.0       8.0000       7.6636       9.9052     1.292496     0.807657     1.043893
    50.0      15.0000      14.7686      18.4695     1.250592     0.812151     1.015670
   100.0      25.0000      26.0813      30.1265     1.155098     0.829835     0.958541
   200.0      46.0000      43.3656      50.1910     1.157393     0.916499     1.060749
   500.0      95.0000      93.7489     101.7975     1.085854     0.933225     1.013346
  1000.0     168.0000     169.9750     177.6097     1.044916     0.945895     0.988381

square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]
max |pi_RVM - li| / B = 2.7213   (at x = 9.73)
max |pi - li| / B     = 2.4490   (at x = 11.00)
max |pi_RVM - li| / B on [50, 1000] = 1.3455
max |pi_RVM - li| / B on [500, 1000] = 0.7216
total run time 1.0 s
```

B.8. **s07_product_identity.**

```
N = 2001052 ordinates; varpi_RVM = 0.9619415807
         K             P_K    P_K - varpi
        10     0.9625640008      6.224e-04
       100     0.9624787401      5.372e-04
      1000     0.9622891707      3.476e-04
     10000     0.9619513787      9.798e-06
    100000     0.9619301865     -1.139e-05
   1000000     0.9619418189      2.382e-07
   2001052     0.9619415390     -4.166e-08
K >=   10000: min P_K = 0.9618023968, max P_K = 0.9620641927, mean = 0.9619418623
K >=  100000: min P_K = 0.9619210208, max P_K = 0.9619605076, mean = 0.9619417591
K >= 1000000: min P_K = 0.9619389488, max P_K = 0.9619439825, mean = 0.9619416633

flow along the lambda-family
```

```
Upsilon(0.00) = 0.0436949578
Upsilon(0.25) = 0.0412445375
Upsilon(0.50) = 0.0387987180
Upsilon(0.75) = 0.0363572218
Upsilon(1.00) = 0.0339197739
sum_k 2 eta_k^2 / gamma_k^2 = 0.0098210688
integral_0^1 Upsilon = sum_k log((1/4+tau_k^2)/(1/4+gamma_k^2)) = 0.0388016004
log L = log(1/varpi_RVM)                                = 0.0388015571
total run time 1.5 s
```

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